



Power Electronics

alexis.martin@ensea.fr
Desk D216



Layout

- Non isolated choppers
 - Introduction
 - Direct conversion
 - Buck
 - Boost
 - Closed loop chopper control
 - Reversible choppers
 - Half bridge
 - H – bridge
 - Inverter
 - Energy storage based chopper
 - Buck-boost
 - Cuck
 - SEPIC
- Isolated power supplies
- Power components
- PFC

Progress

- Lesson : 8 x 2h
- Tutorials : 9 x 2h
- Labs : 4 x 4h
 - Buck chopper
 - Reversible chopper for DC motor
 - Switch mode power supply Flyback
 - Switch mode power supply Forward
- Marks : Final exam - Labs

Prerequisites

- Kirchhoff's circuit laws
- Average value of a signal, integral calculus, RMS value
- AC and DC real power, power factor, complex representation, phasor
- Transformers (magnetic field, flux)
- Linear control (Laplace transform, Bode diagram, gain/phase margin, PI controller)

Introduction

Direct conversion

Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Introduction

Introduction

Direct conversion

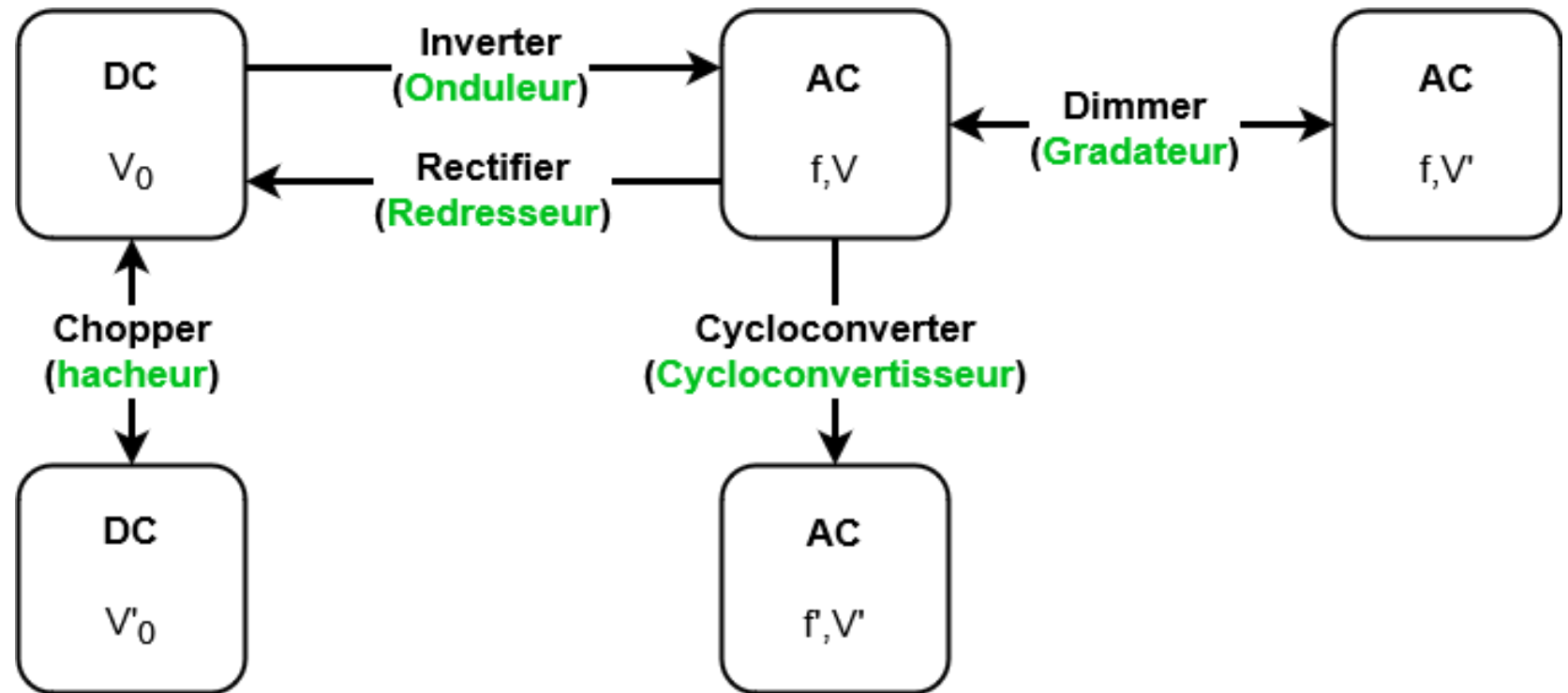
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Energy conversion: principles



Introduction

Direct conversion

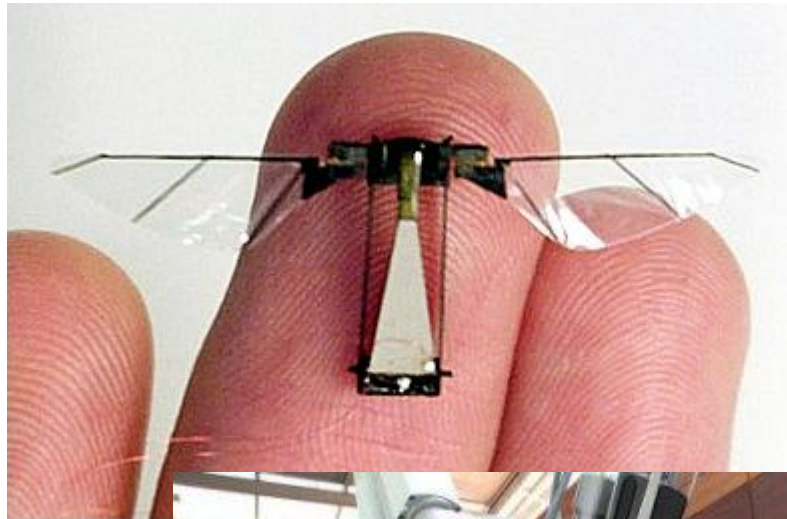
Reversible
choppers

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Energy storage
based chopper

Micro-mechanisms



1-100 W

Introduction

Direct conversion

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based chopper

Use : automatisms



0,1 à 10 kW

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Use : rail traction



100 kW

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High speed rail traction



10 MW

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Direct conversion

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Photovoltaic plant



300 MW_c

Introduction

Direct conversion

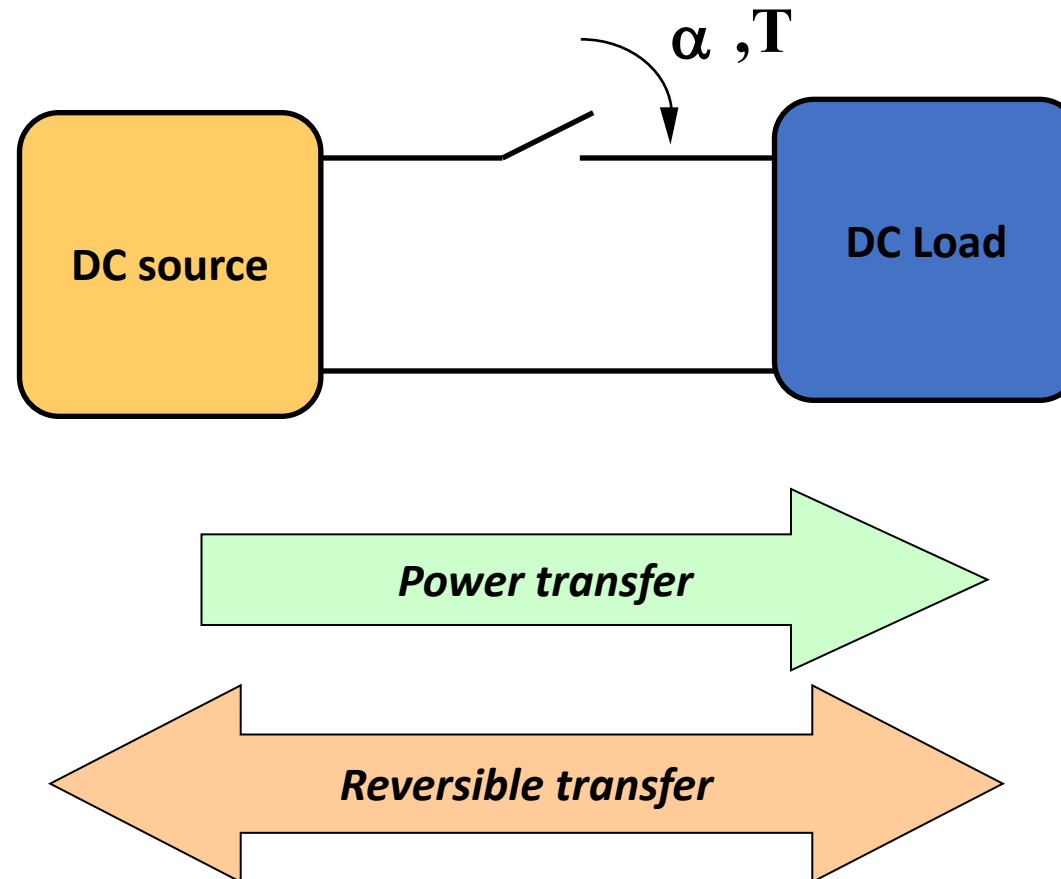
Reversible
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«Chopper » function : a switching electronic



Introduction

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Link power sources

Voltage Source

Active
(Energy generator)

Current Source

Passive
(Energy receiver)

Theorem

*Only two sources with different category
can be directly connected.*

Introduction

Direct conversion

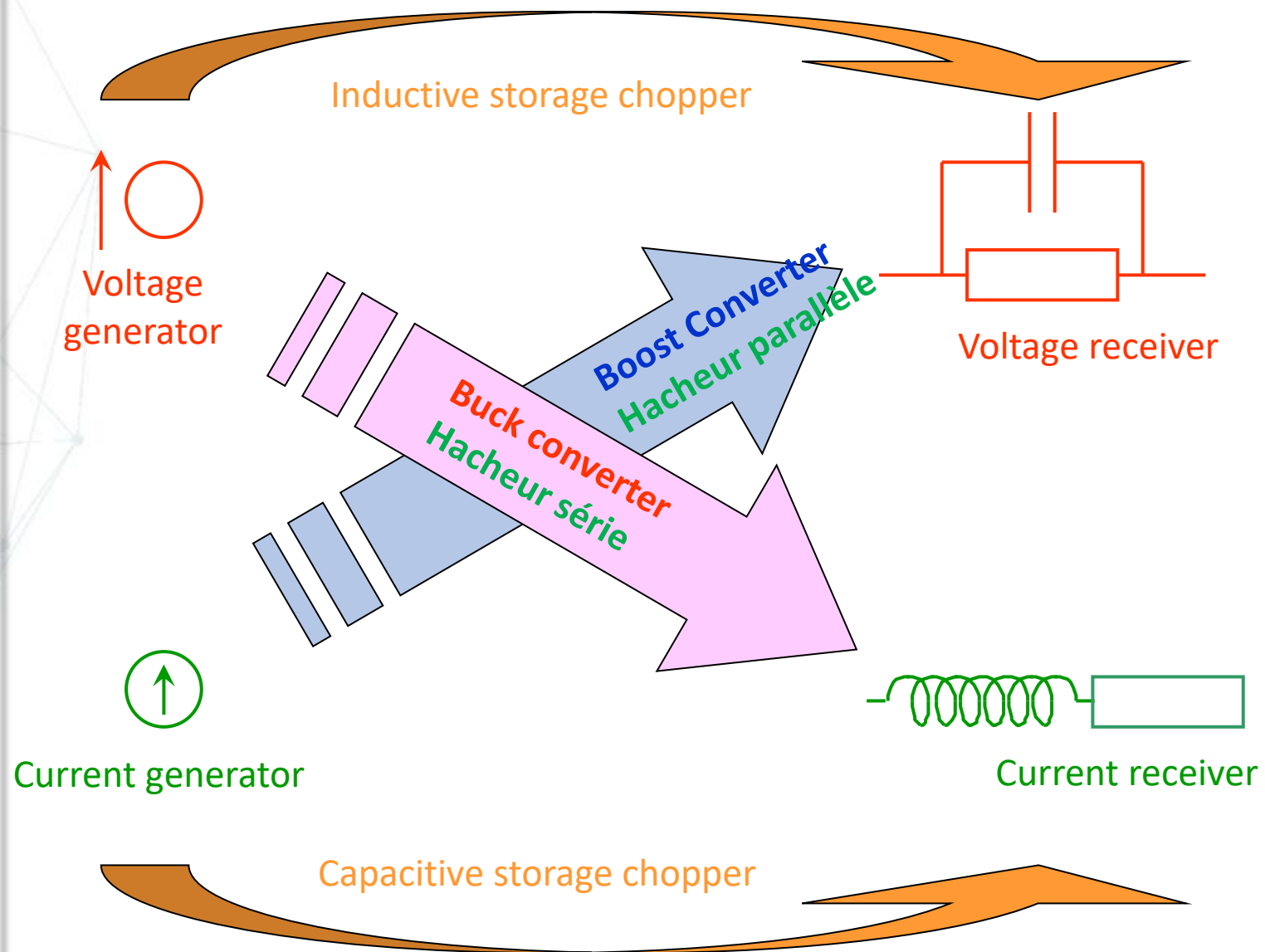
Reversible
choppers

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Energy storage
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Non reversible chopper



Introduction

Direct conversion

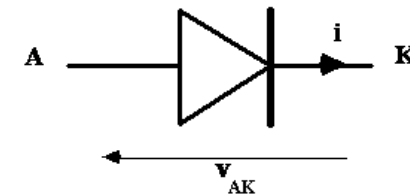
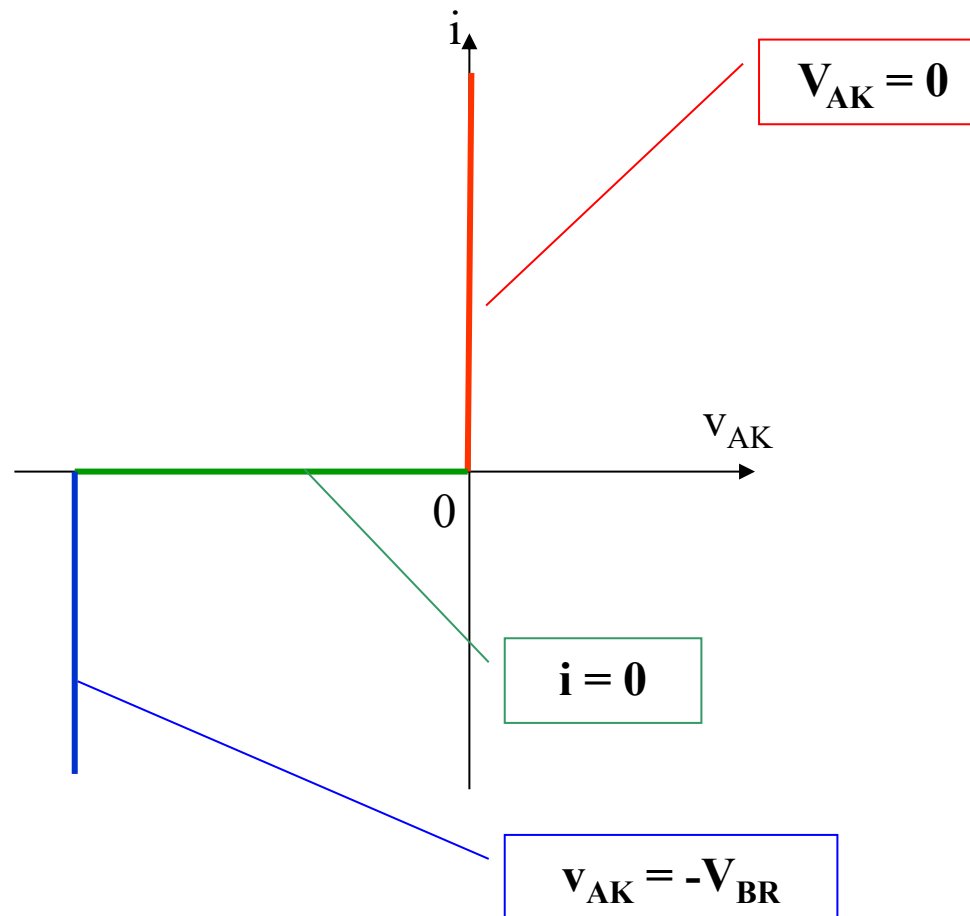
Reversible
choppers

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Closed loop
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Energy storage
based chopper

« Perfect » diode



Introduction

Direct conversion

Reversible
choppers

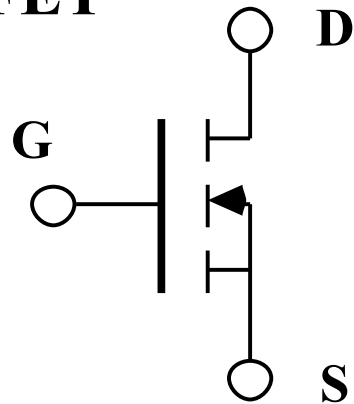
Inverters

Closed loop
chopper control

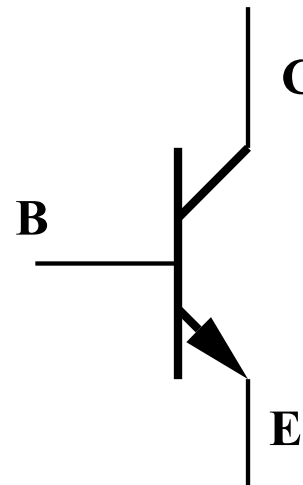
Energy storage
based chopper

Controlled switch

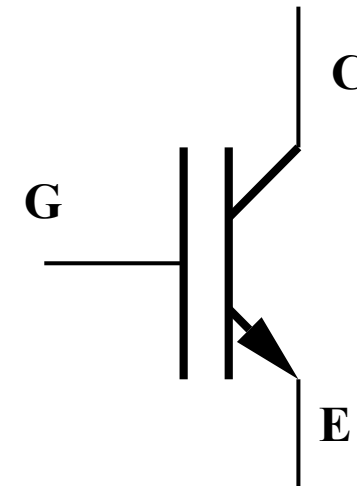
MOSFET



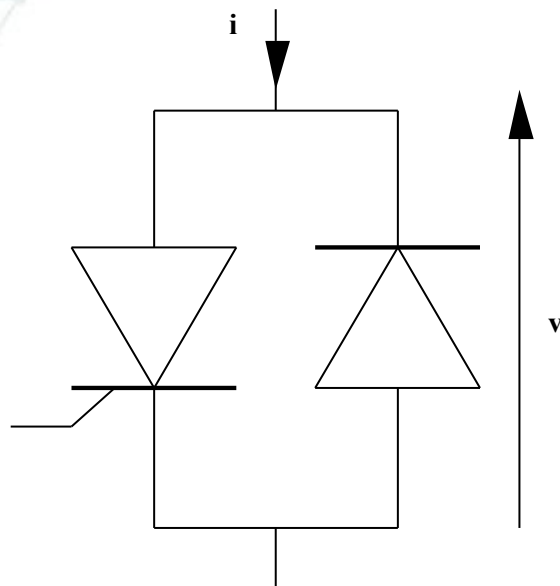
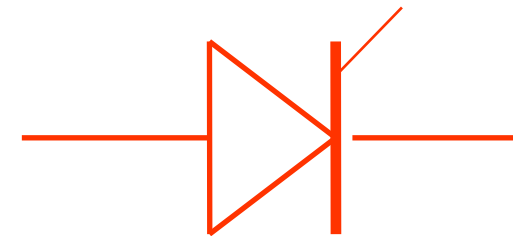
**Bipolar
transistor**



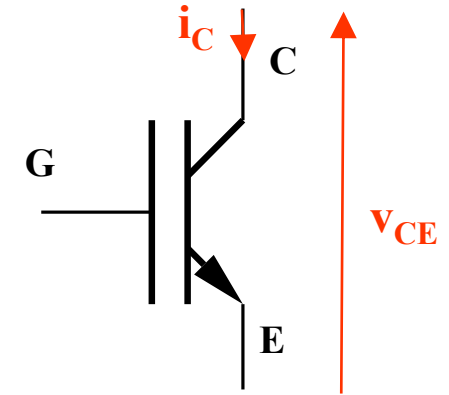
IGBT



Thyristor



« Perfect » controlled switch ex : IGBT



- On state: « short circuit »
 - $v_{CE} = 0$, i_C fixed by circuit
- Off state: « open circuit »
 - $i_C = 0$, v_{CE} fixed by circuit

Introduction

Direct conversion

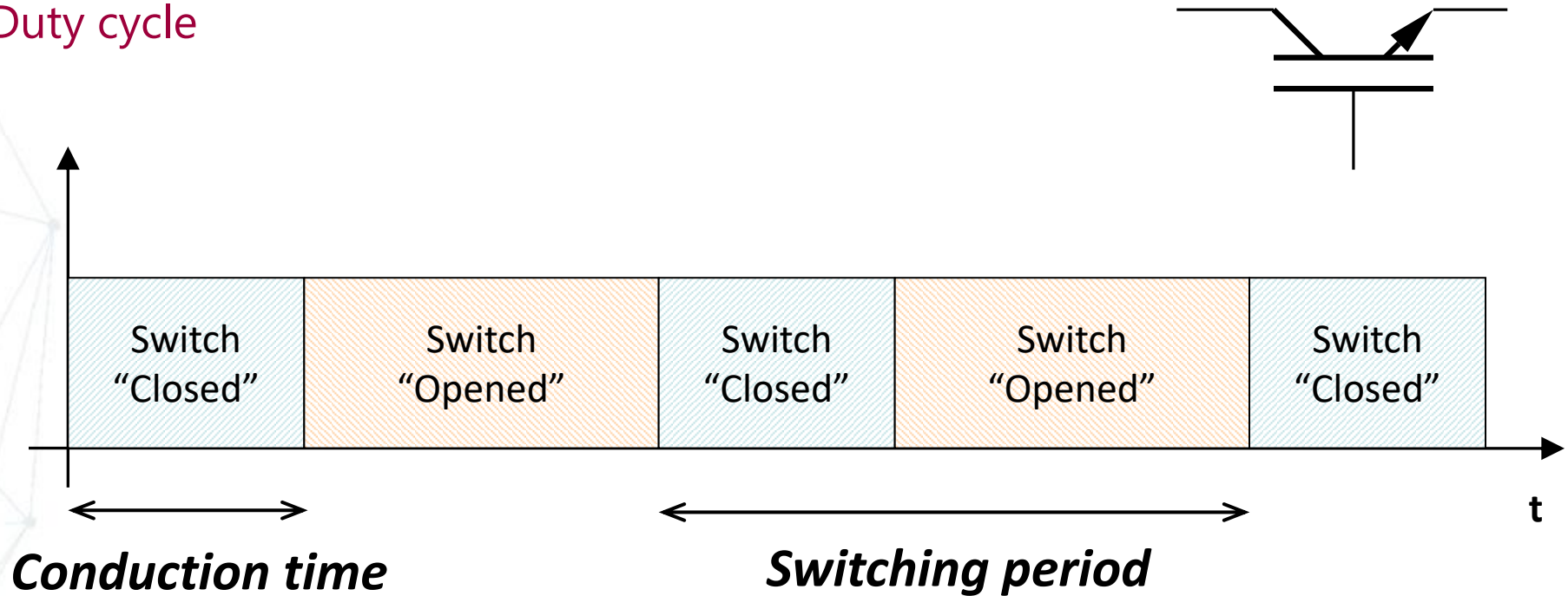
Reversible
choppers

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Duty cycle



$$\alpha = \frac{\text{Conduction time}}{\text{Switching period}}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

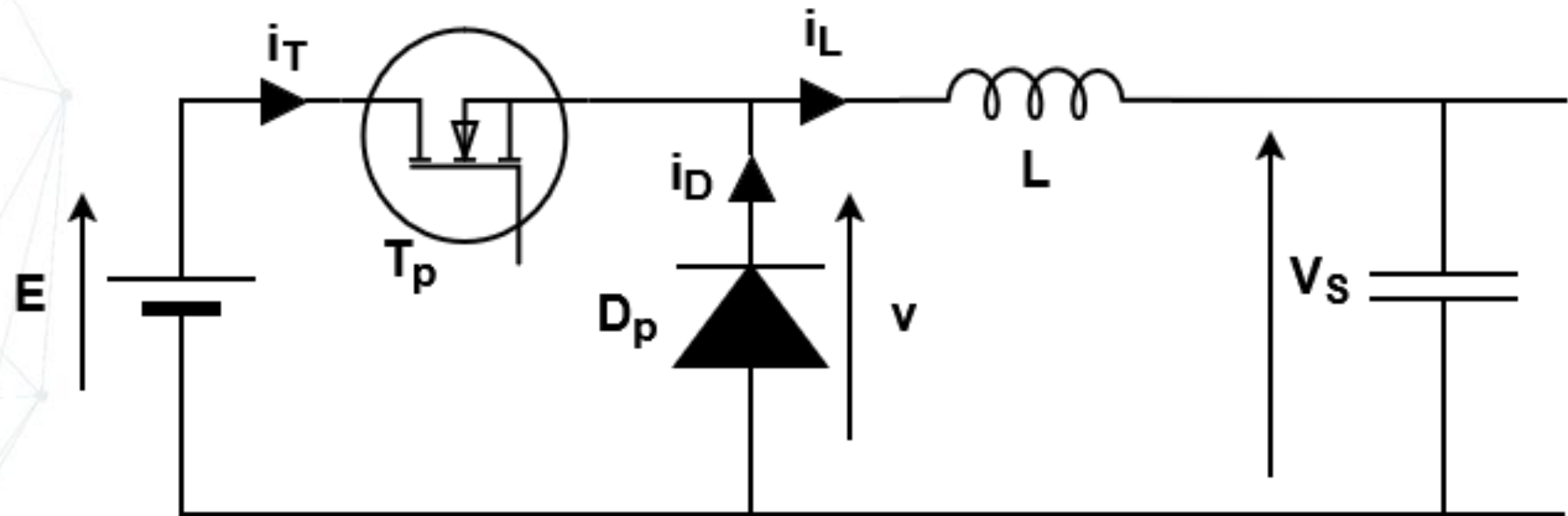
Inverters

Closed loop
chopper control

Energy storage
based chopper

Buck Chopper

Buck chopper : principle



Hypothesis : $V_s = \text{constant}$

Introduction

Direct conversion

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Boost chopper

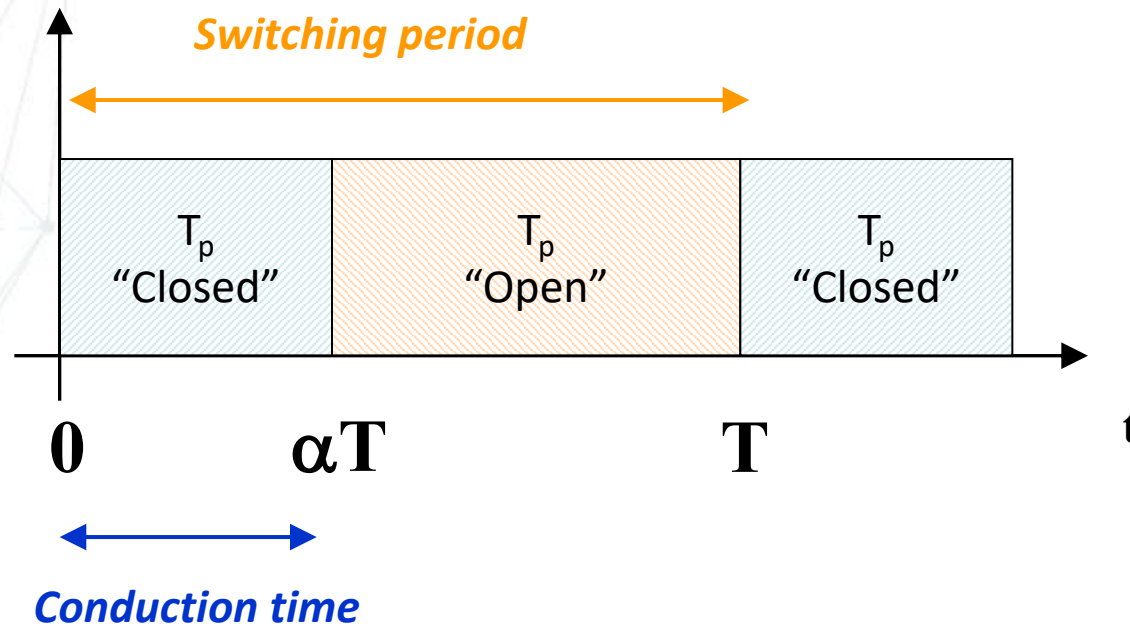
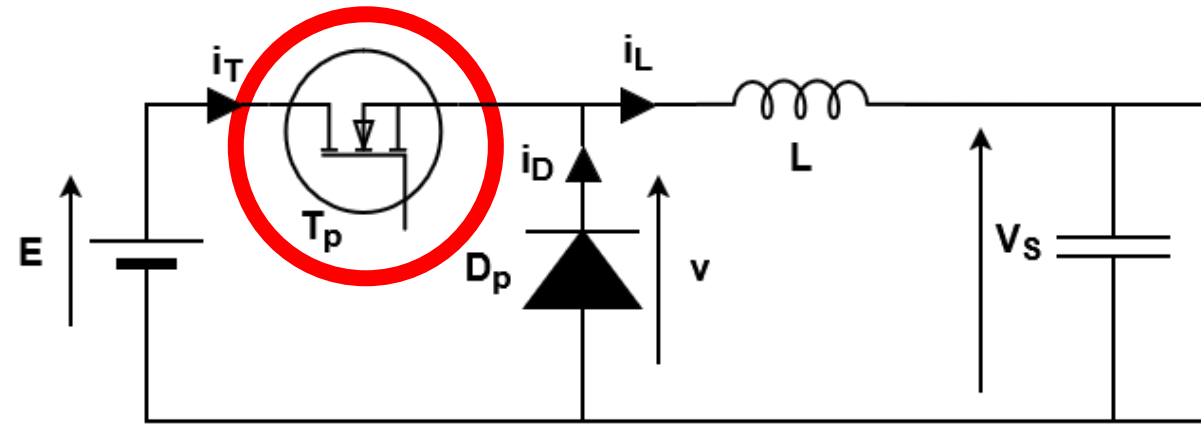
Reversible
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Control



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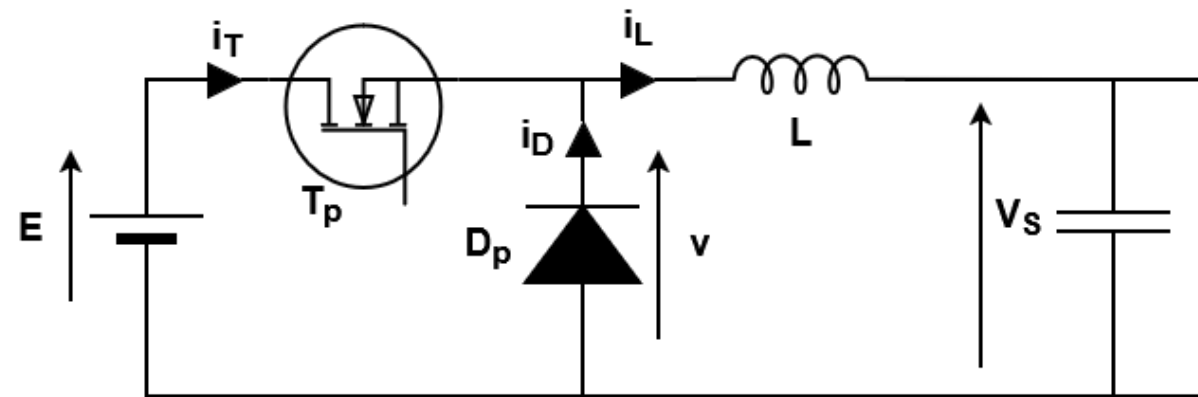
Reversible choppers

Inverters

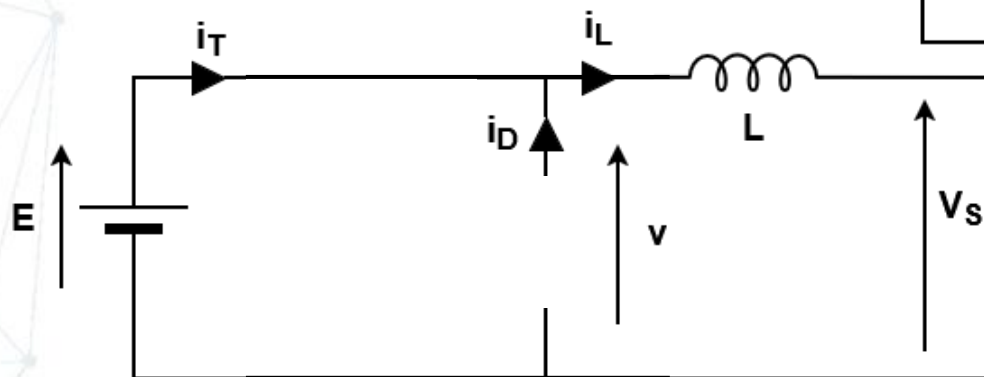
Closed loop chopper control

Energy storage based chopper

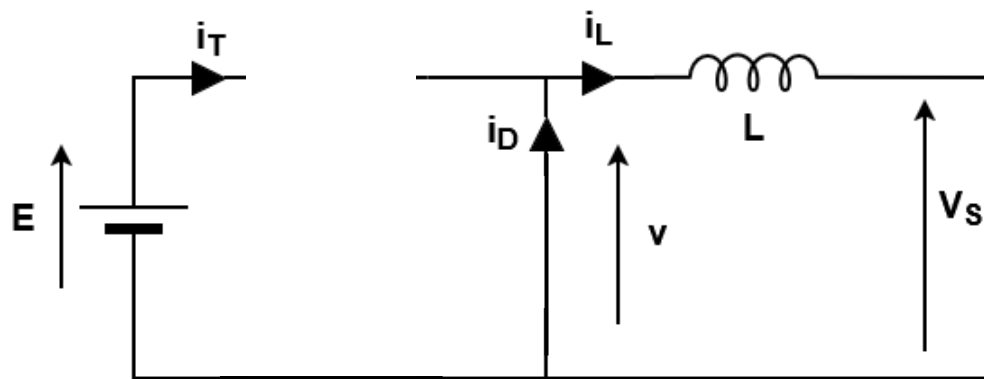
Operation



$$0 < t < \alpha T$$

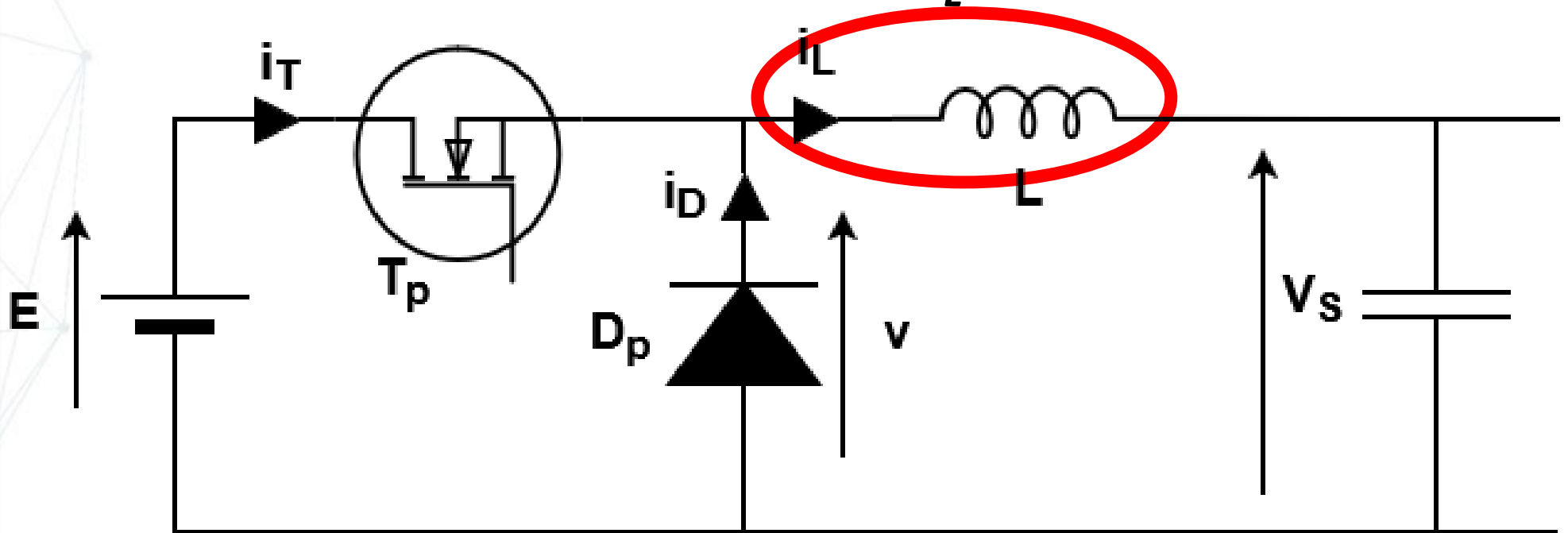


$$\alpha T < t < T$$



Modes

- ***Continuous conduction mode: $i_L > 0$***



- ***Discontinuous conduction mode: $i_L(t_0) = 0 ; t_0 < T$***

Introduction

Direct conversion

Buck chopper

Boost chopper

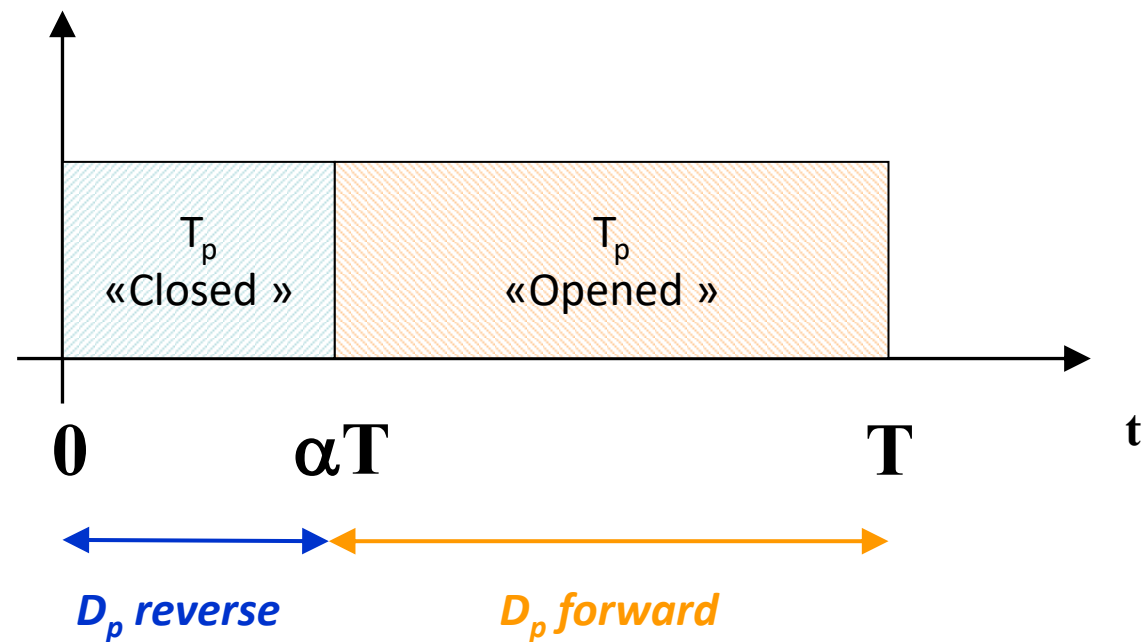
Reversible
choppers

Inverters

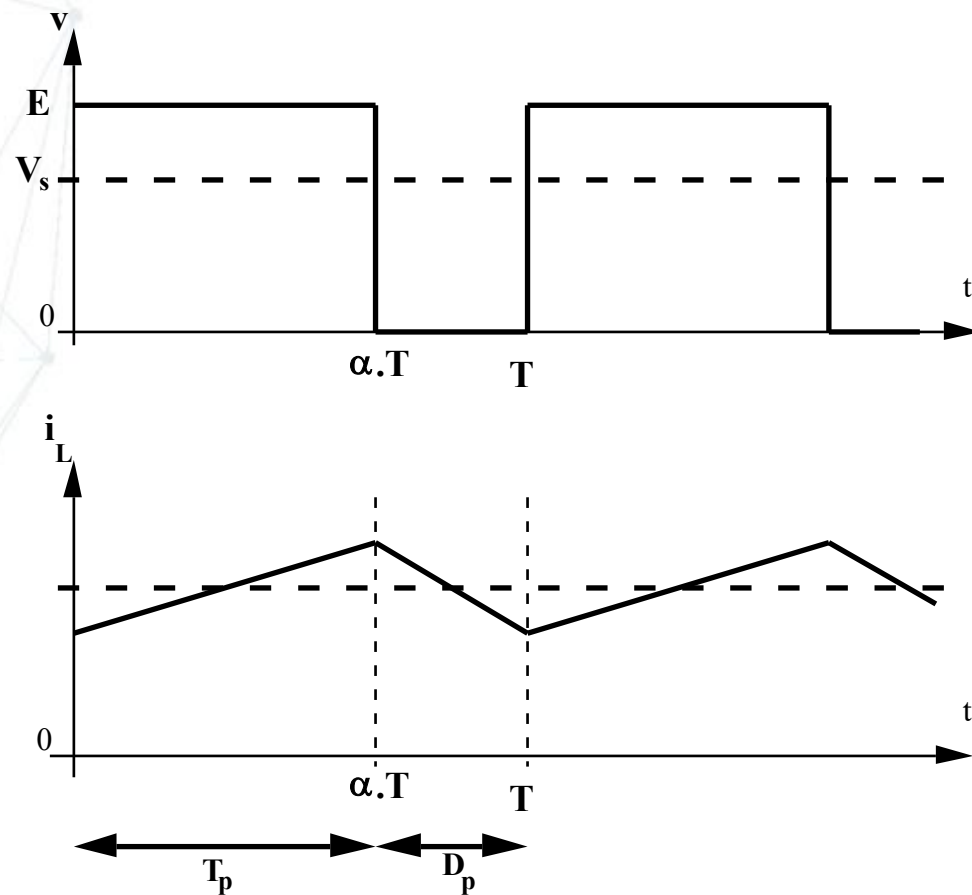
Closed loop
chopper control

Energy storage
based chopper

Continuous mode



Continuous mode: Chronograms



- $0 < t < \alpha T$

$$\frac{di}{dt} = \frac{E - V_s}{L}$$

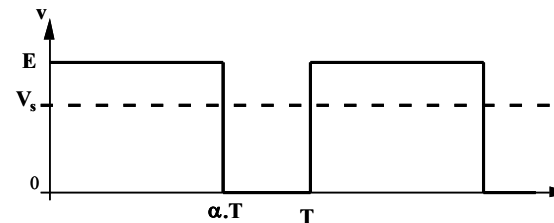
- $\alpha T < t < T$

$$\frac{di}{dt} = -\frac{V_s}{L}$$

$$E > V_s$$

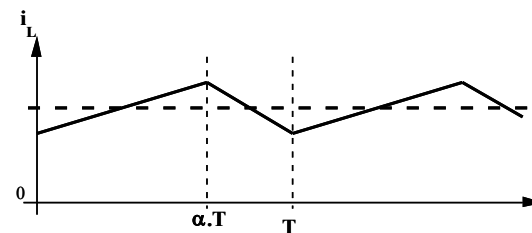
Continuous mode (Calculation)

Output voltage



$$\langle v \rangle = \alpha \cdot E = V_s$$

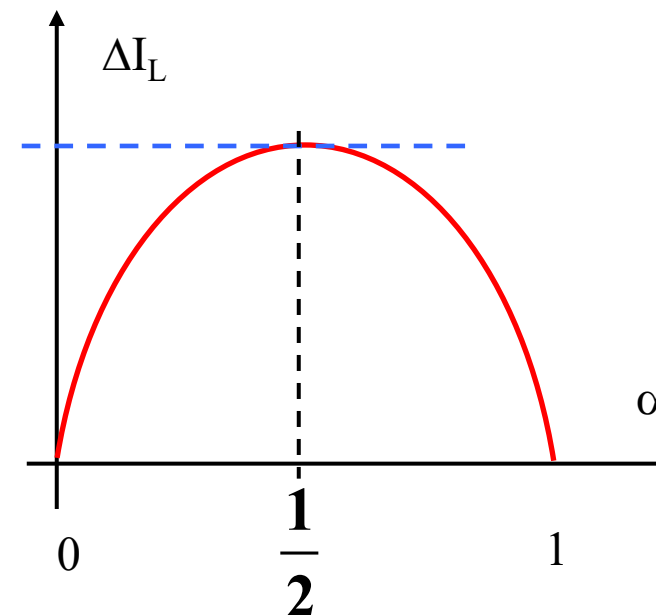
Current ripple i_L



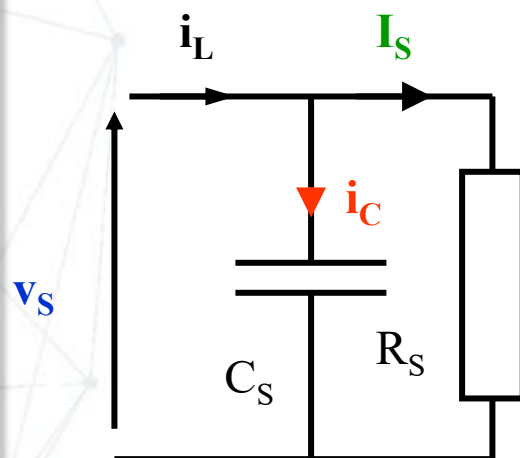
$$\Delta I_L = \alpha \cdot (1 - \alpha) \cdot \frac{E \cdot T}{L}$$

Maximum ripple

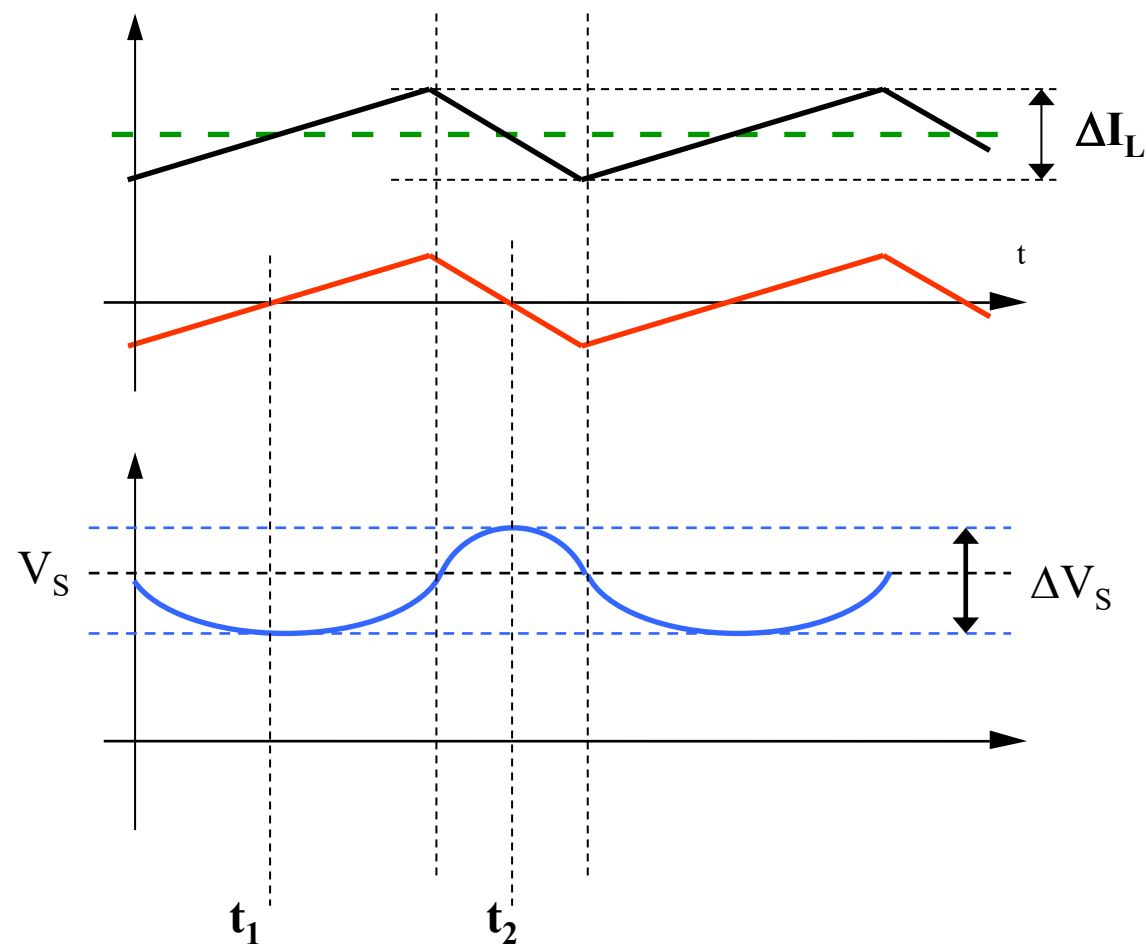
$$\Delta I_{L \max} = \frac{E \cdot T}{4 L}$$



Continuous mode: Output filtering principle



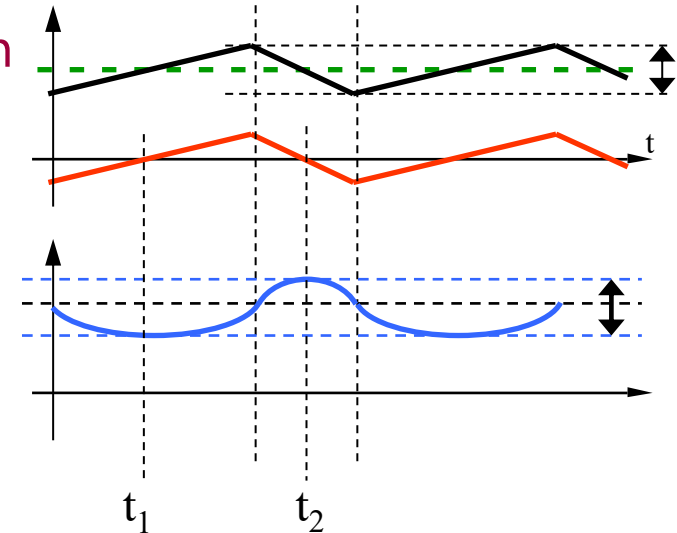
$$i_c = C_s \cdot \frac{dv_s}{dt}$$



Continuous mode: Output filtering calculation

$$t_1 = \alpha \frac{T}{2}$$

$$t_2 = \frac{(1+\alpha).T}{2}$$

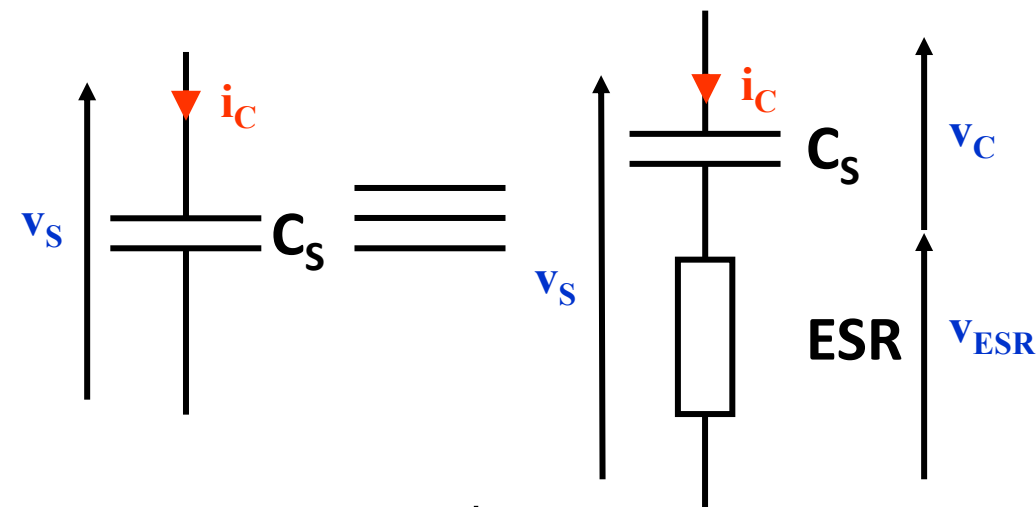
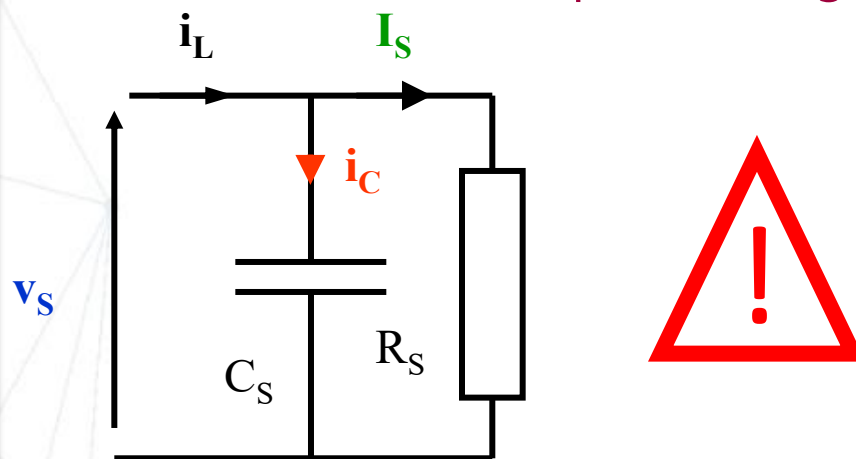


$$\Delta I_L = \alpha \cdot (1 - \alpha) \cdot \frac{E \cdot T}{L}$$

$$\Delta V_S = \frac{1}{C_S} \cdot \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{1}{C_S} \cdot \frac{1}{2} \cdot \frac{\Delta I_L}{2} \cdot (t_2 - t_1) = \frac{\Delta I_L \cdot T}{8 \cdot C_S}$$

$$C_S = \frac{\alpha \cdot (1 - \alpha) \cdot E \cdot T^2}{8 \cdot L \cdot \Delta V_S}$$

Continuous mode: Output filtering calculation

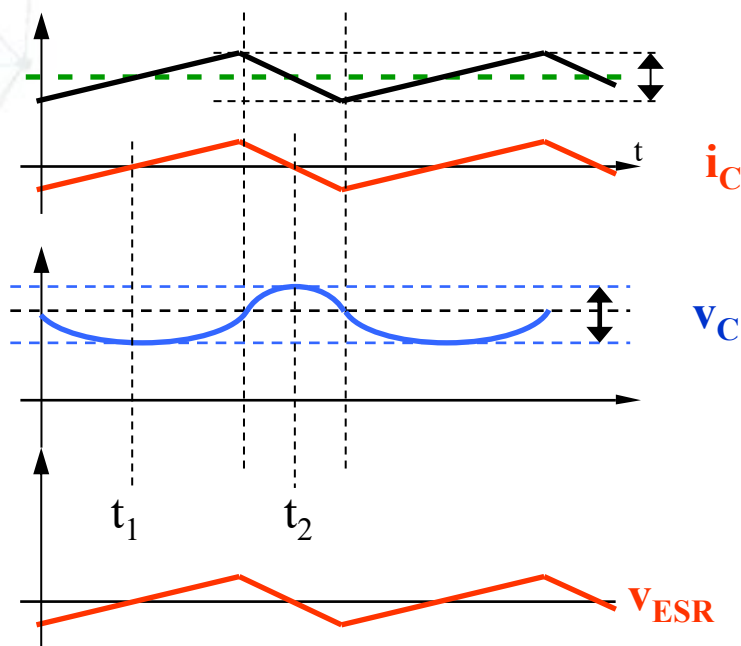


$$v_s = v_C + v_{ESR}$$

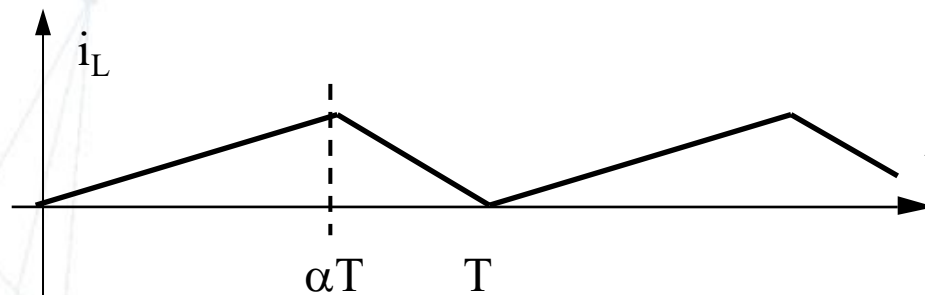
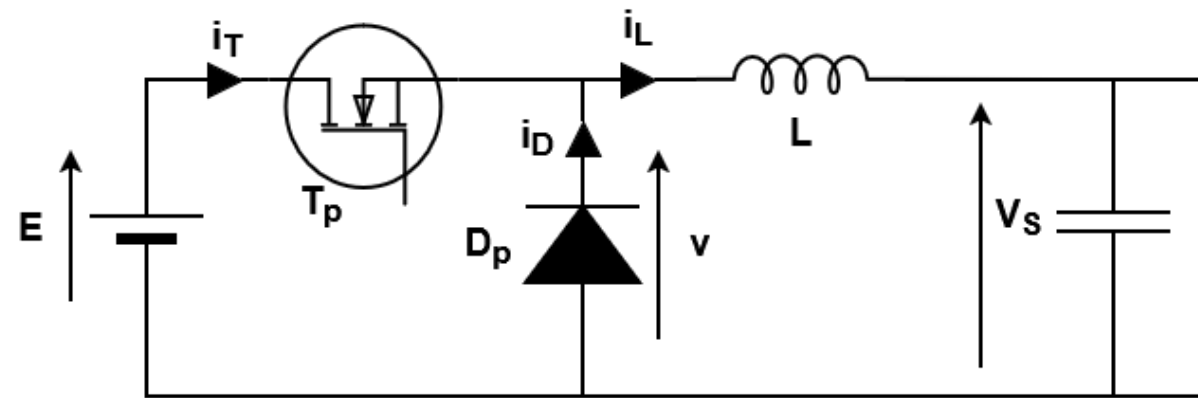
$$\Delta v_s < \Delta v_C + \Delta v_{ESR}$$

For high value of C_s ,
 $\Delta v_s \approx \Delta v_{ESR} = ESR \cdot \Delta i_L$

$$ESR < \frac{\Delta v_s}{\Delta i_L}$$



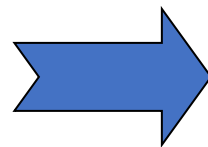
Boundary mode



$$i_L(T) = 0$$

$$\langle I_L \rangle = \frac{1}{2} \cdot \left(\frac{E - V_s}{L} \cdot \alpha T \right) = \frac{1}{2} \cdot \frac{\alpha(1-\alpha) \cdot E \cdot T}{L}$$

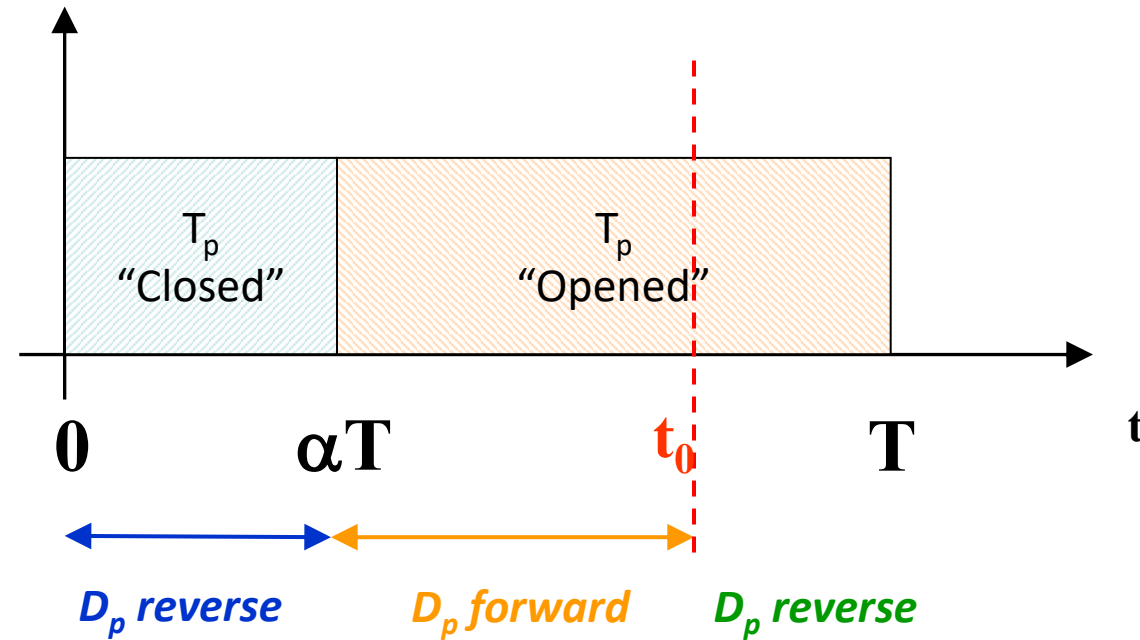
$$V_s = \alpha \cdot E$$



$$\langle I_L \rangle = \frac{T}{2L} \cdot V_s \left(1 - \frac{V_s}{E} \right)$$

Curve of mode
separation (V_s, I_s)

Discontinuous mode



At $t_0 = a' T$: diode switches to reverse mode

Introduction

Direct conversion

Buck chopper

Boost chopper

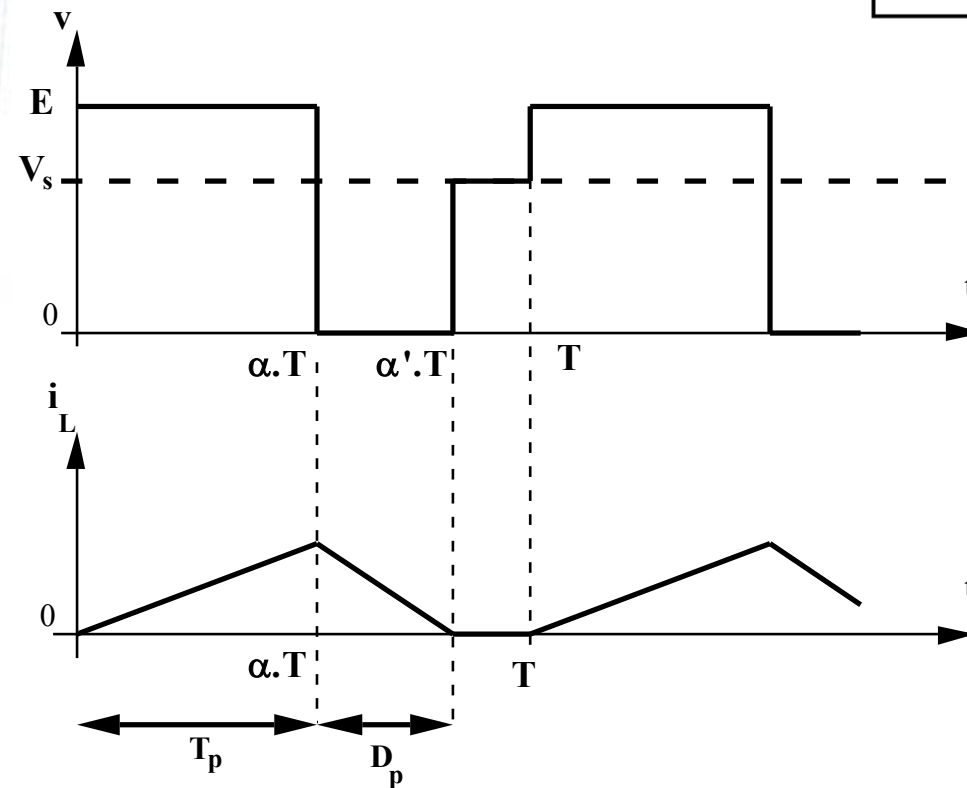
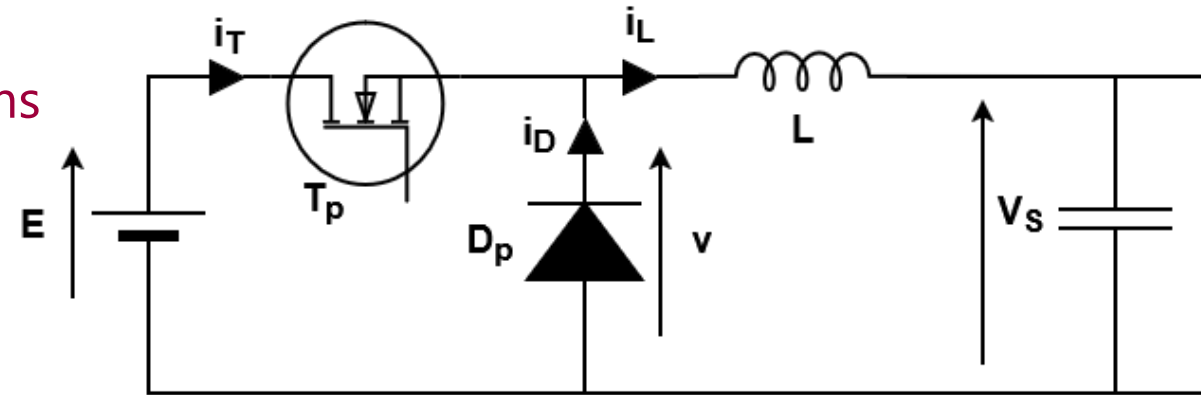
Reversible
choppers

Inverters

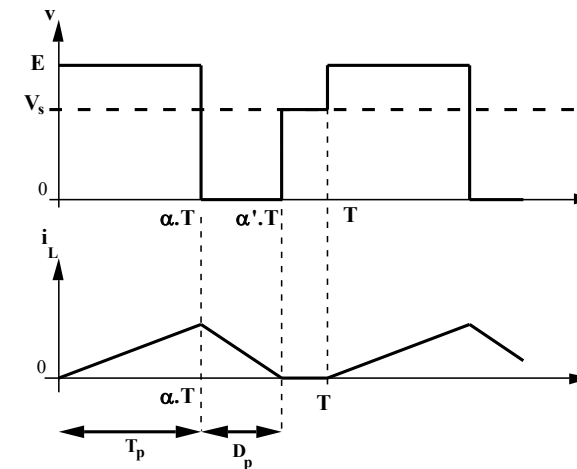
Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode: Chronograms



Discontinuous mode: calculation



$$I_{max} = \frac{E - V_S}{L} \cdot \alpha T = \frac{V_S}{L} \cdot (\alpha' T - \alpha T) \quad \Rightarrow \quad \frac{\alpha}{\alpha'} = \frac{V_S}{E}$$

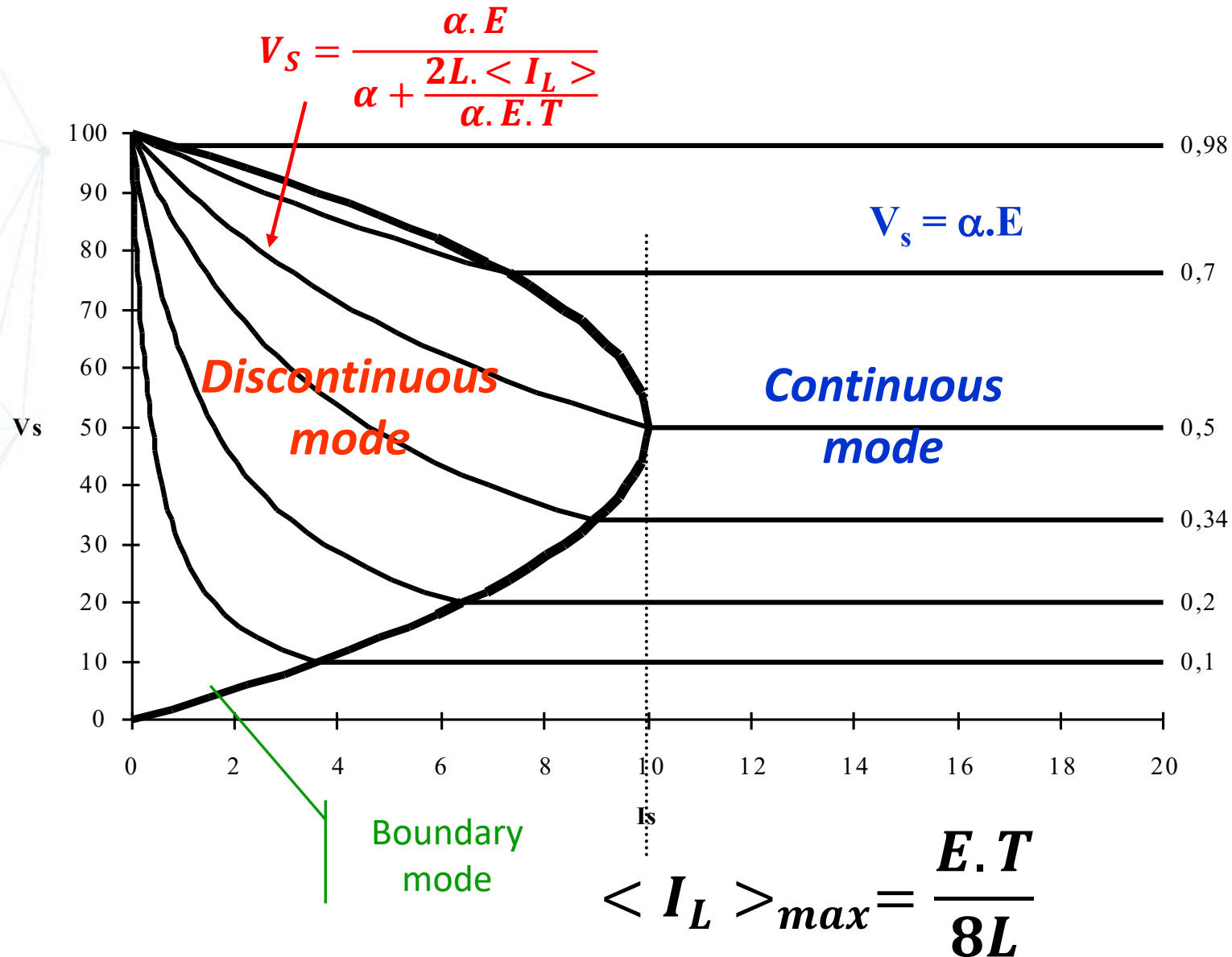
$$\langle I_L \rangle = \frac{1}{2} \cdot I_{max} \cdot \alpha' = \frac{E - V_S}{2L} \cdot \alpha \cdot \alpha' \cdot T = \frac{\alpha \cdot E \cdot T}{2L} (\alpha' - \alpha)$$

$$\Rightarrow \alpha' = \alpha + \frac{2L \cdot \langle I_L \rangle}{\alpha \cdot E \cdot T}$$

$$\Rightarrow V_S = \frac{\alpha \cdot E}{\alpha + \frac{2L \cdot \langle I_L \rangle}{\alpha \cdot E \cdot T}}$$

Output current: $\langle I_L \rangle = I_S$

Output characteristics



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Direct conversion

Buck chopper

Boost chopper

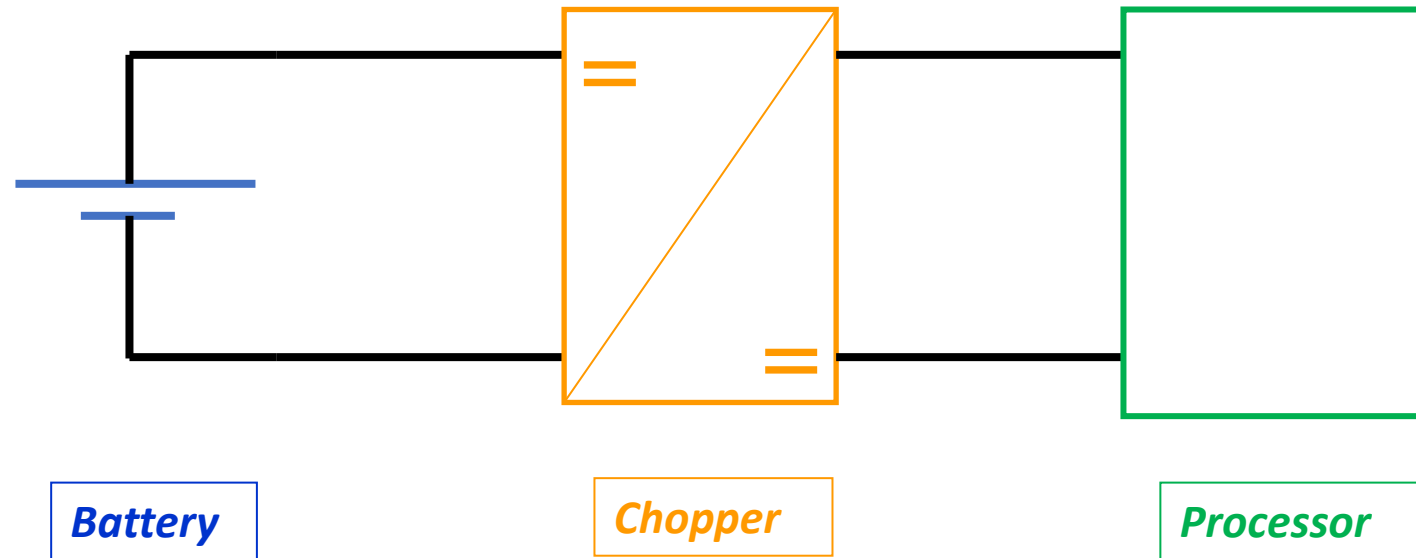
Reversible
choppers

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Closed loop
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Energy storage
based chopper

Example



=> DC supply from battery

Introduction

Direct conversion

Buck chopper

Boost chopper

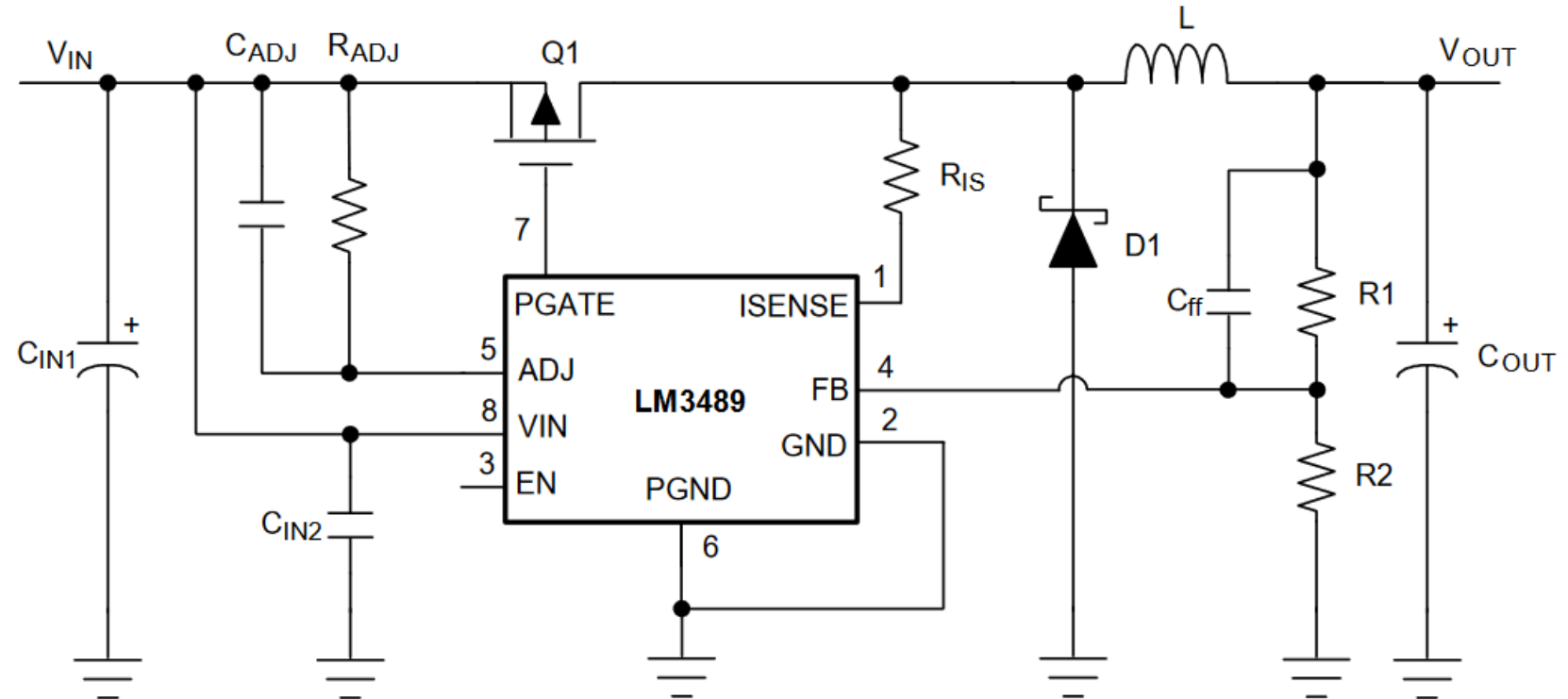
Reversible choppers

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Closed loop chopper control

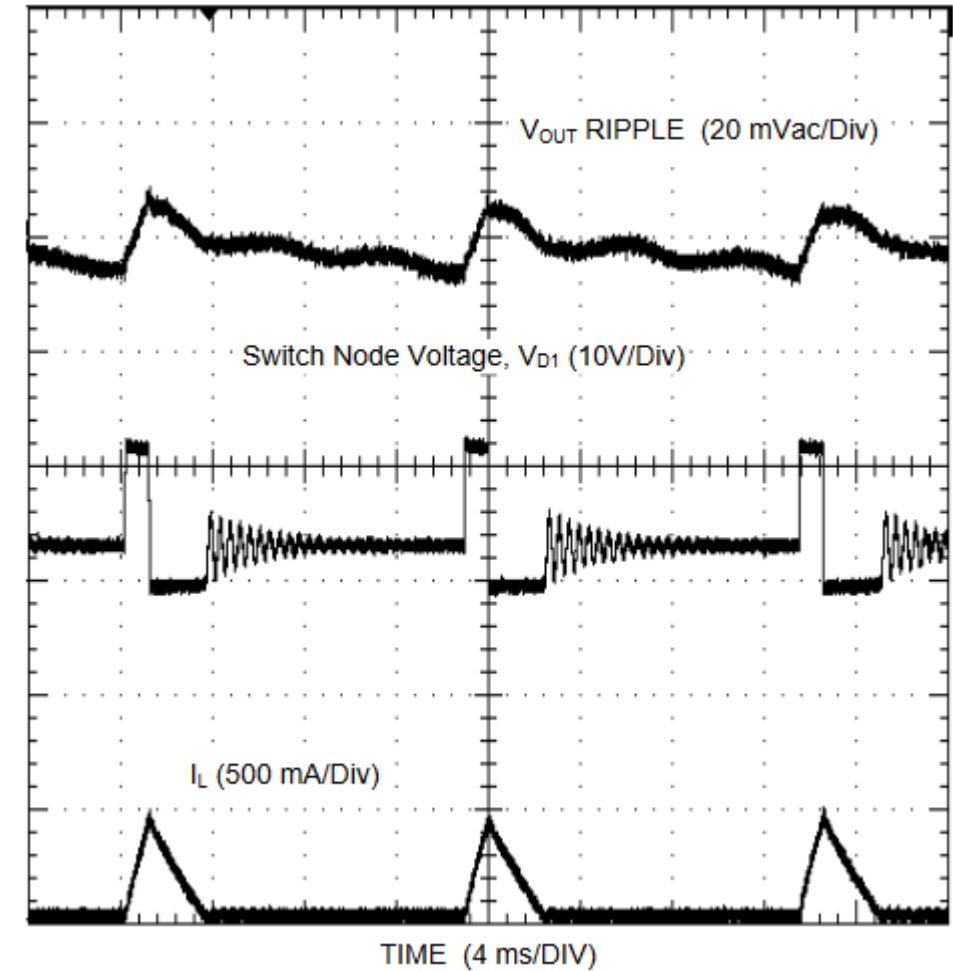
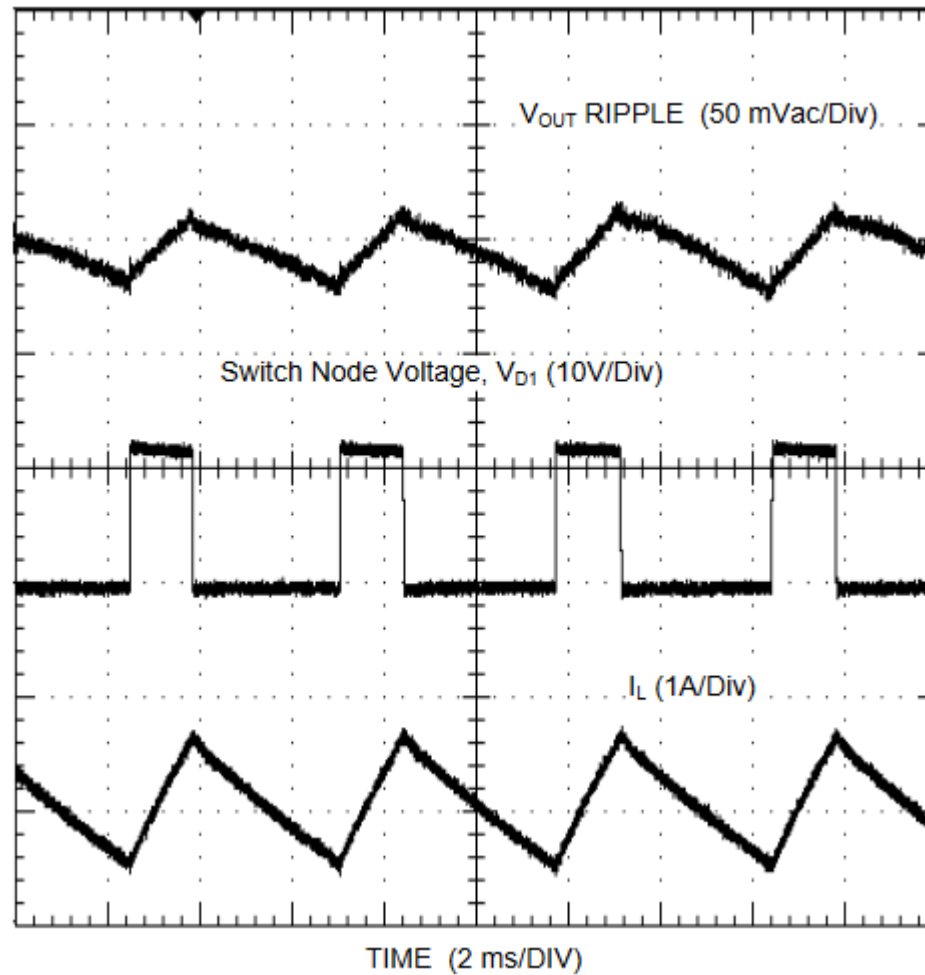
Energy storage based chopper

Example : LM3489



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Example : LM3489



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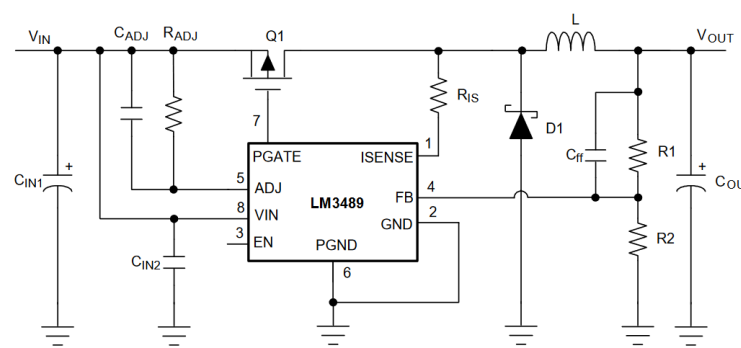
Reversible
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Closed loop
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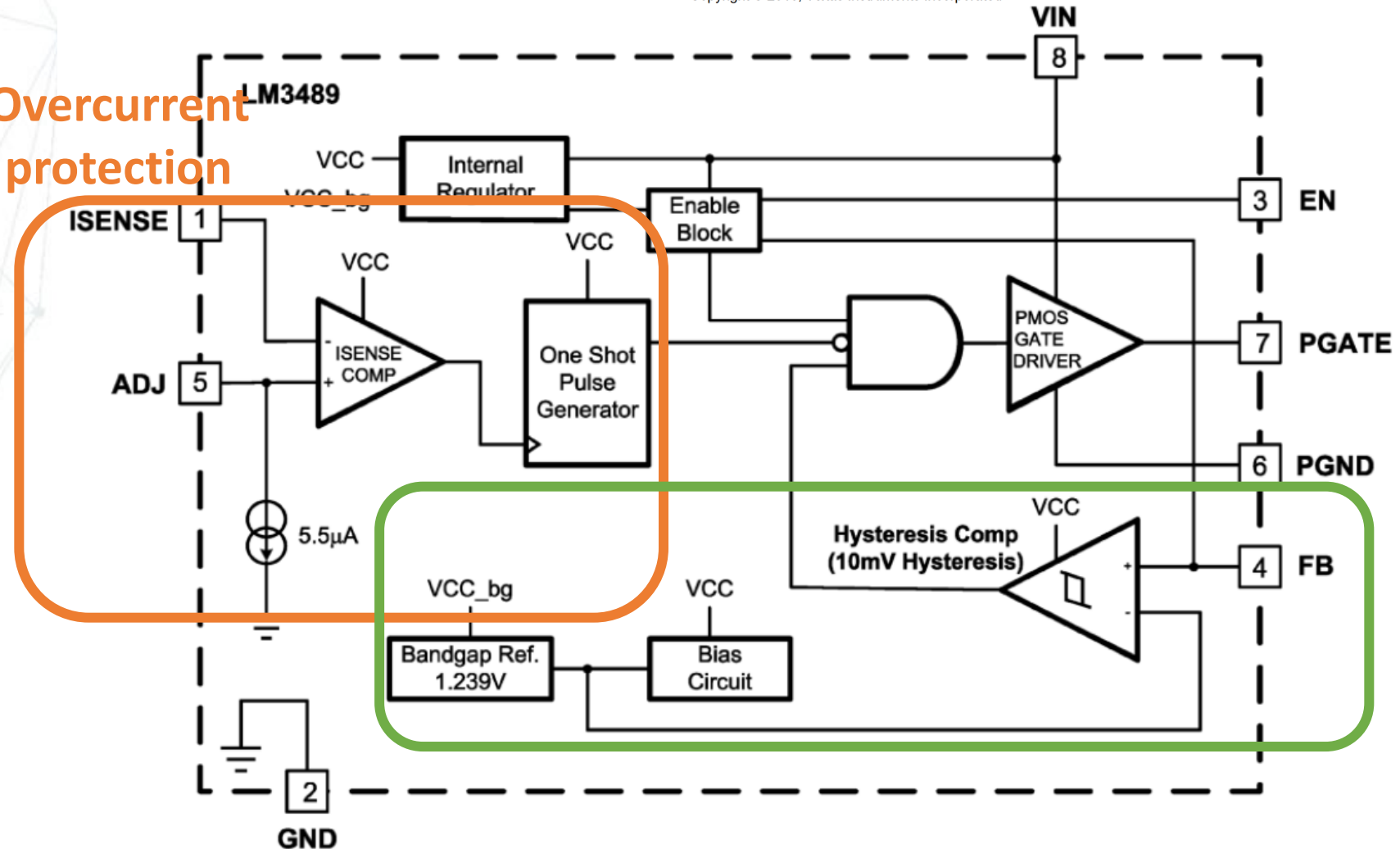
Energy storage
based chopper

Example : LM3489



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Overcurrent
protection



Voltage
hysteresis
control

Introduction

Direct conversion

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Boost chopper

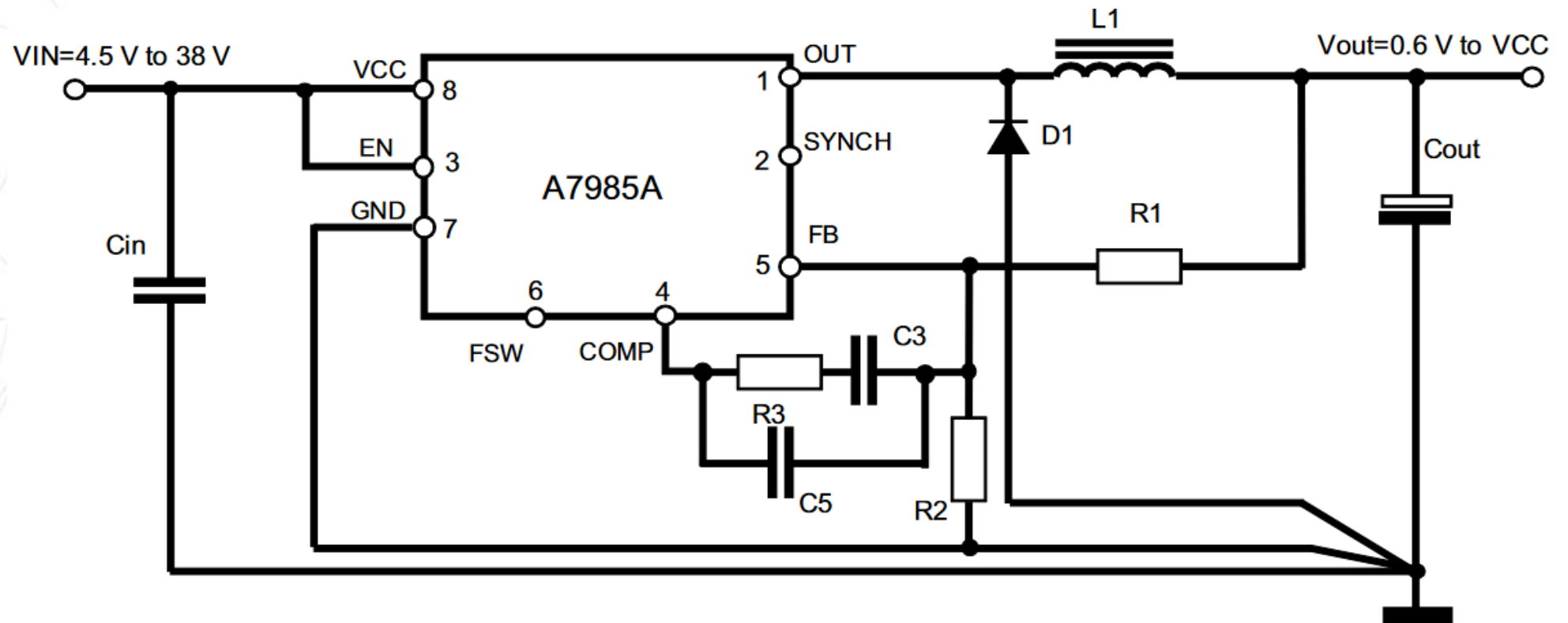
Reversible choppers

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Closed loop chopper control

Energy storage based chopper

Example : A7985A



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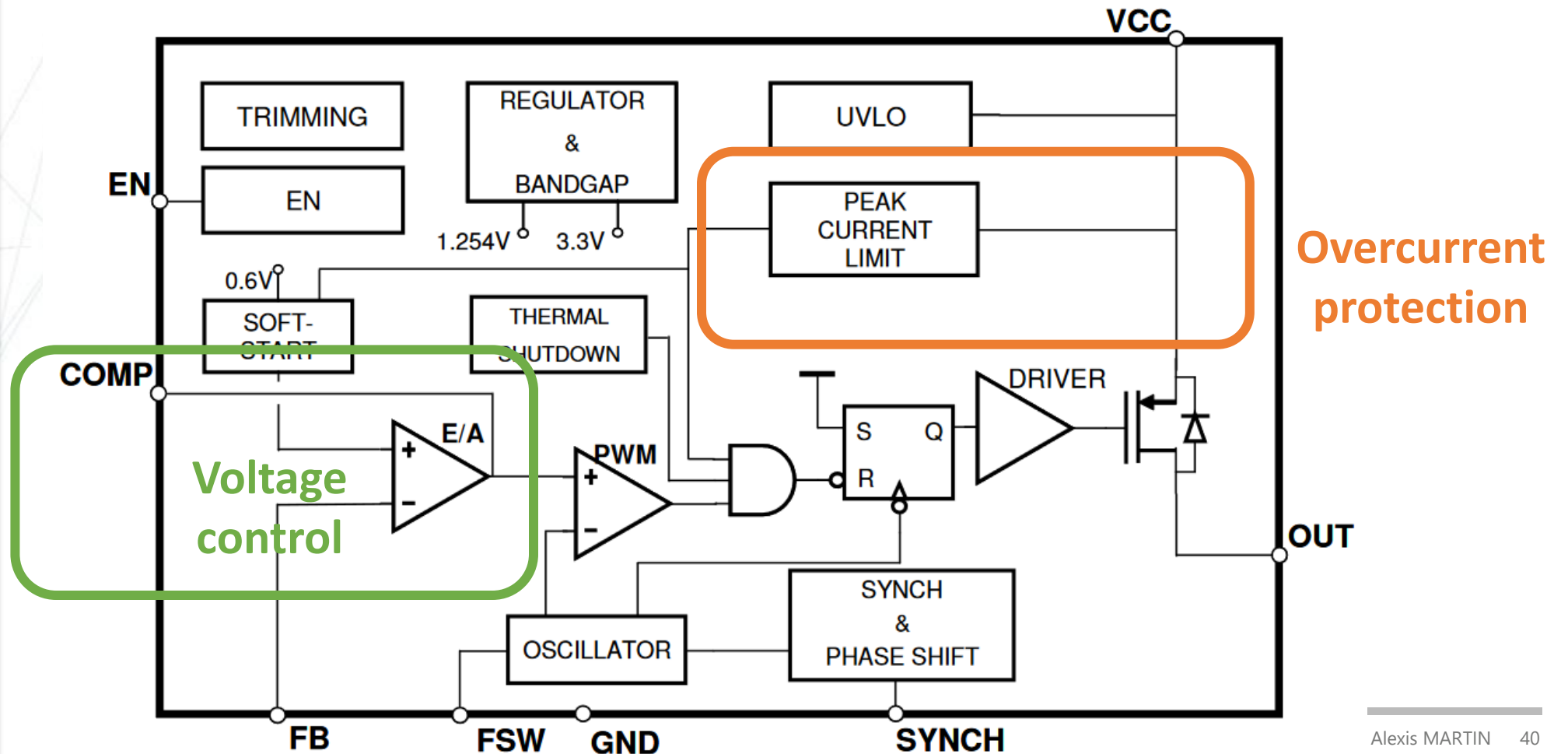
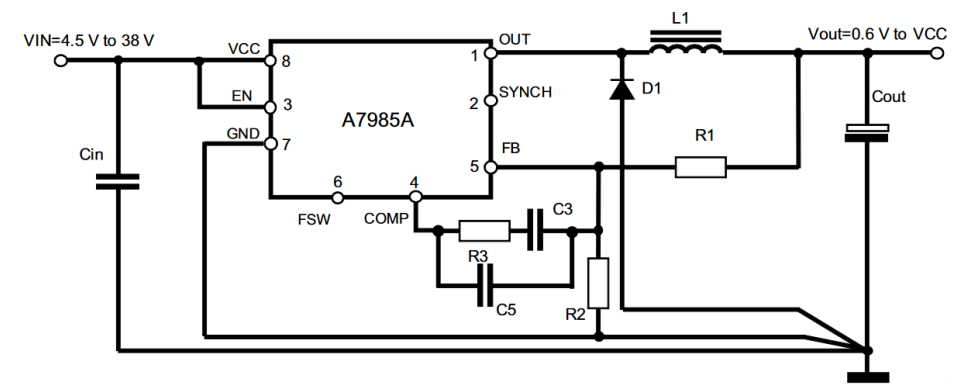
Reversible choppers

Inverters

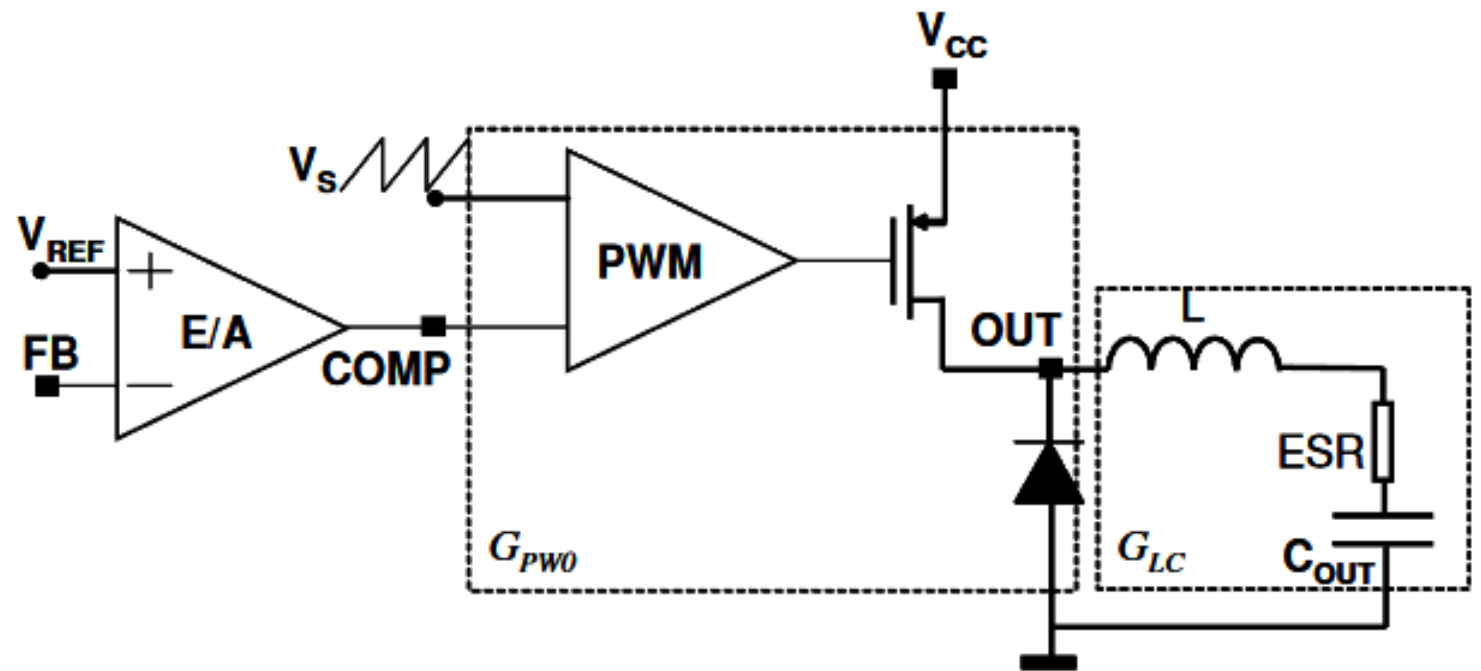
Closed loop chopper control

Energy storage based chopper

Example : A7985A



Example : A7985A



$$G_{LC}(s) = \frac{1 + \frac{s}{2\pi \cdot f_{zESR}}}{1 + \frac{s}{2\pi \cdot Q \cdot f_{LC}} + \left(\frac{s}{2\pi \cdot f_{LC}}\right)^2}$$

$$f_{LC} = \frac{1}{2\pi \cdot \sqrt{L \cdot C_{OUT}} \cdot \sqrt{1 + \frac{ESR}{R_{OUT}}}}, \quad f_{zESR} = \frac{1}{2\pi \cdot ESR \cdot C_{OUT}}$$

$$Q = \frac{\sqrt{R_{OUT} \cdot L \cdot C_{OUT} \cdot (R_{OUT} + ESR)}}{L + C_{OUT} \cdot R_{OUT} \cdot ESR}$$

Usually : $f_{LC} < f_{zESR}$

→ Slightly depends on the load

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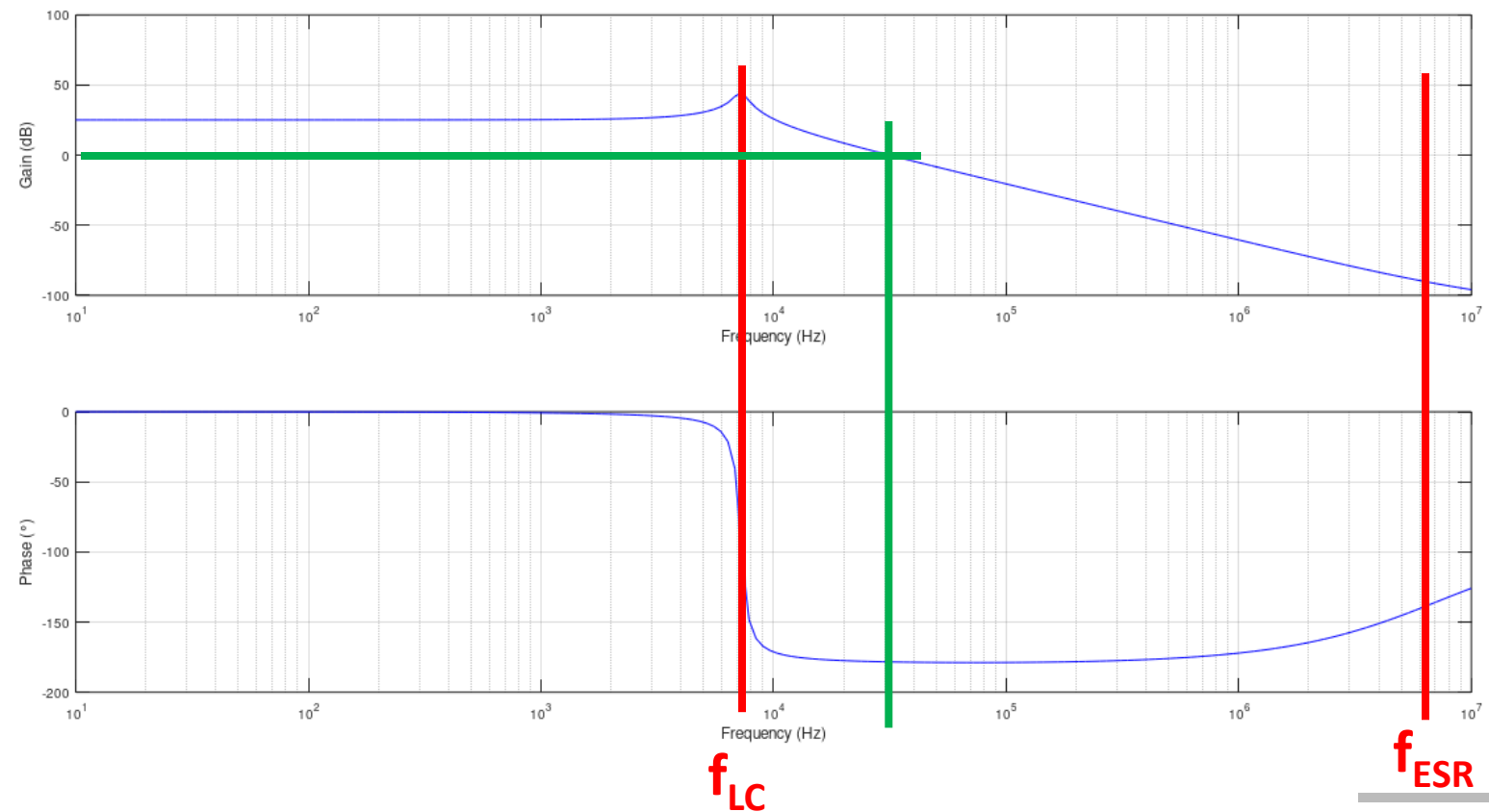
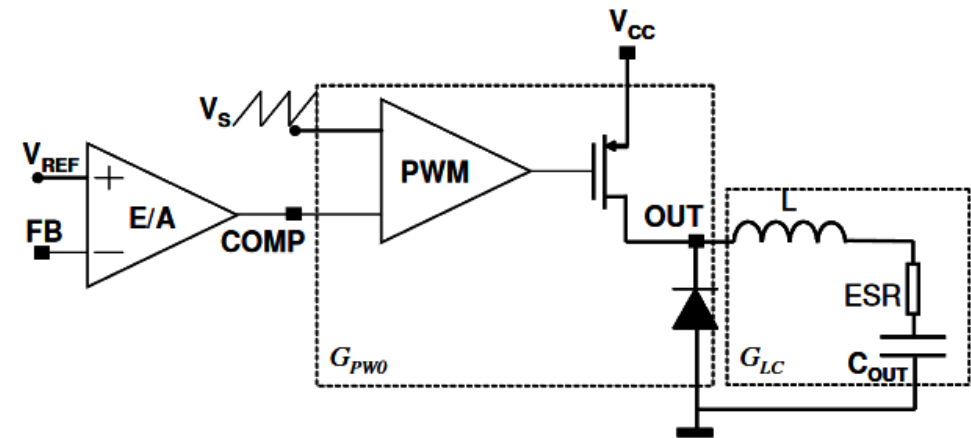
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Example : A7985A



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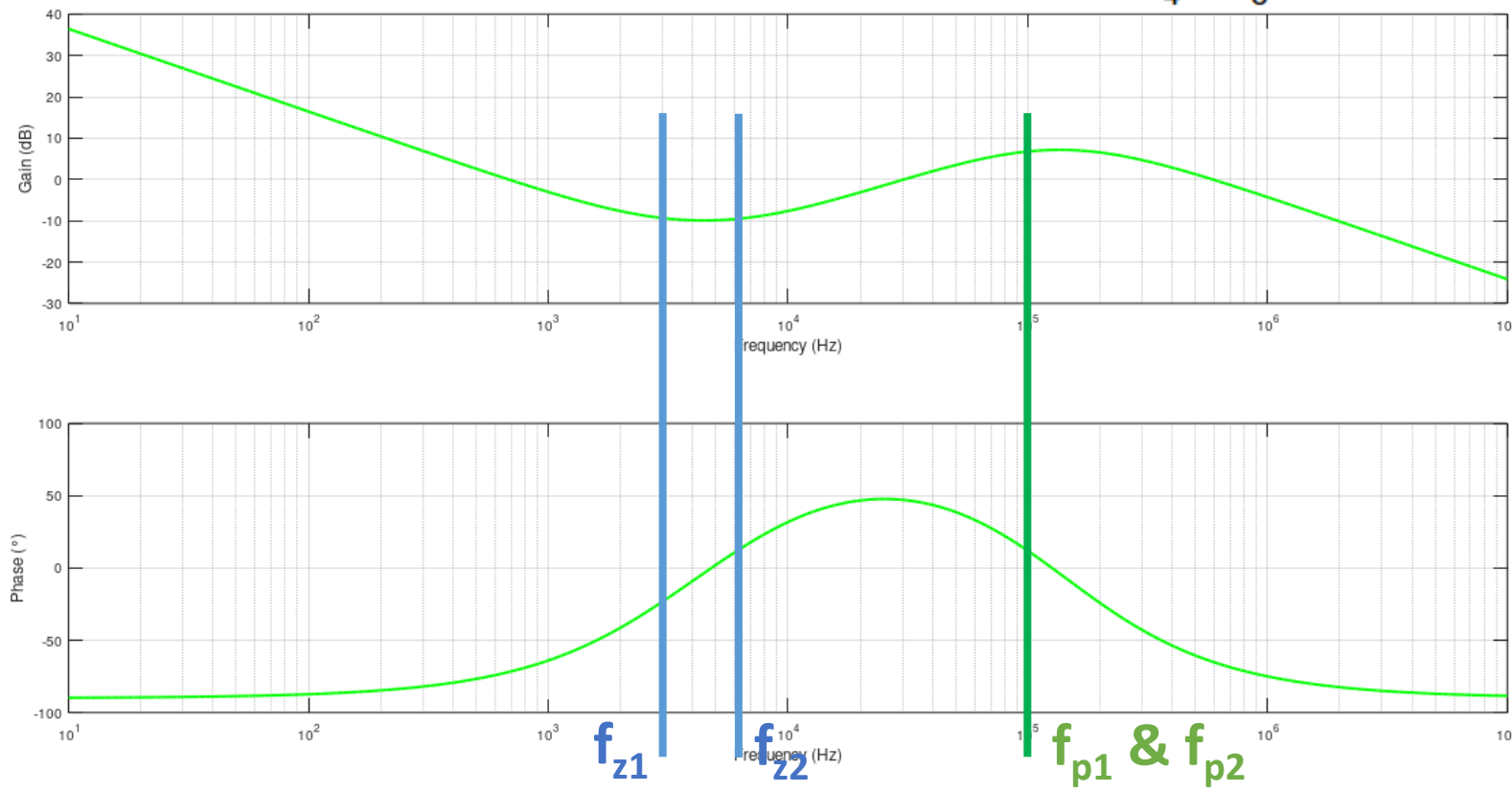
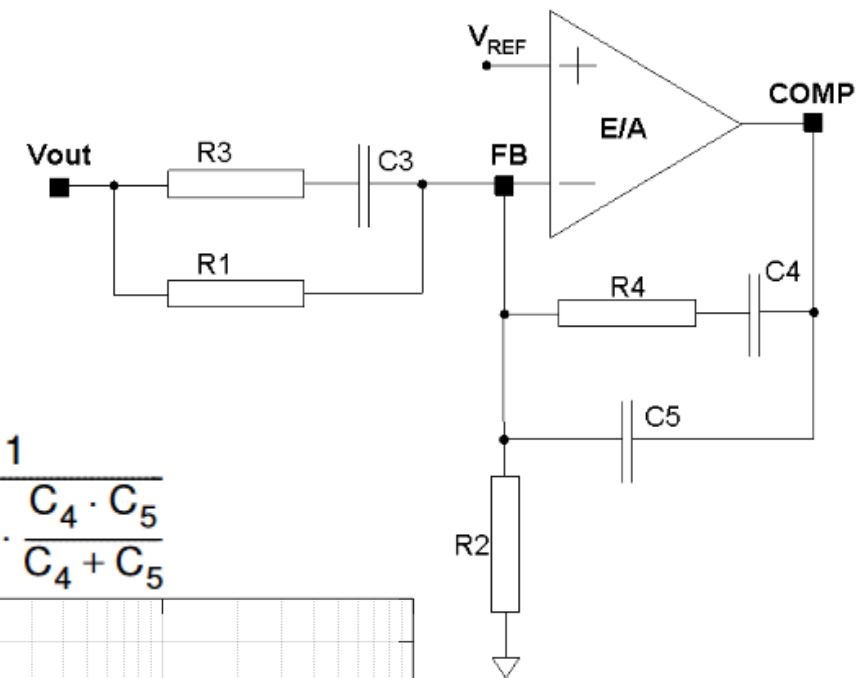
Energy storage based chopper

Example : A7985A

If $f_{zESR} \gg BW$ (desired)
(ESR < 1mΩ)

Zeroes : $f_{z1} = \frac{1}{2\pi \cdot C_3 \cdot (R_1 + R_3)}$, $f_{z2} = \frac{1}{2\pi \cdot R_4 \cdot C_4}$

Poles : $f_{p0} = 0$, $f_{p1} = \frac{1}{2\pi \cdot R_3 \cdot C_3}$, $f_{p2} = \frac{1}{2\pi \cdot R_4 \cdot \frac{C_4 \cdot C_5}{C_4 + C_5}}$



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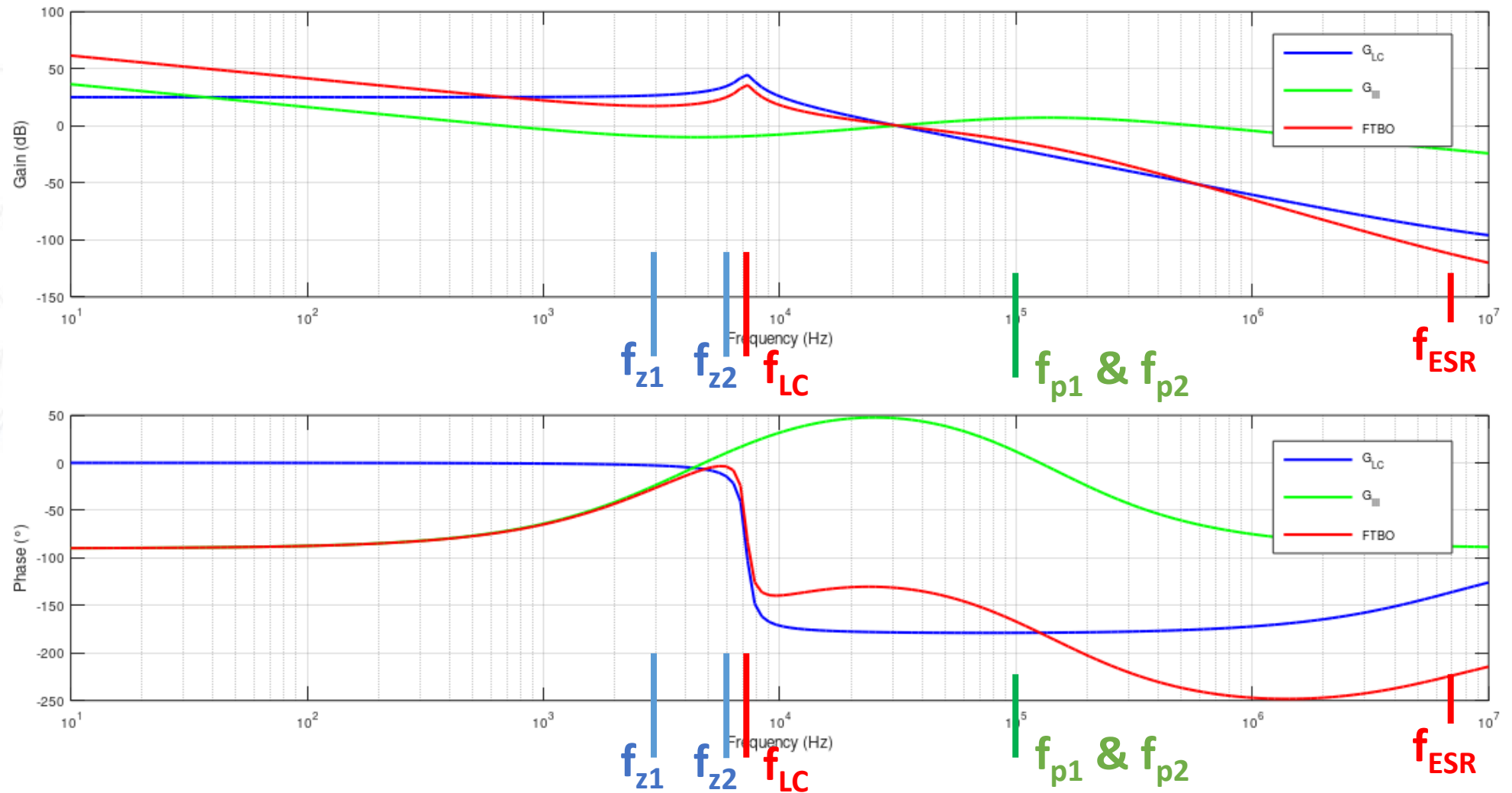
Reversible choppers

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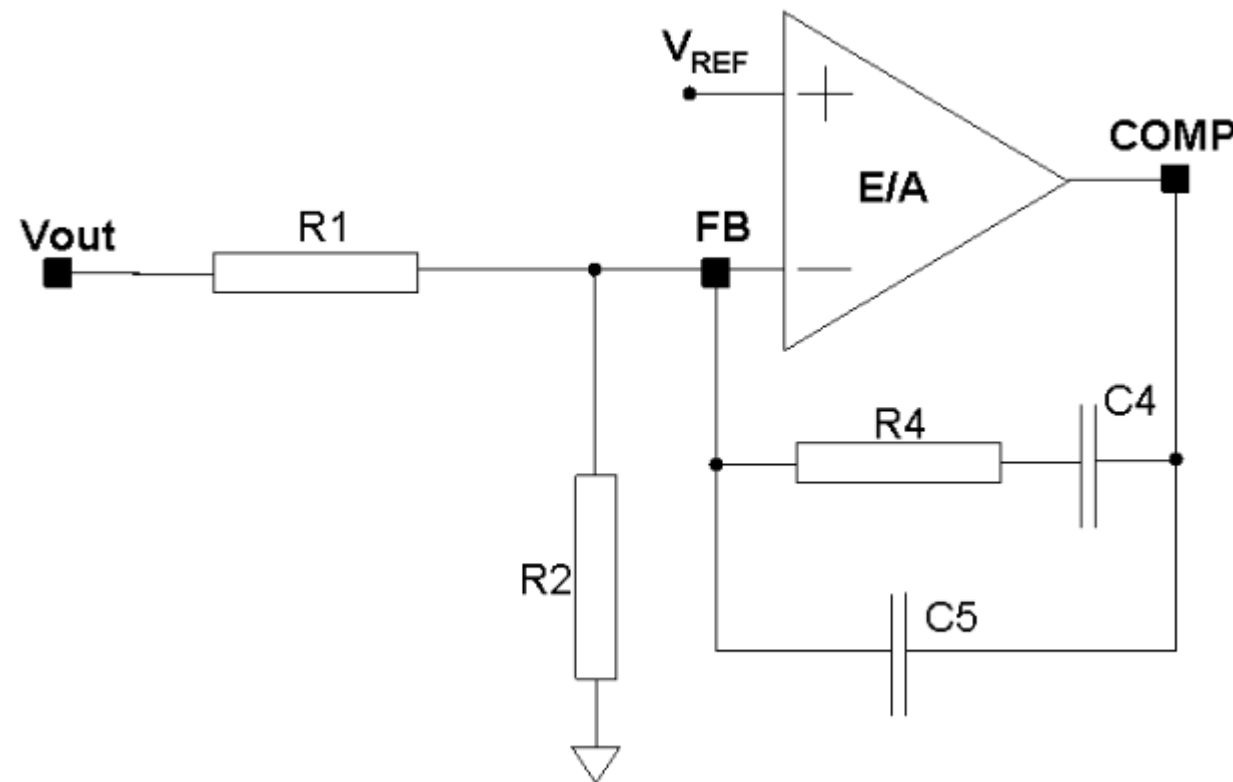
Example : A7985A



Example : A7985A

If $f_{zESR} \approx BW$ (desired)
ESR > 30 mΩ

→ Use of the zero of ESR to stabilize the loop



Zero : $f_{z1} = \frac{1}{2\pi \cdot R_4 \cdot C_4},$

Poles : $f_{p0} = 0,$

$$f_{p1} = \frac{1}{2\pi \cdot R_4 \cdot \frac{C_4 \cdot C_5}{C_4 + C_5}}$$

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Boost Chopper

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Boost chopper

Buck chopper : Voltage source



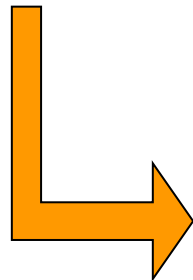
Current receiver

Boost chopper : Current source



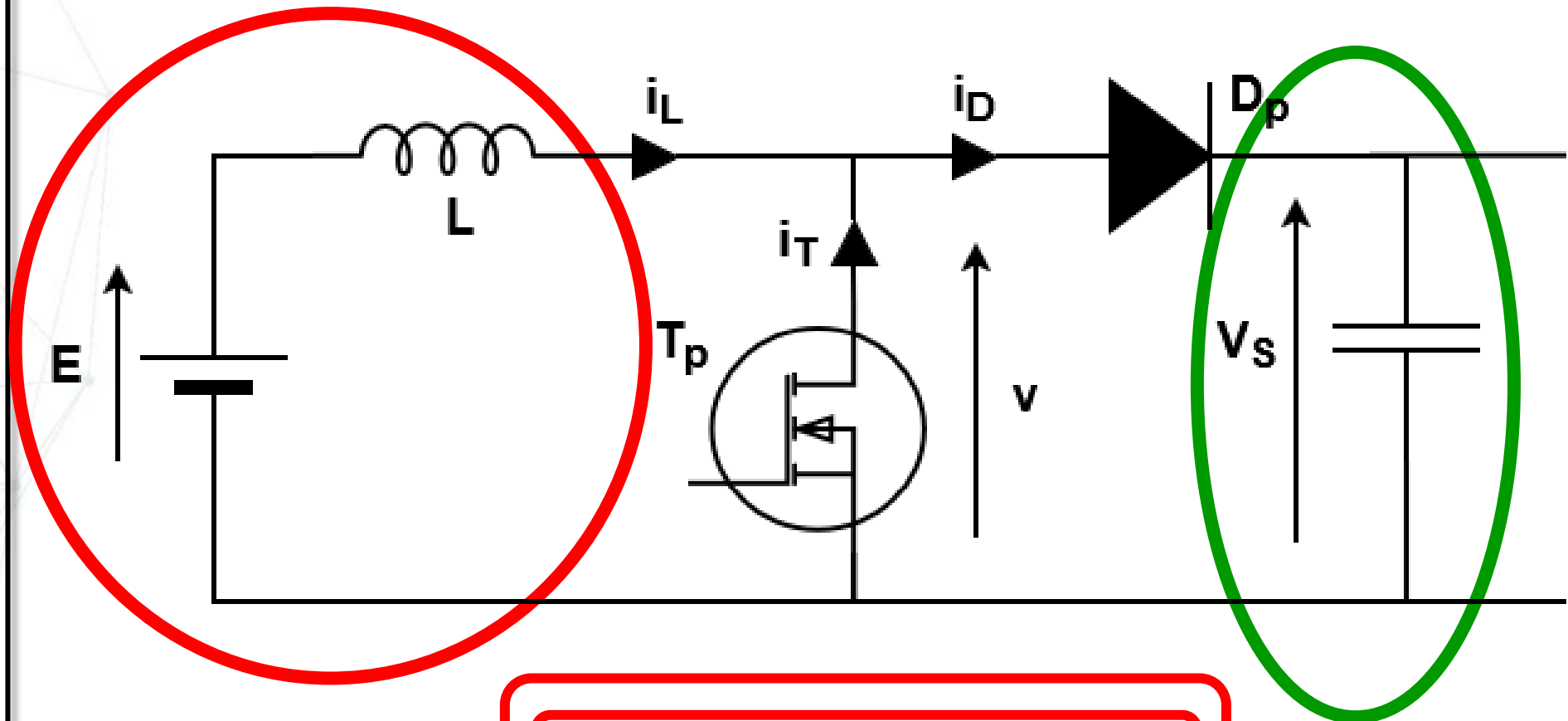
Voltage receiver

Current source $\Leftrightarrow$ Voltage source + inductor



Voltage receiver $\Leftrightarrow$ Battery or capacitive receiver

Boost chopper: principle



Hypothesis : $V_s = \text{constant}$

Introduction

Direct conversion

Buck chopper

Boost chopper

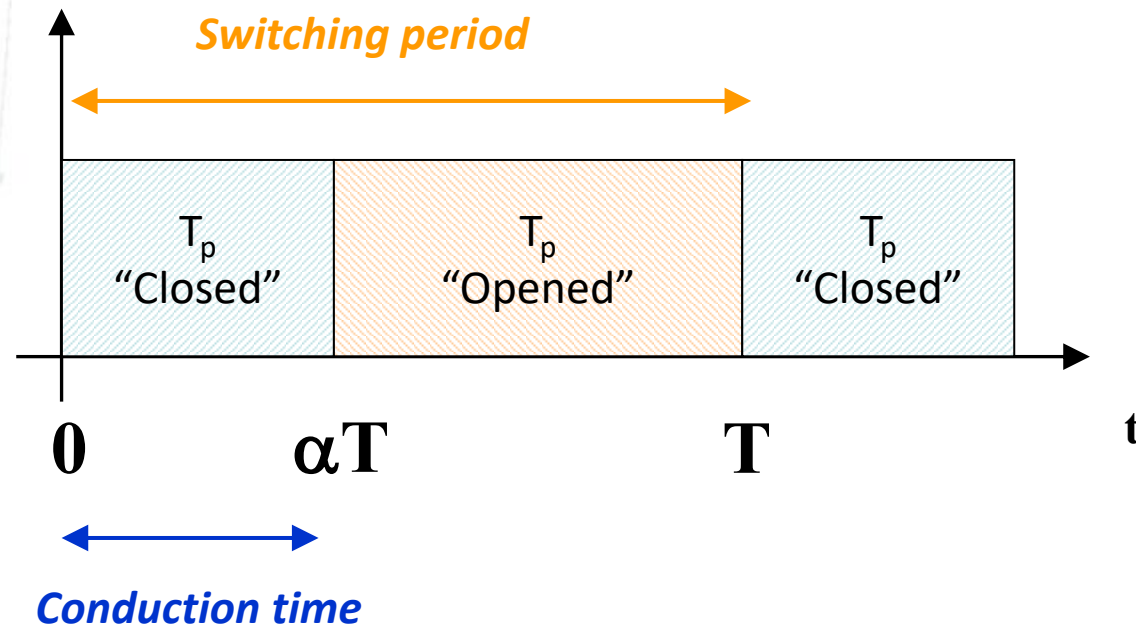
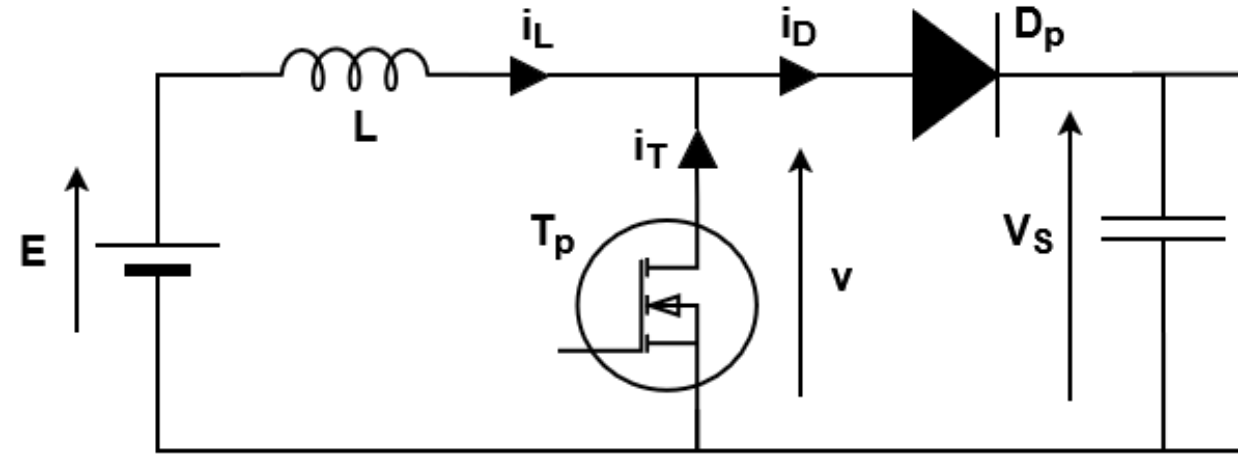
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Control



Direct conversion

Buck chopper

Boost chopper

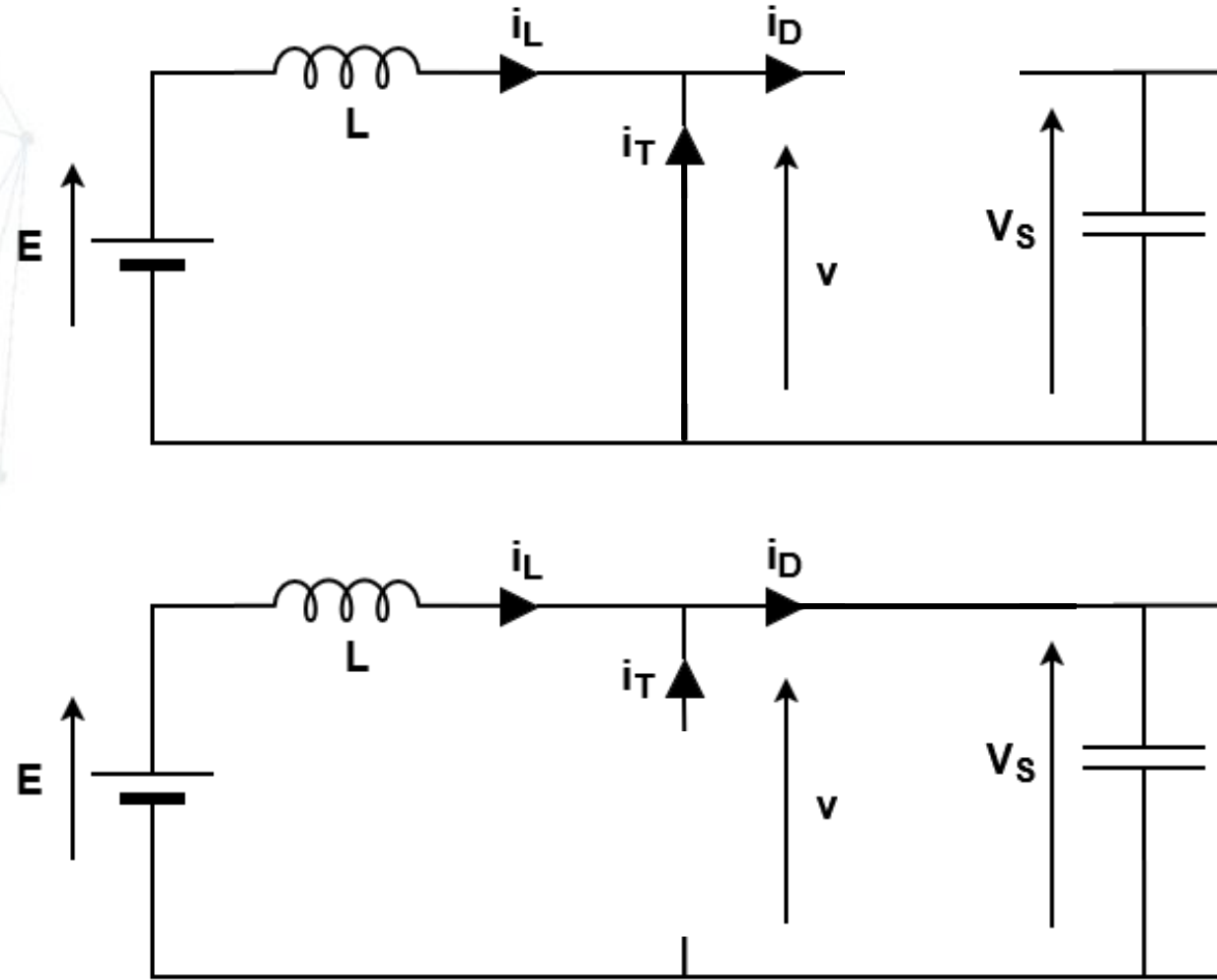
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Principle

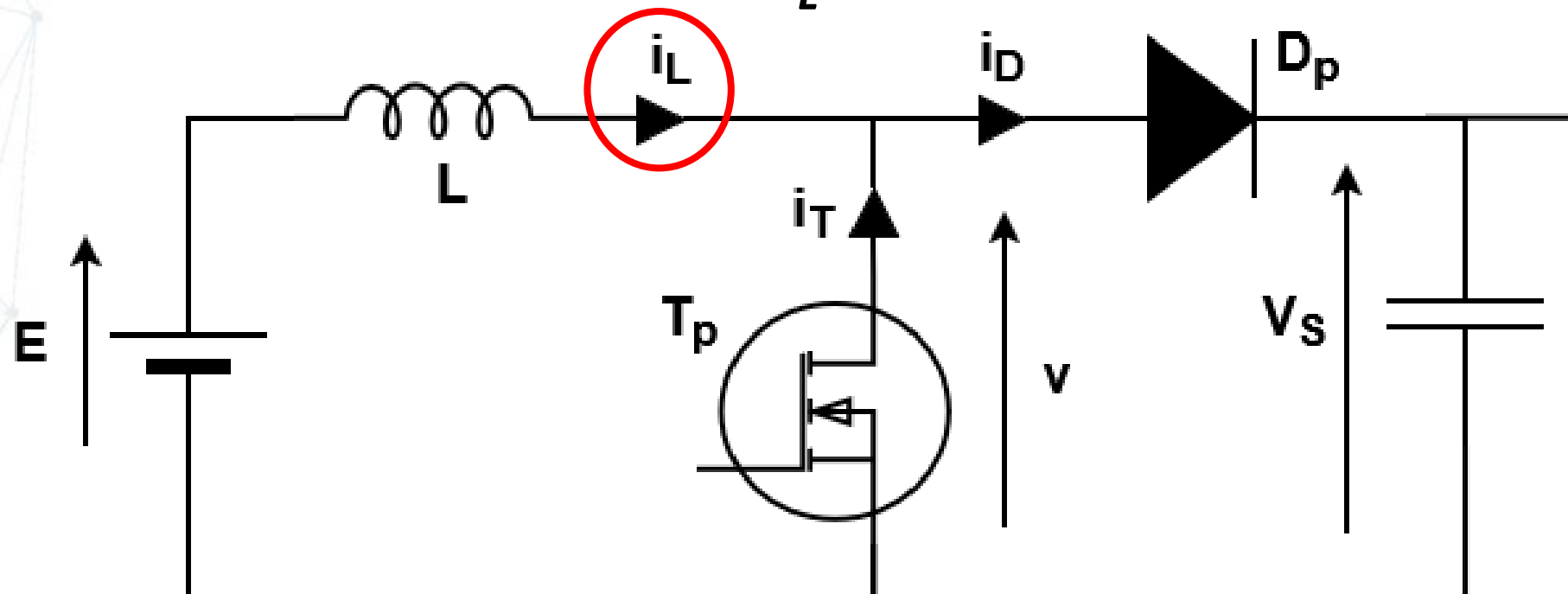


$$0 < t < \alpha T$$

$$\alpha T < t < T$$

Modes

- **Continuous conduction: $i_L > 0$**



- **Discontinuous conduction: $i_L(t_0) = 0 ; t_0 < T$**

Introduction

Direct conversion

Buck chopper

Boost chopper

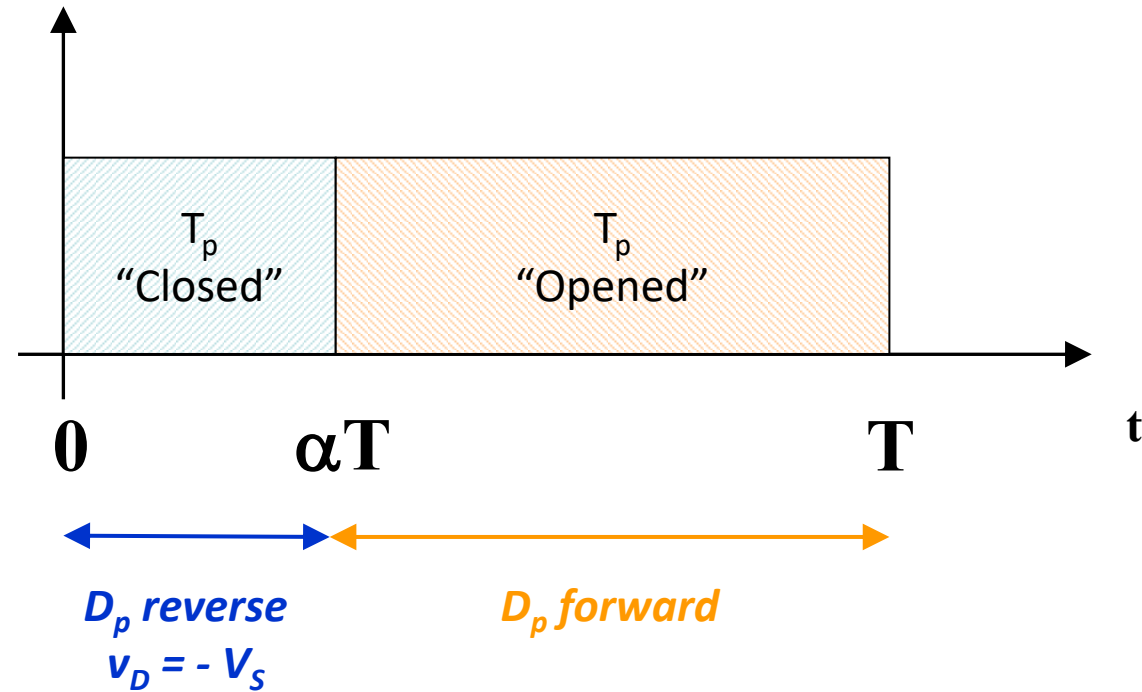
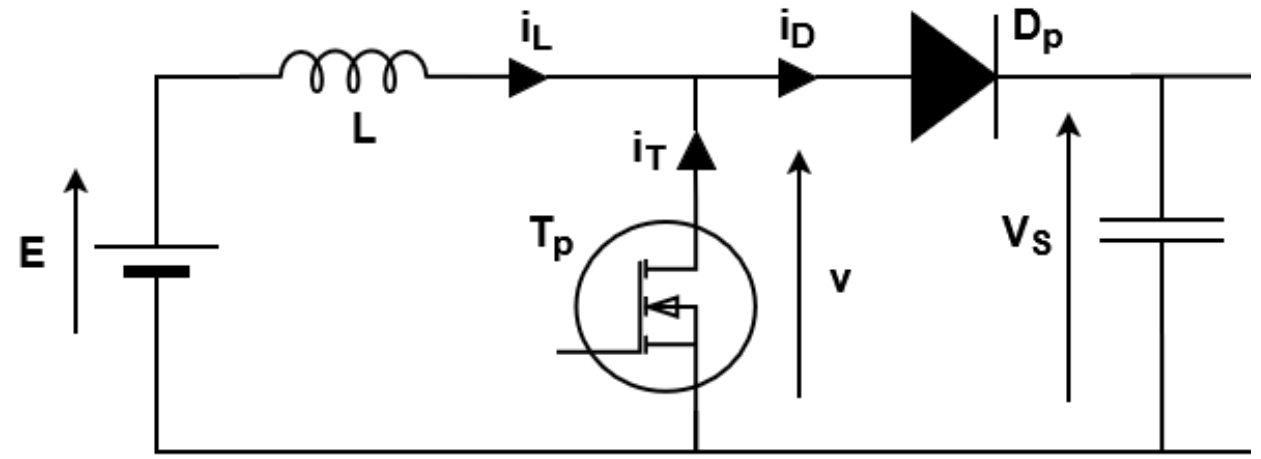
Reversible
choppers

Inverters

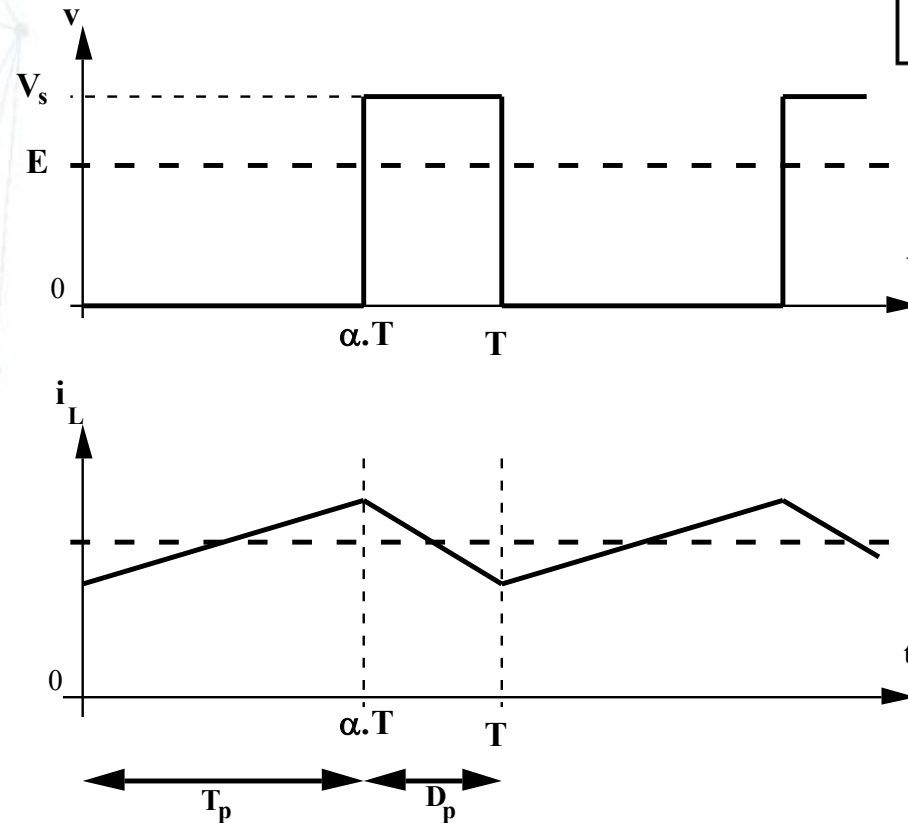
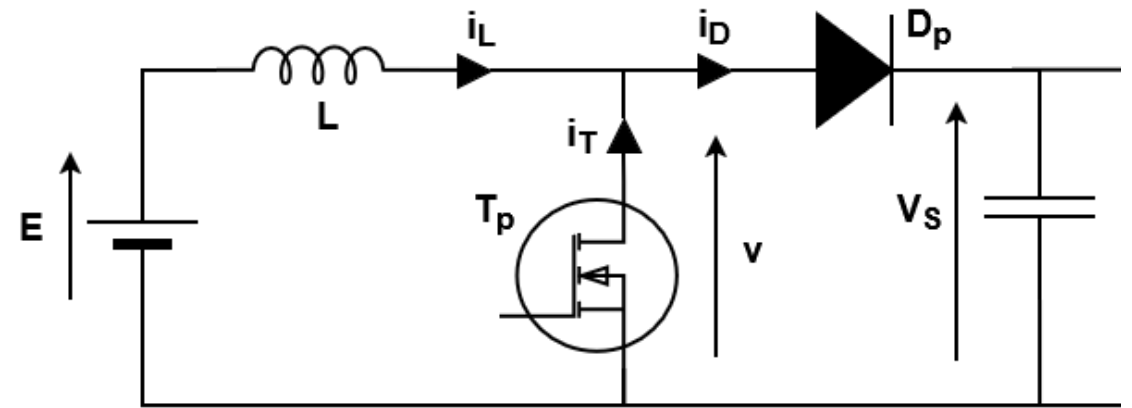
Closed loop
chopper control

Energy storage
based chopper

Continuous mode



Continuous mode: Chronograms



$$E < V_s$$

- $0 < t < \alpha T$

$$\frac{di}{dt} = \frac{E}{L}$$

- $\alpha T < t < T$

$$\frac{di}{dt} = \frac{E - V_s}{L}$$

BOOST

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

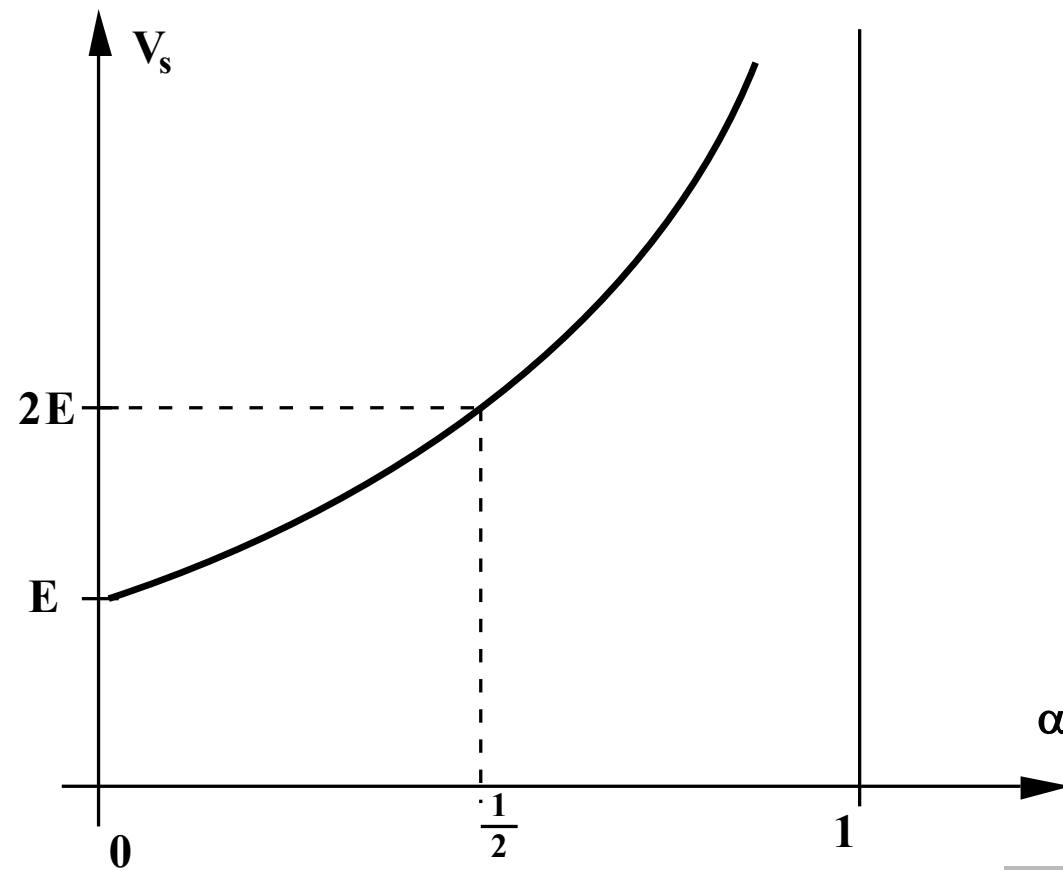
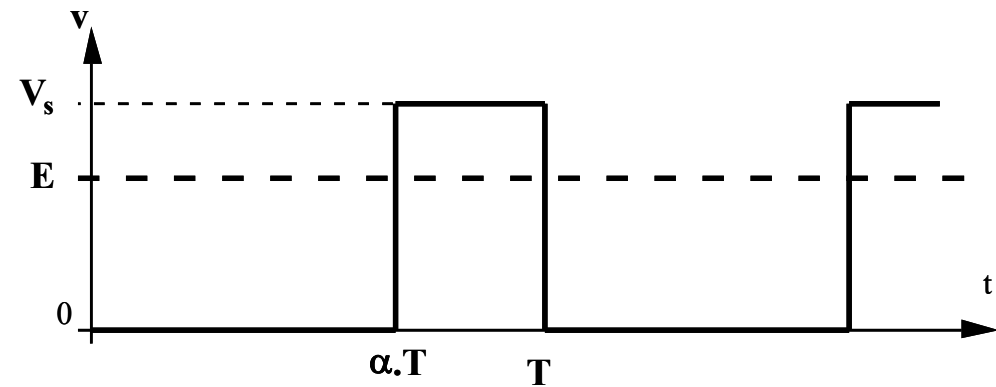
Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode:
Output voltage calculation

$$V_s = \frac{E}{1 - \alpha}$$



Direct conversion

Buck chopper

Boost chopper

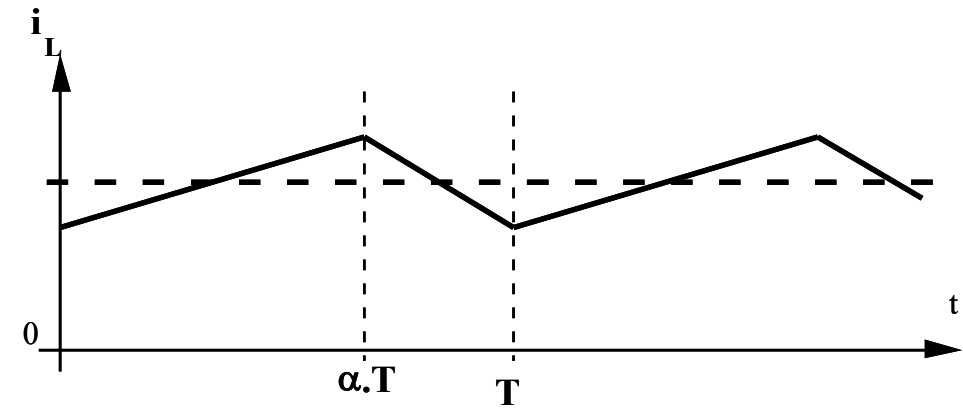
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode:
Current ripple calculation



Current ripple i_L

$$\Delta I_L = \alpha \cdot \frac{E \cdot T}{L}$$

Maximum ripple ($\alpha = 1$)

$$\Delta I_{L \max} = \frac{E \cdot T}{L}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

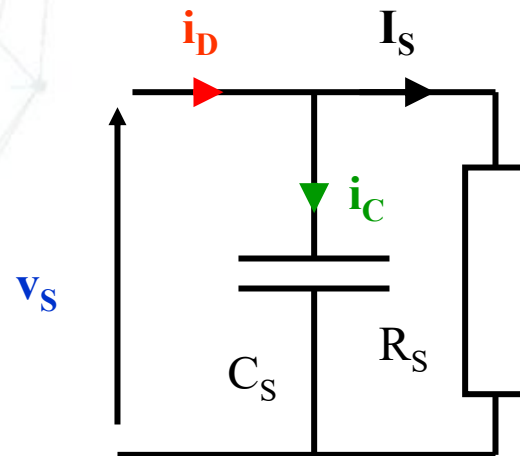
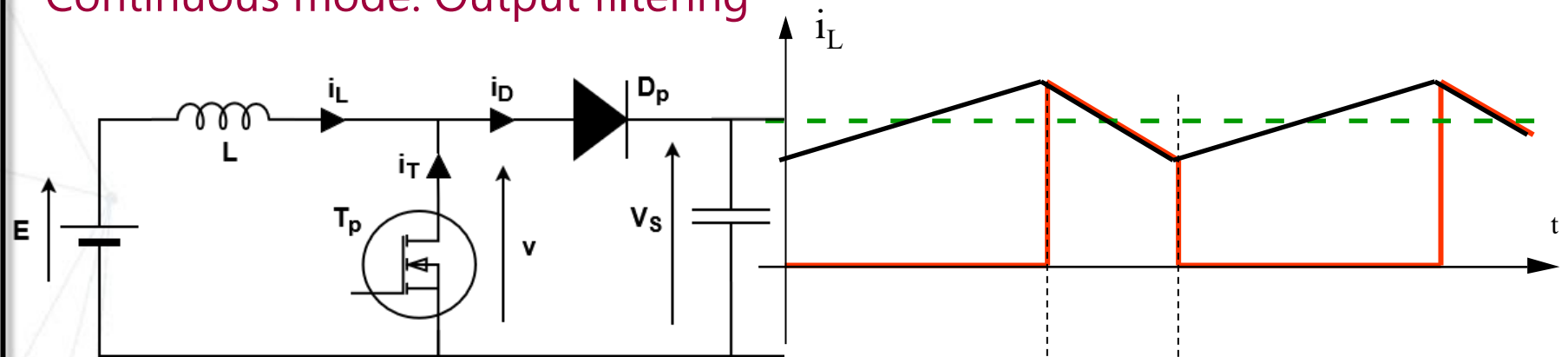
Reversible
choppers

Inverters

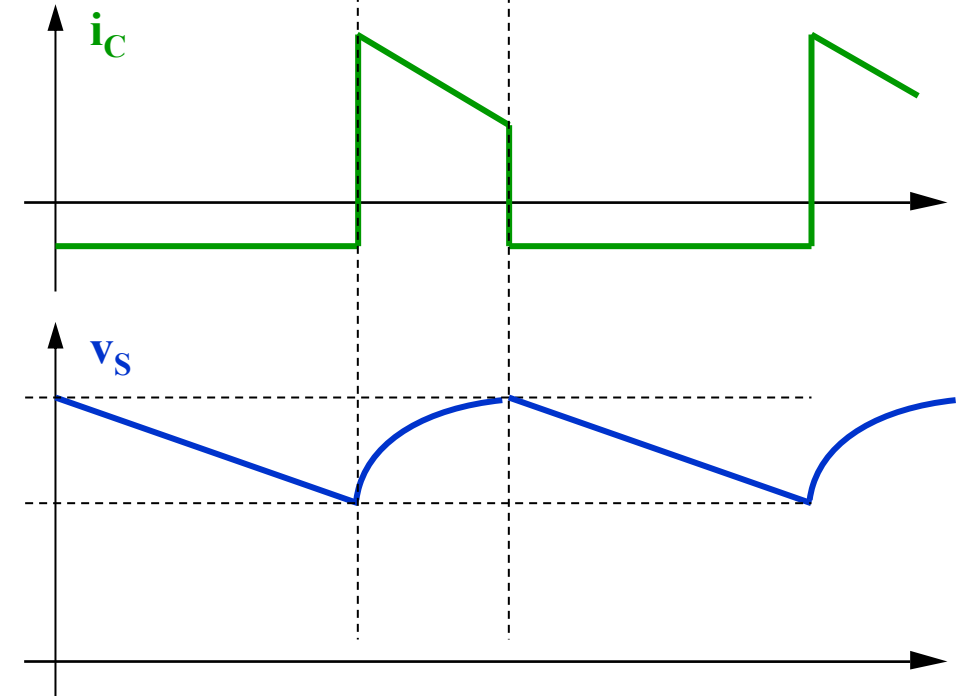
Closed loop
chopper control

Energy storage
based chopper

Continuous mode: Output filtering



$$i_c = C_s \cdot \frac{dv_s}{dt}$$



Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

Inverters

Closed loop
chopper control

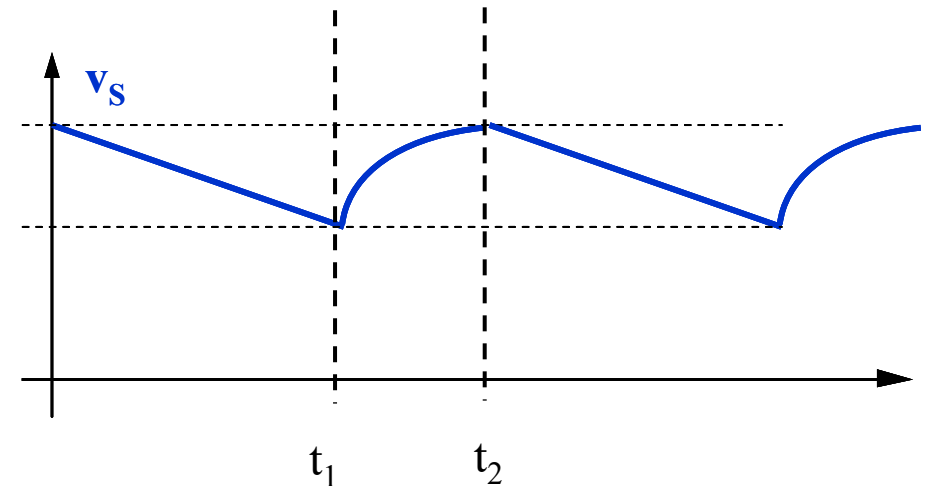
Energy storage
based chopper

Continuous mode: Output filtering

$$\Delta V_S = \frac{1}{C_S} \cdot \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{I_S \cdot \alpha T}{C_S}$$

$$\text{or } I_S = \frac{V_S}{R_S} = \frac{E}{(1-\alpha)R_S}$$

$$\Delta V_S = \frac{ET}{R_S C_S} \cdot \frac{\alpha}{1-\alpha}$$



$$C_S = \frac{\alpha \cdot E \cdot T}{(1-\alpha) \cdot R_S \cdot \Delta V_S}$$

Direct conversion

Buck chopper

Boost chopper

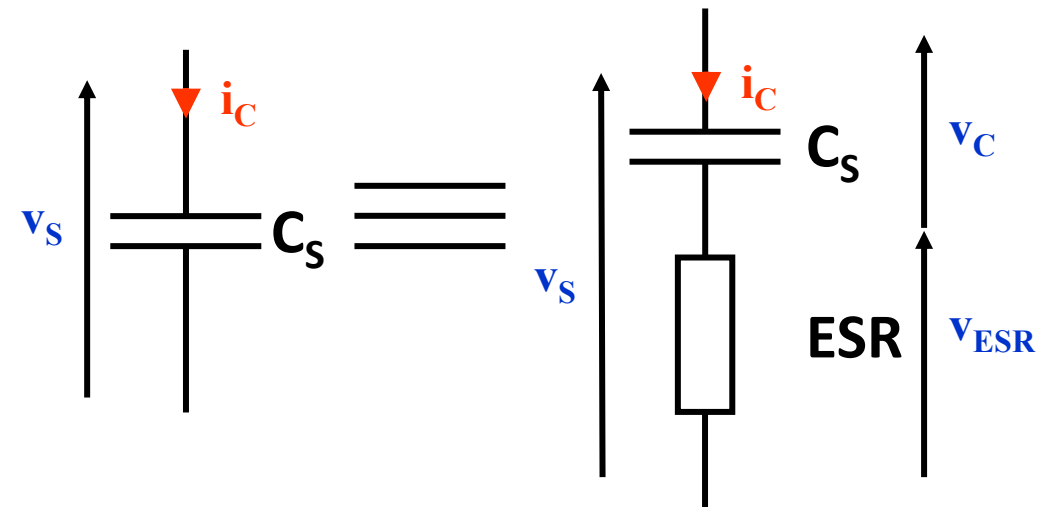
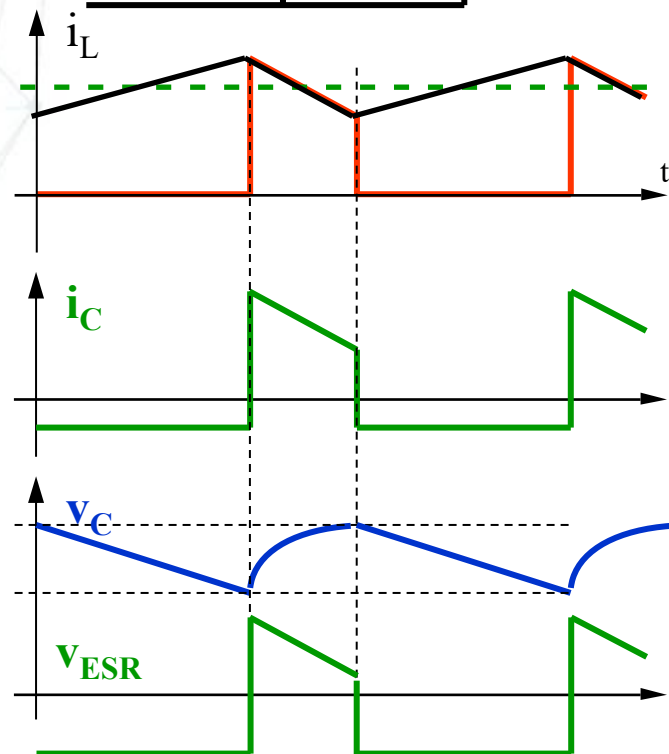
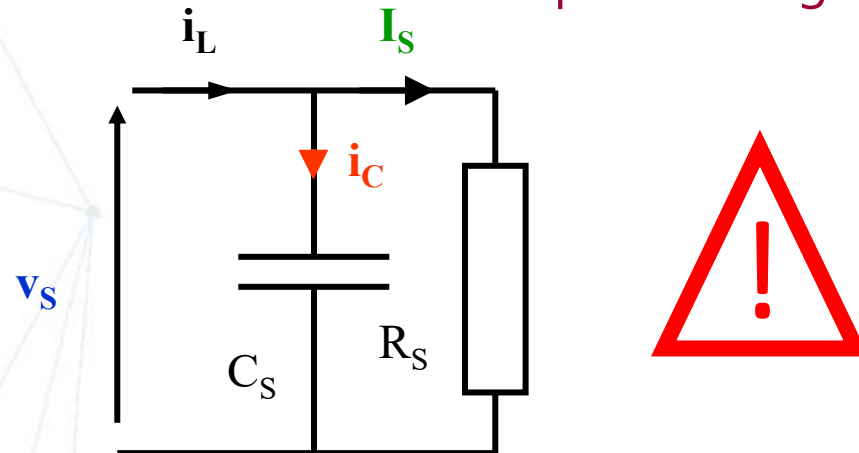
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode: Output filtering



$$v_s = v_C + v_{ESR}$$

$$\Delta v_s < \Delta v_C + \Delta v_{ESR}$$

For high value of C_s ,

$$\Delta v_s \approx \Delta v_{ESR} = ESR \cdot \Delta i_L$$

$$ESR < \frac{\Delta v_s}{\Delta i_L}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

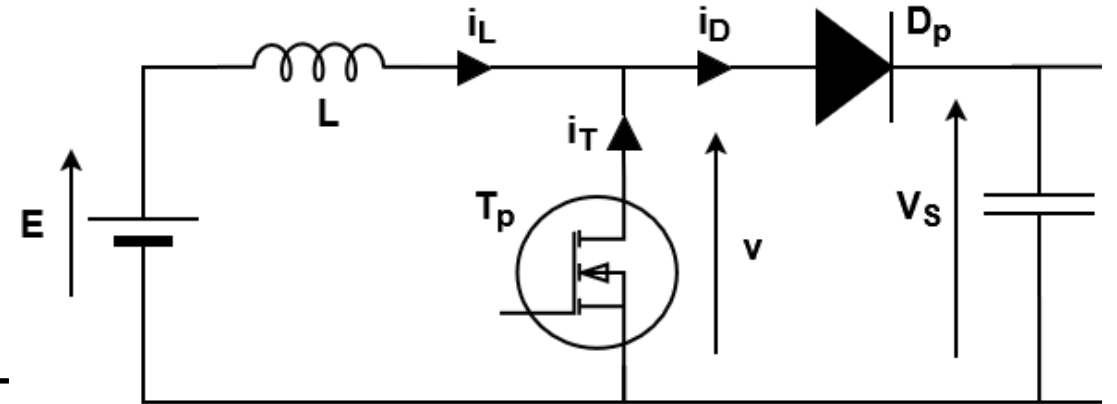
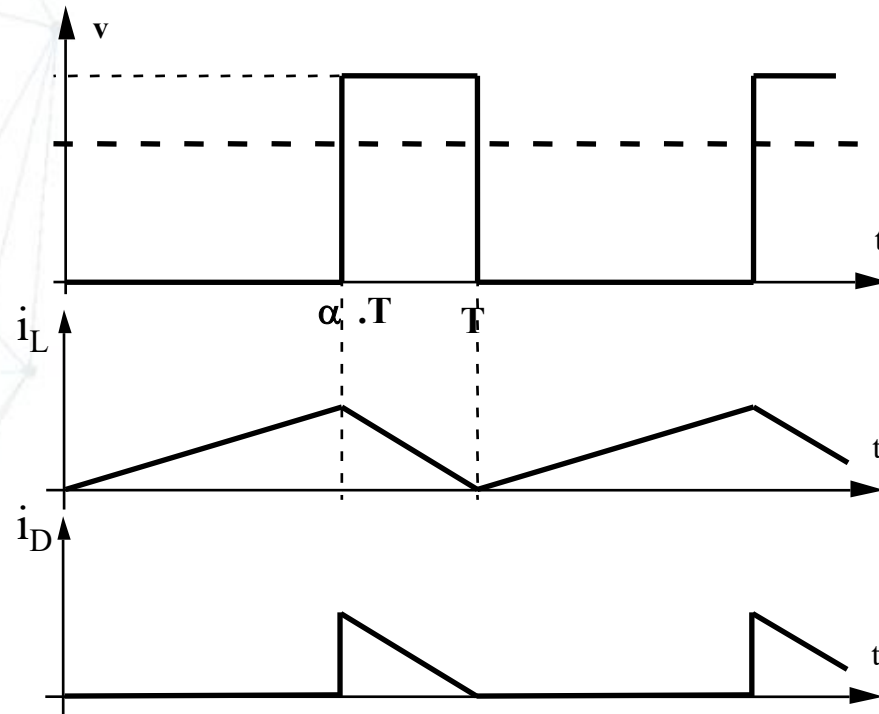
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Boundary mode: $i_L(T) = 0$

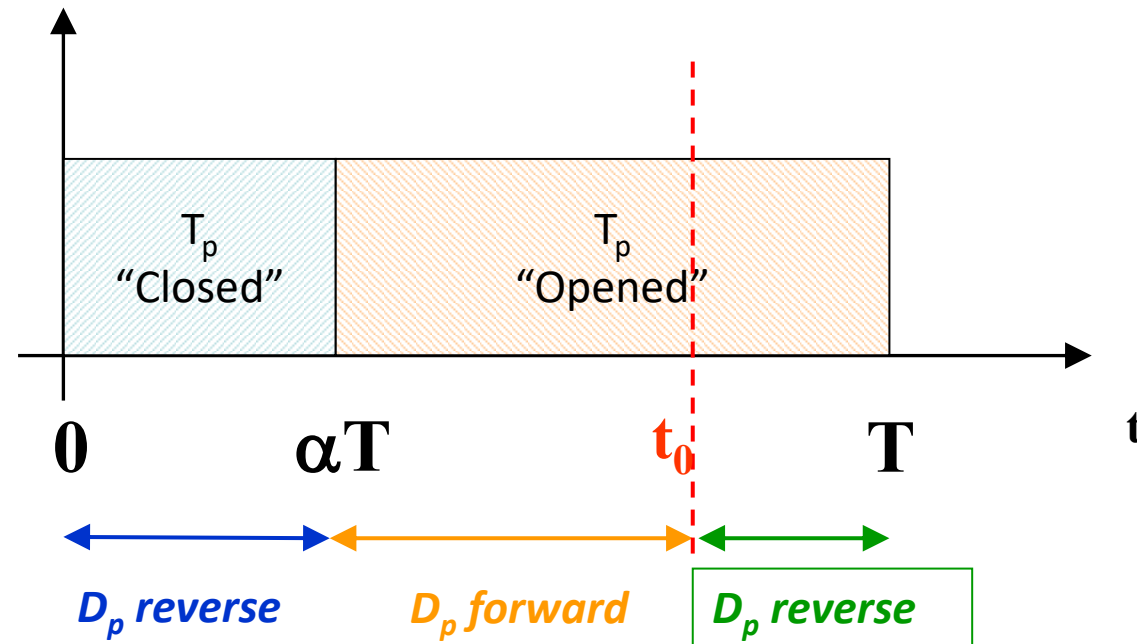


$$I_S = \langle i_D \rangle = \frac{1}{2} \cdot I_{L_{\max}} \cdot (1 - \alpha) = \frac{ET}{2L} \cdot \alpha(1 - \alpha)$$

$$V_S = \frac{E}{1 - \alpha}$$

$$I_S = \frac{ET}{2L} \cdot \frac{E}{V_S} \cdot \left(1 - \frac{E}{V_S}\right)$$

Discontinuous mode



At $t_0 = \alpha'.T$: diode switches to reverse mode

Direct conversion

Buck chopper

Boost chopper

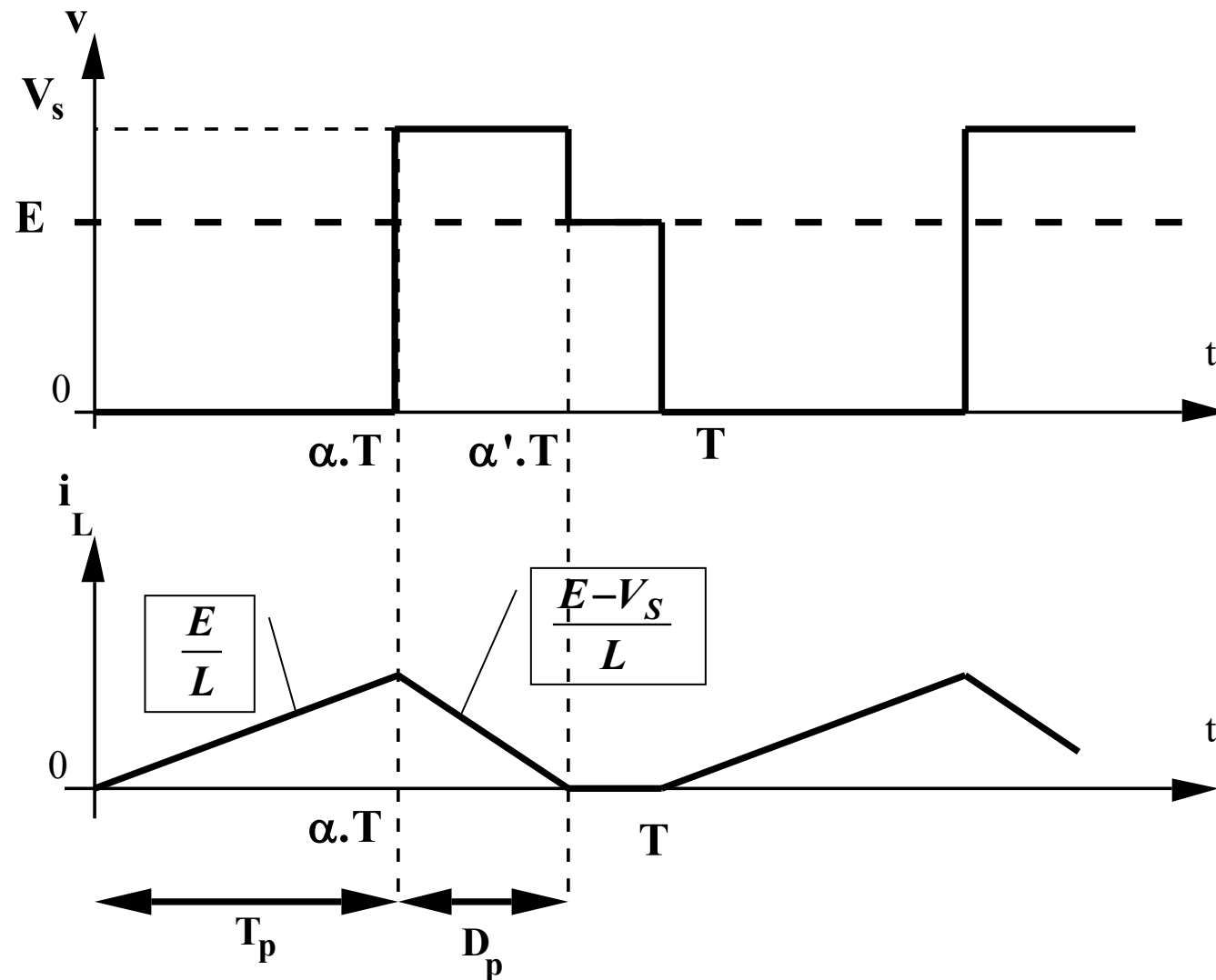
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode: Chronograms



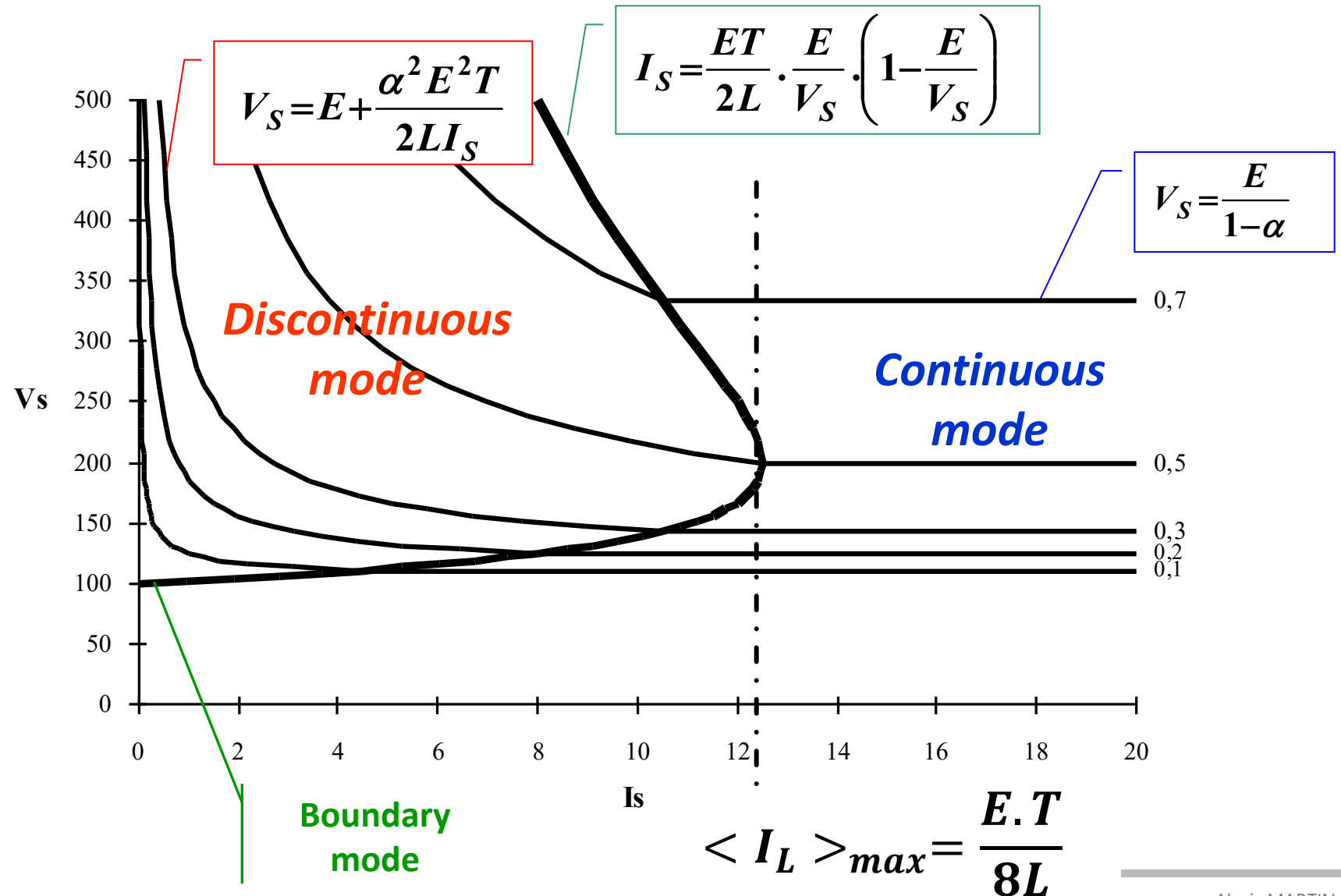
Discontinuous mode: calculation

$$\frac{E}{L} \alpha T = \frac{V_S - E}{L} (\alpha' - \alpha) T \quad \Rightarrow \quad V_S = \frac{\alpha'}{\alpha' - \alpha} E$$

$$I_S = \langle i_D \rangle = \frac{1}{2} \cdot I_{L \max} \cdot (\alpha' - \alpha) = \frac{ET}{2L} \cdot \alpha (\alpha' - \alpha)$$

$$\Rightarrow \quad V_S = E + \frac{\alpha^2 E^2 T}{2L I_S}$$

Discontinuous mode: output characteristics



Introduction

Direct conversion

Buck chopper

Boost chopper

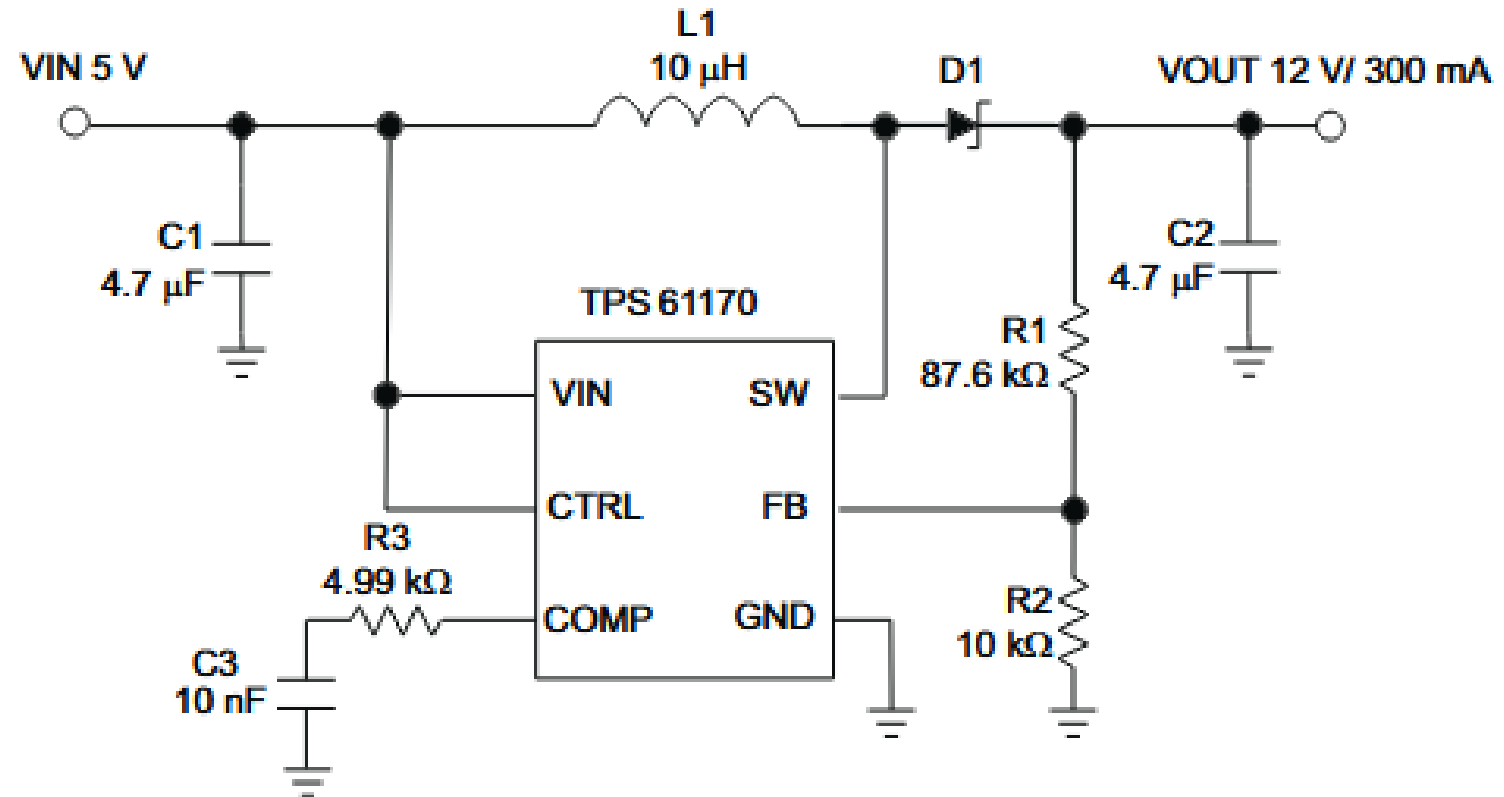
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Example : TPS61170



Introduction

Direct conversion

Buck chopper

Boost chopper

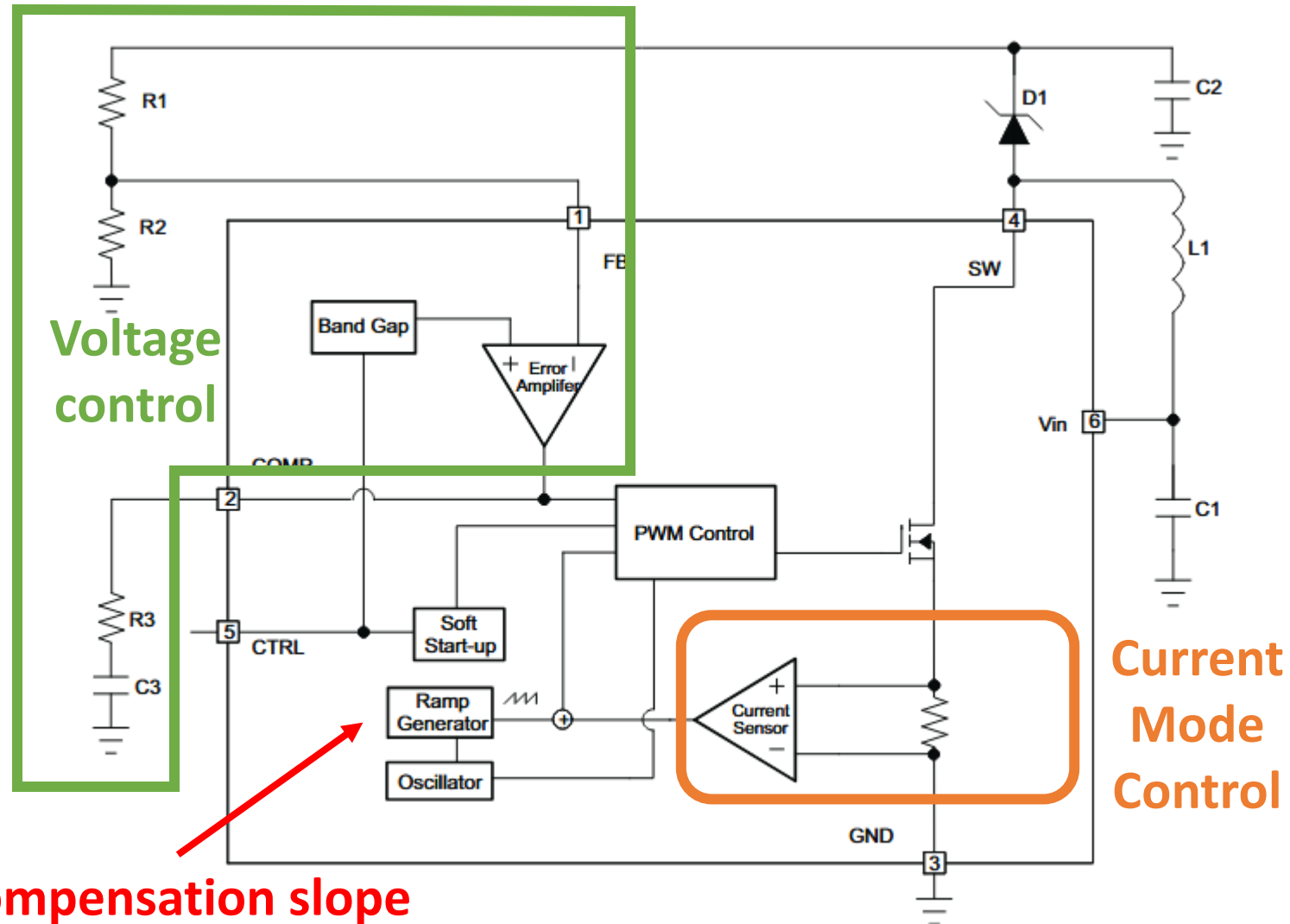
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Example : TPS61170



Compensation slope
(for stabilization)

Current
Mode
Control

Introduction

Direct conversion

Buck chopper

Boost chopper

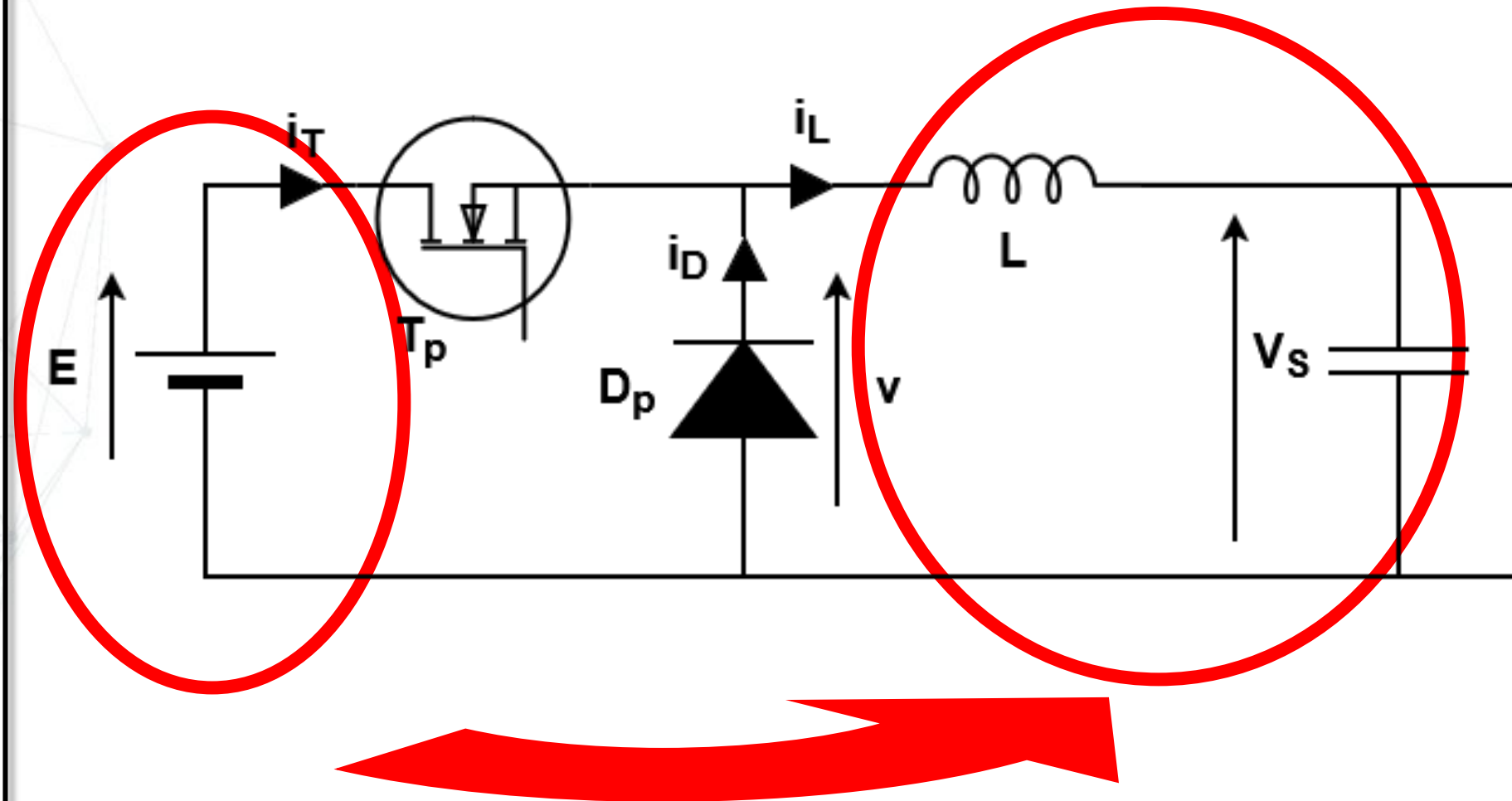
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reminder : buck chopper



Direct conversion

Buck chopper

Boost chopper

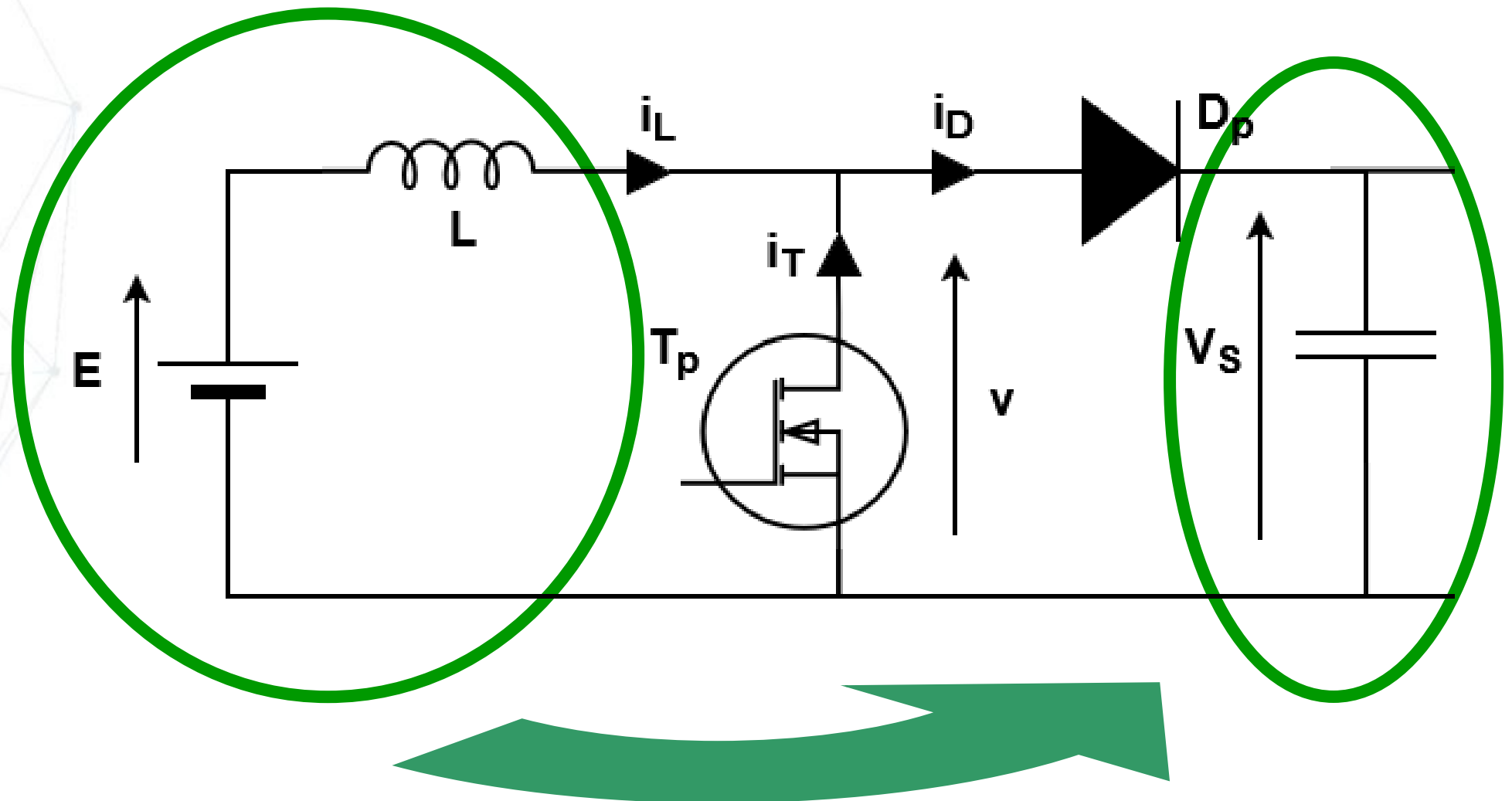
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reminder : boost chopper



Introduction

Direct conversion

Buck chopper

Boost chopper

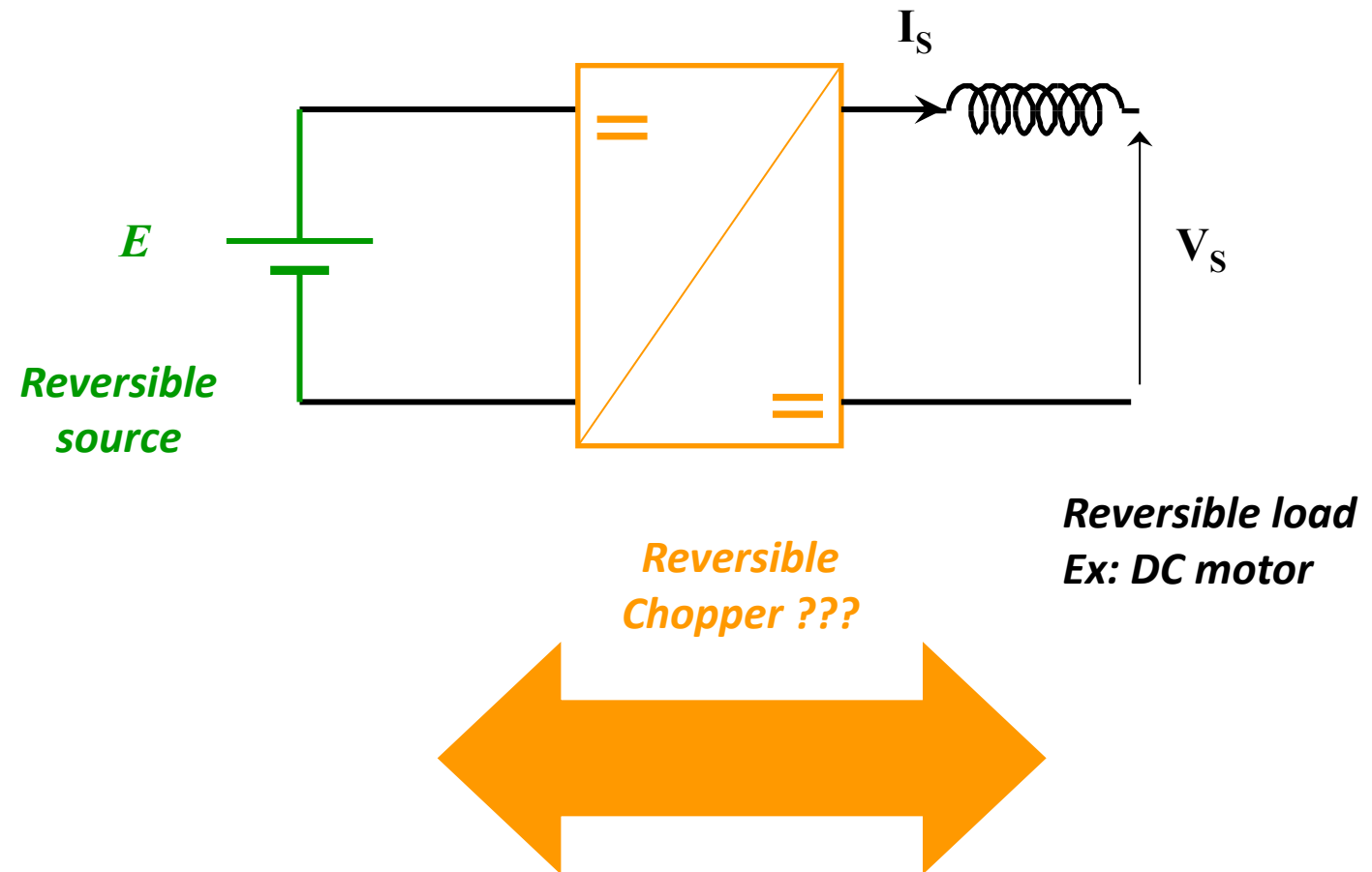
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Linked sources



Introduction

Direct conversion

**Reversible
choppers**

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reversible Chopper

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper

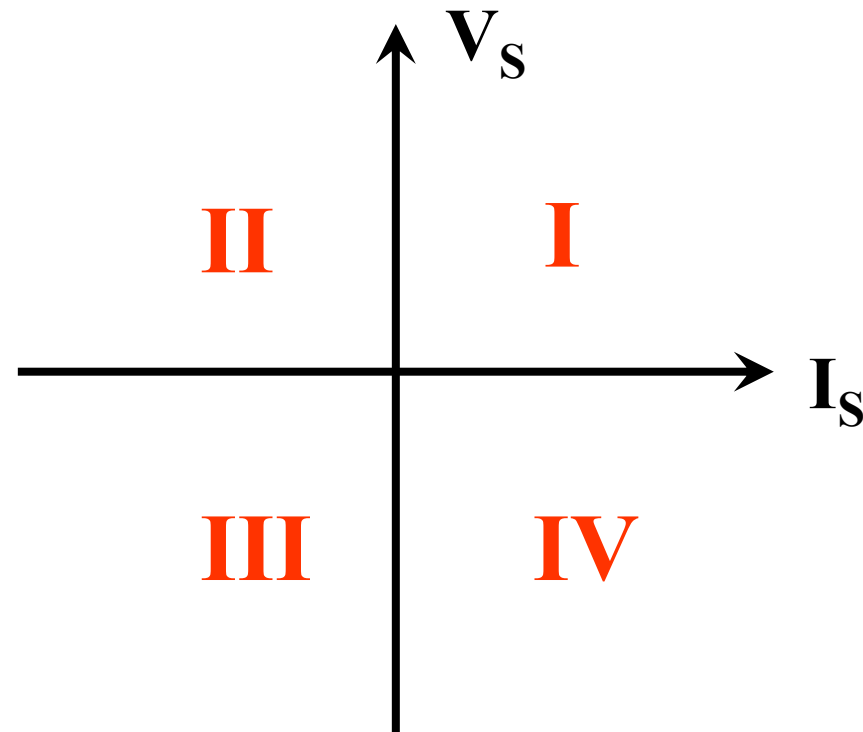
4 quadrants chopper

Inverters

Closed loop chopper control

Energy storage based chopper

Reversible choppers:
quadrants



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

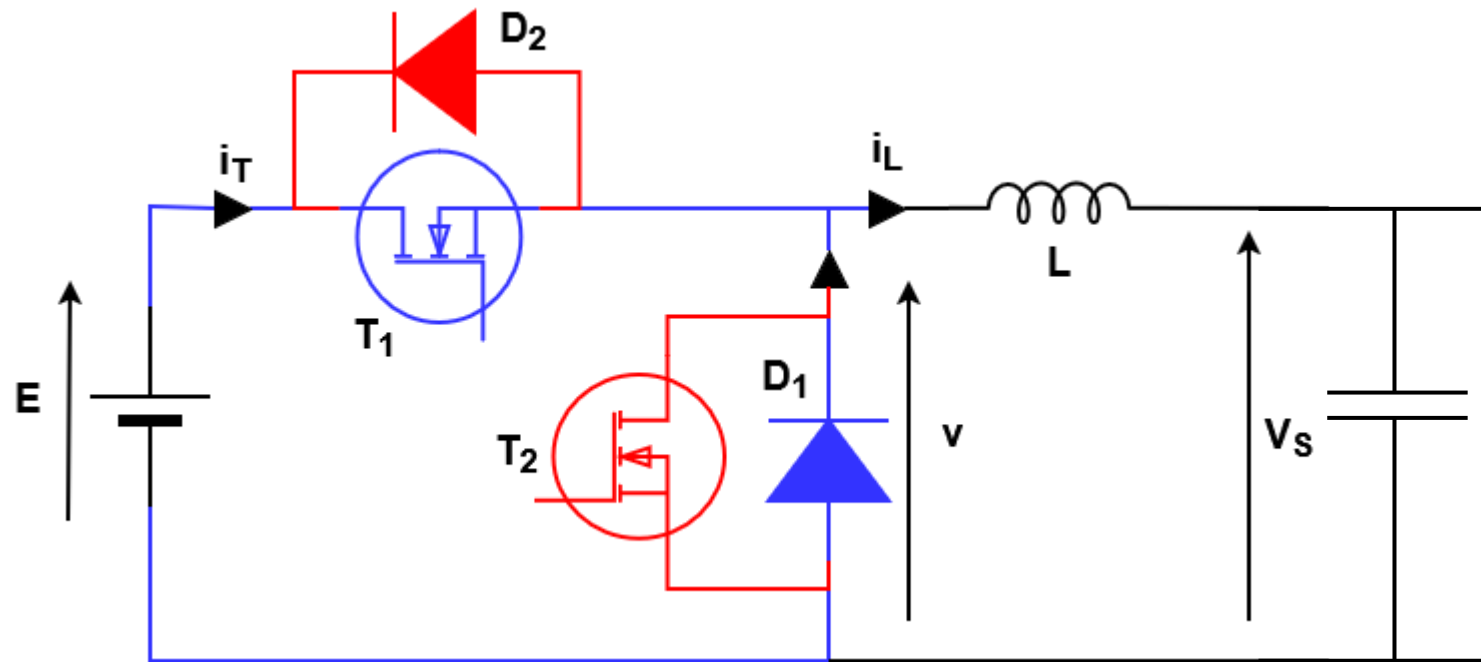
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants reversible chopper



Buck chopper

Boost chopper

Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

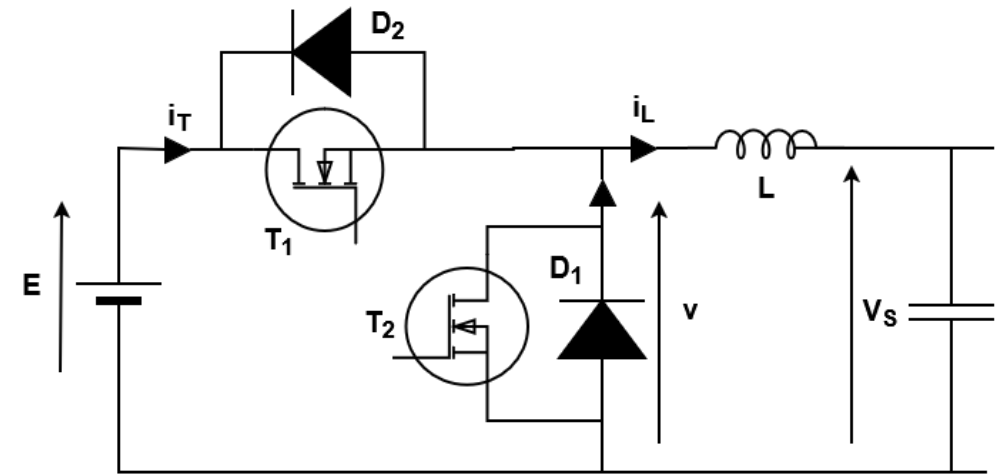
4 quadrants chopper

Inverters

Closed loop
chopper control

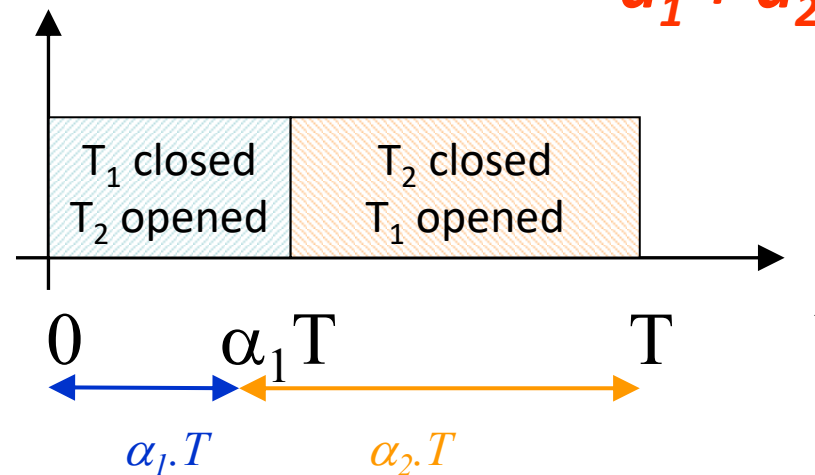
Energy storage
based chopper

Two quadrants chopper:
control



2 switches => 2 duty cycles
Complementary control

$$a_1 + a_2 = 1$$



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

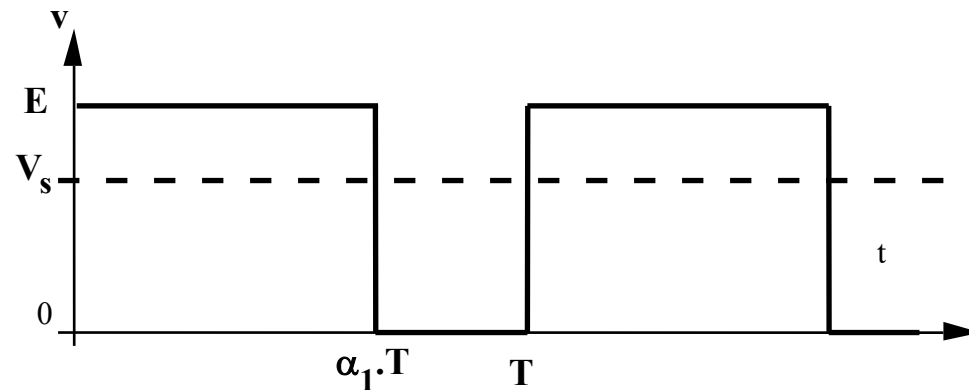
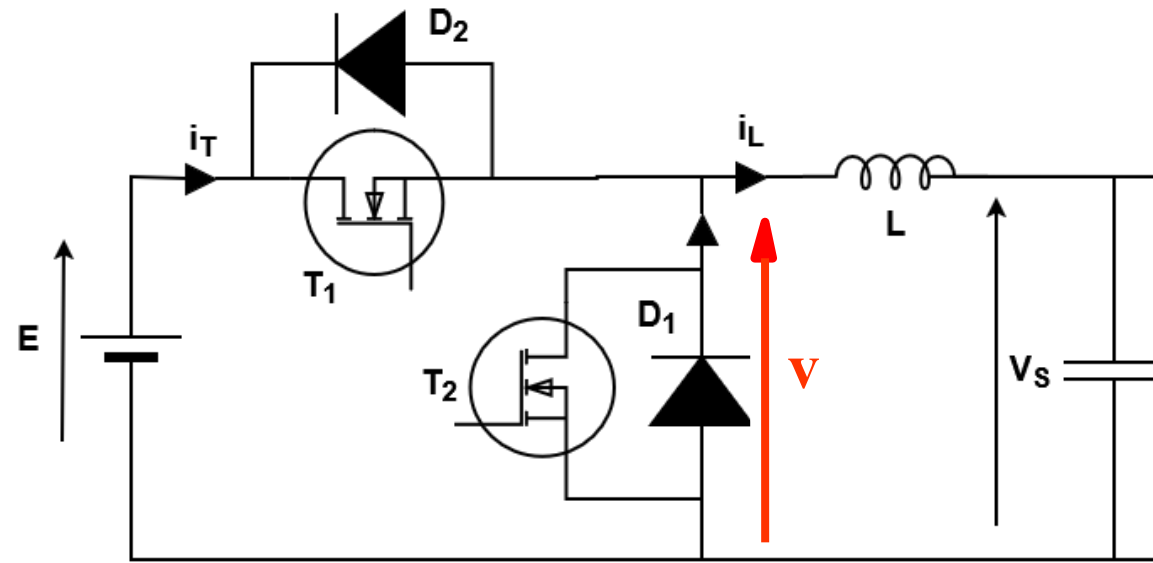
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: output voltage



$$V_s = \alpha_1 \cdot E$$

Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

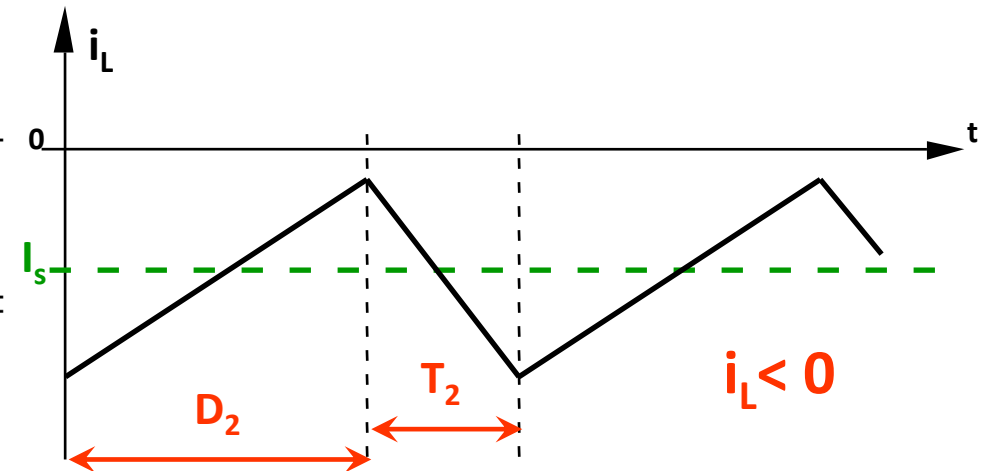
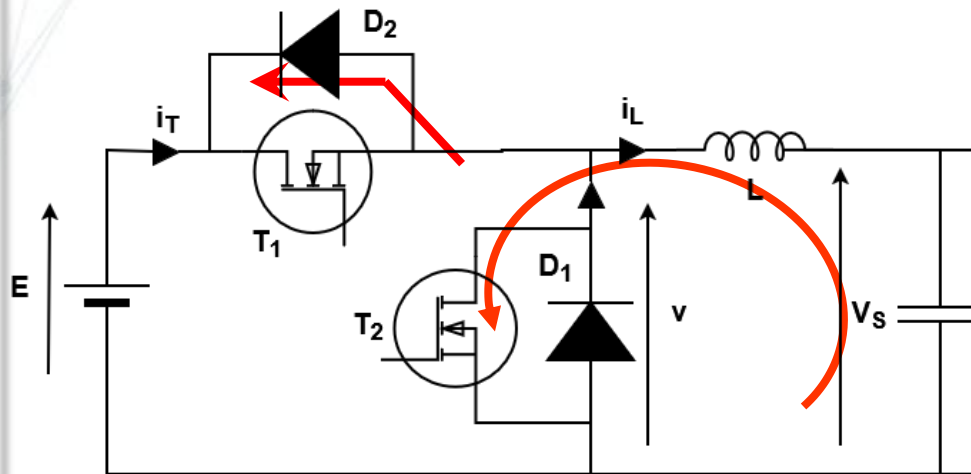
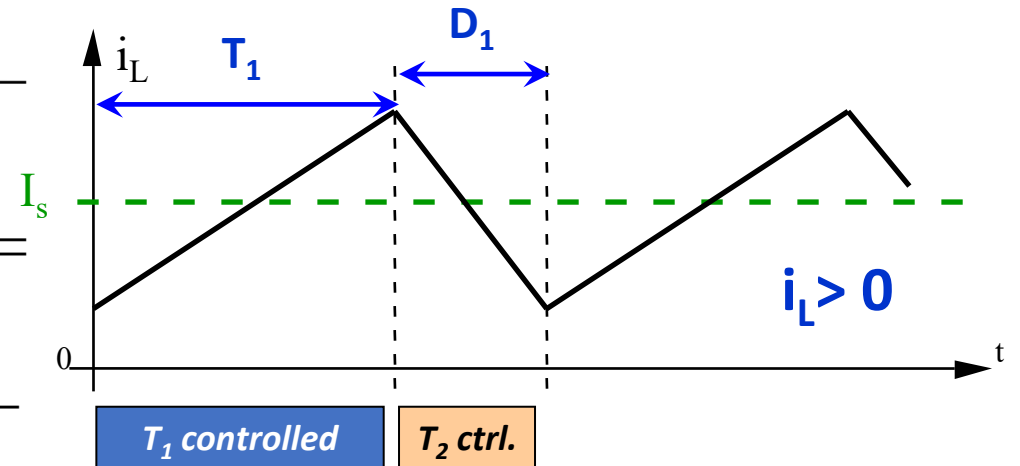
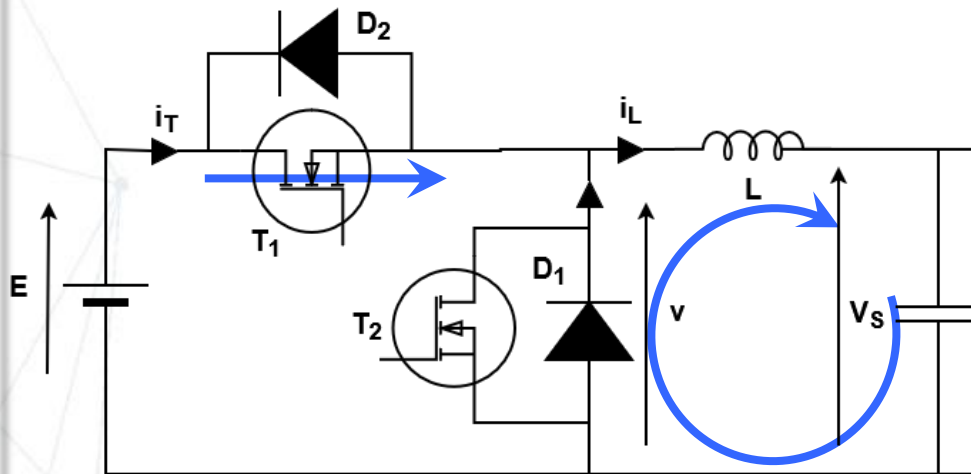
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: output current



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

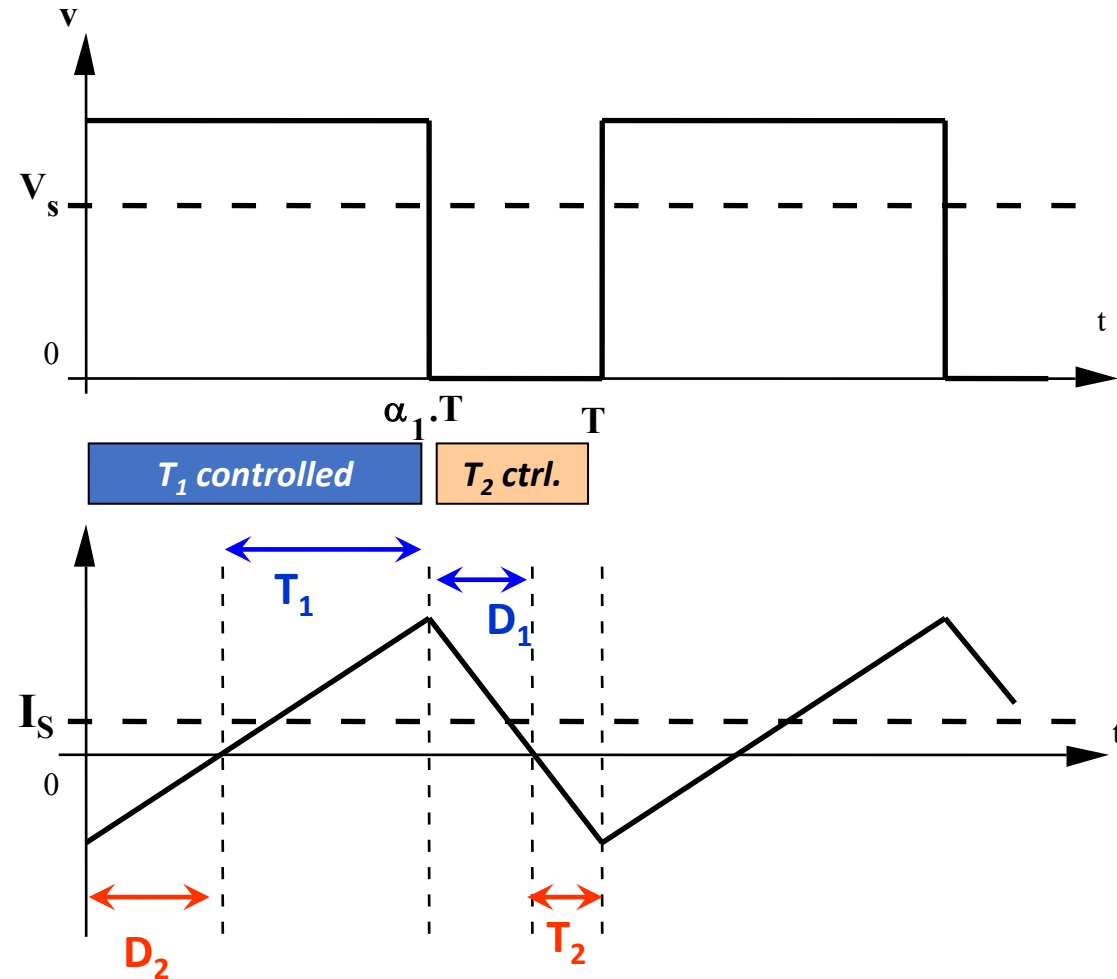
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: Arbitrary output current



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

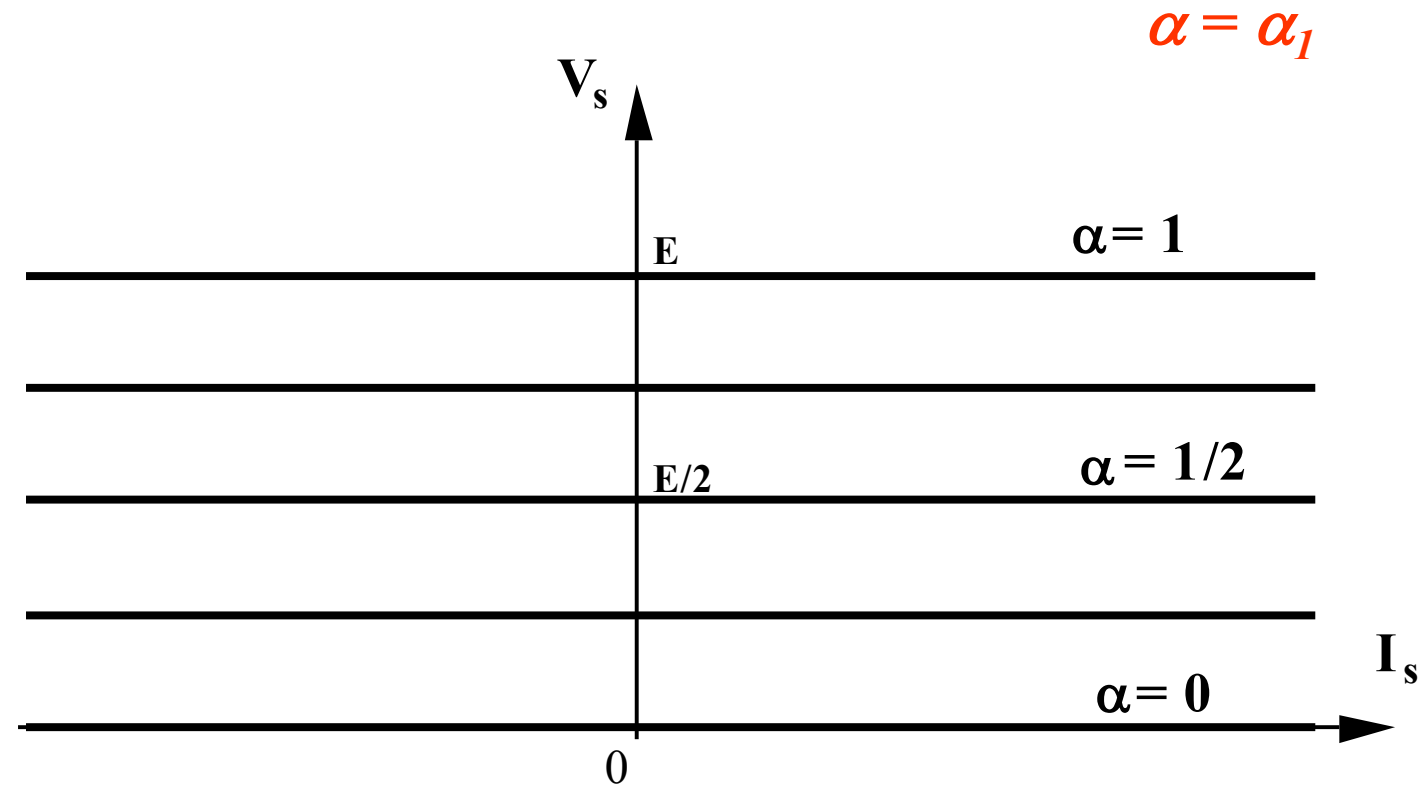
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: Output characteristics grid



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

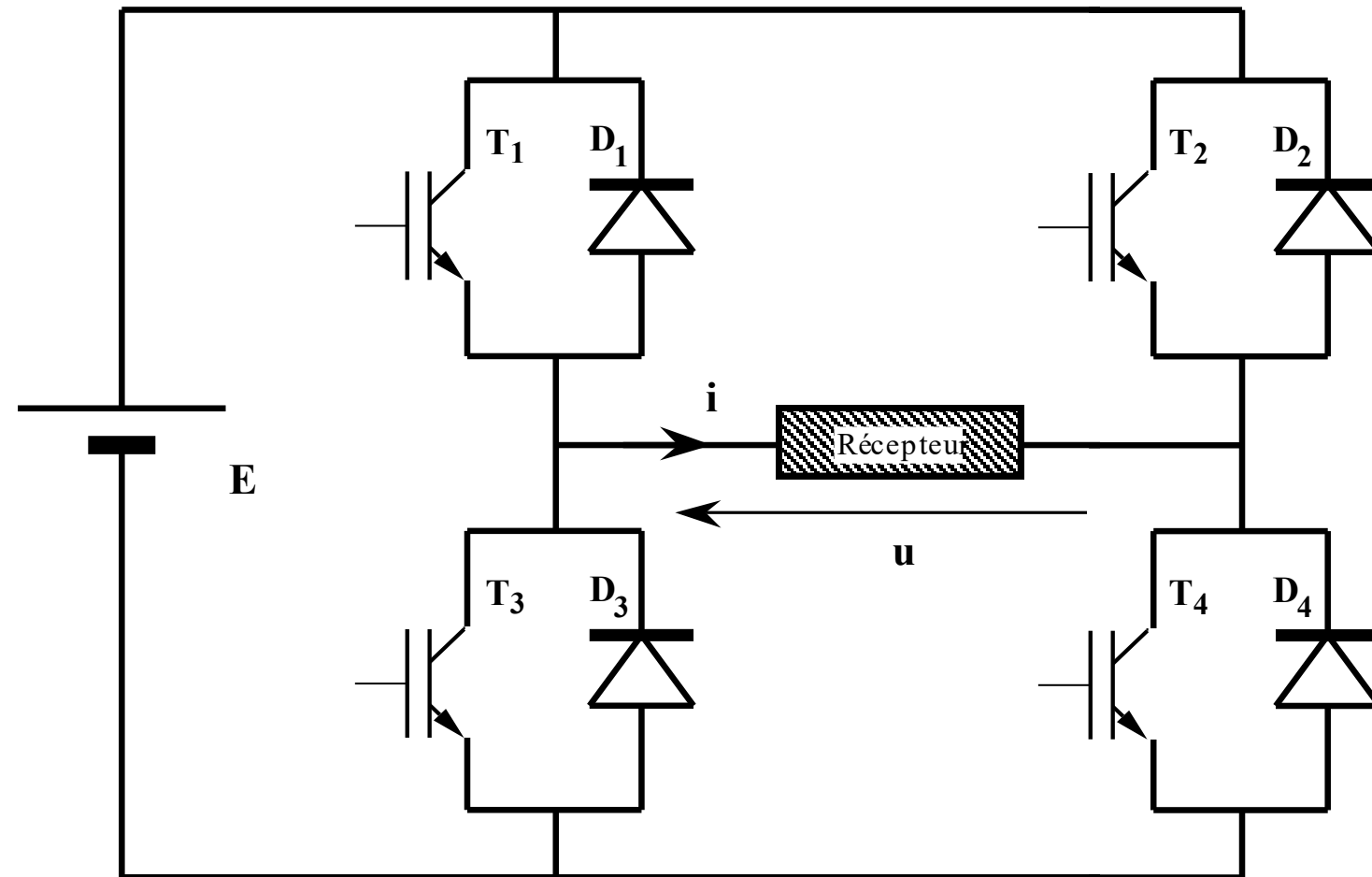
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

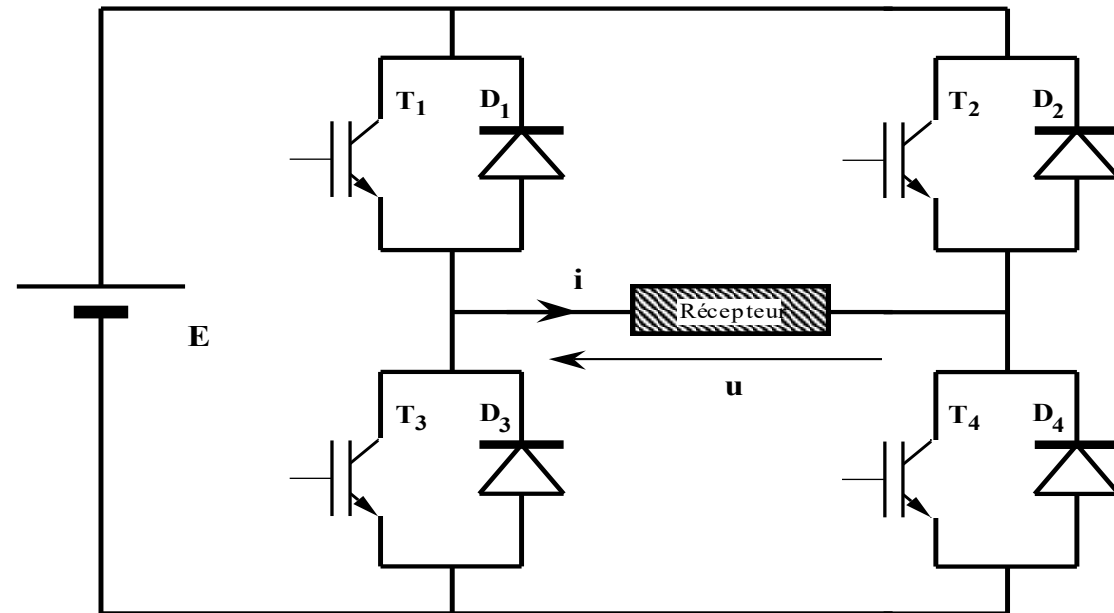
4 quadrants chopper

Inverters

Closed loop
chopper control

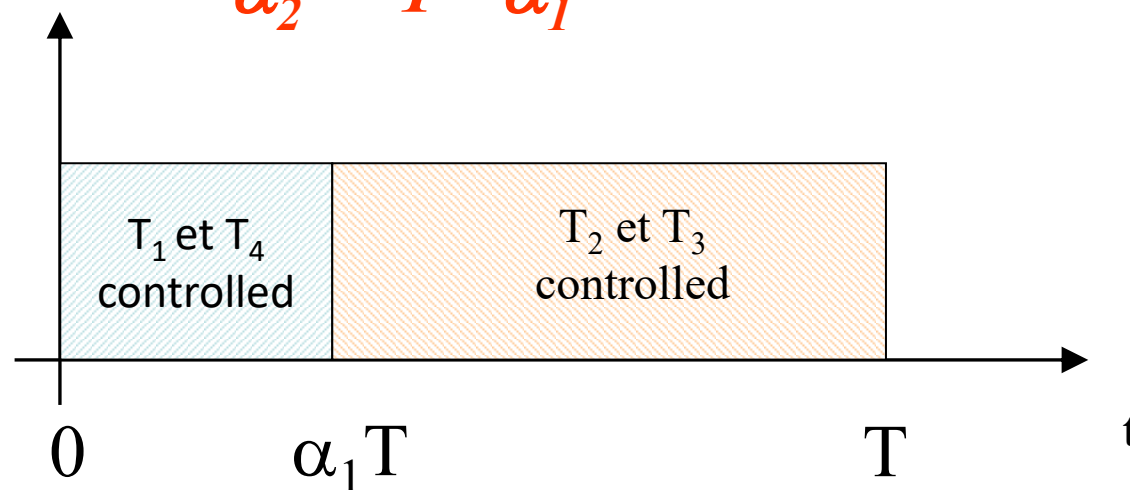
Energy storage
based chopper

4 quadrants chopper:
Synchronous complementary control



$$\alpha_1 = \alpha_4 \text{ et } \alpha_2 = \alpha_3$$

$$\alpha_2 = 1 - \alpha_1$$



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

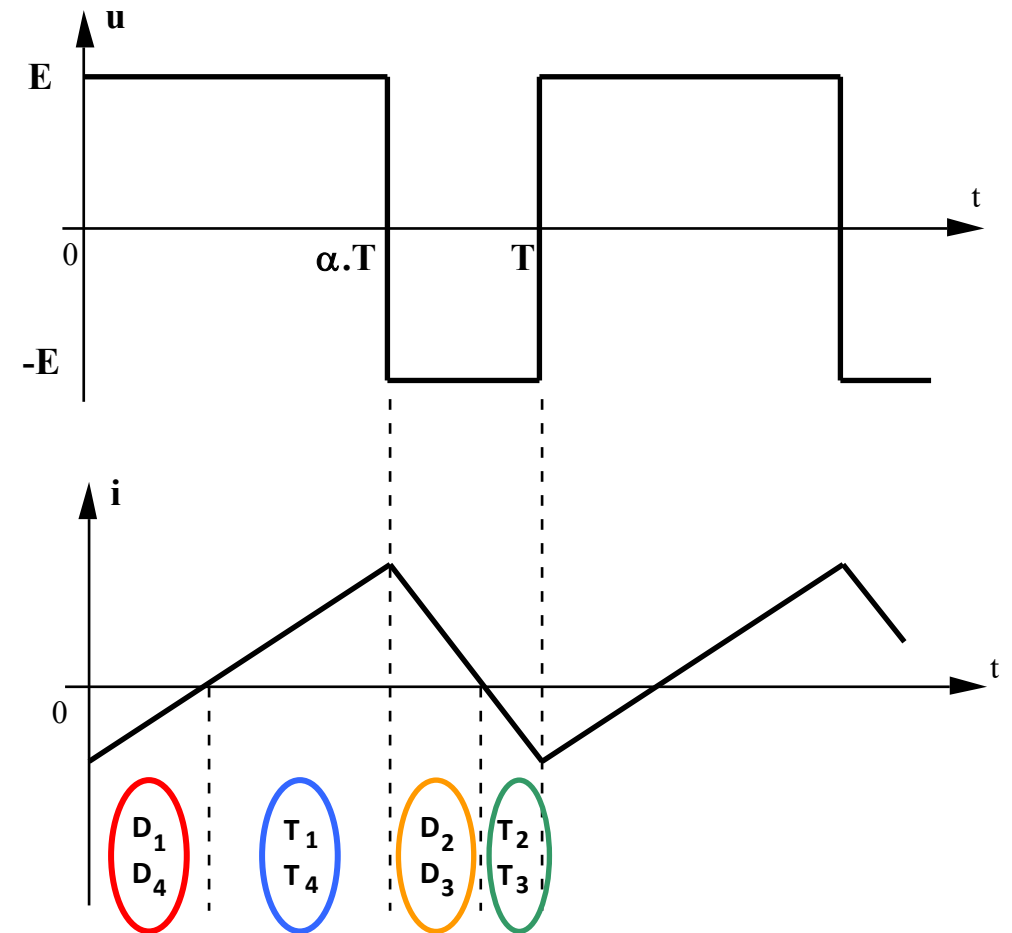
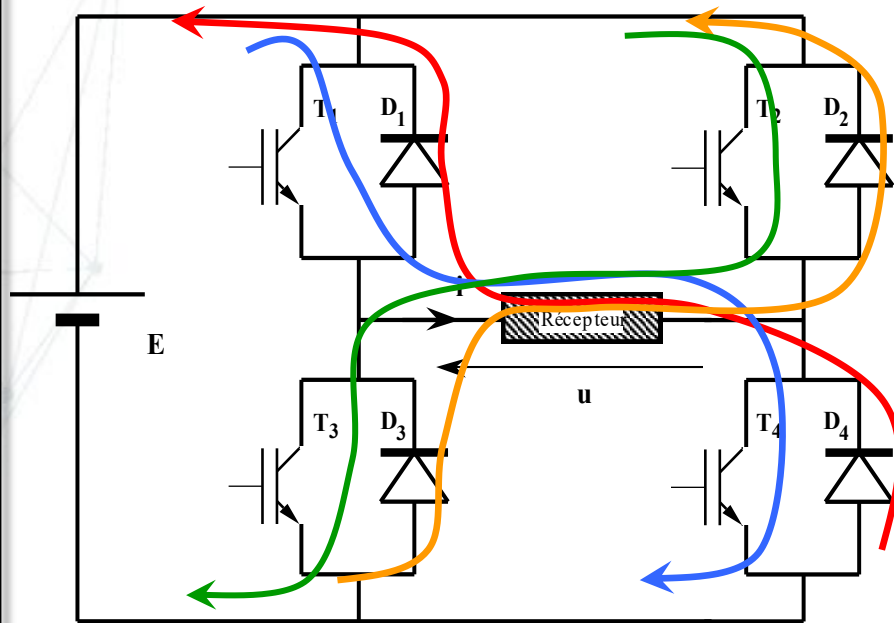
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper:
Synchronous complementary control (chronograms)



Introduction

Direct conversion

Reversible choppers

2 quadrants chopper

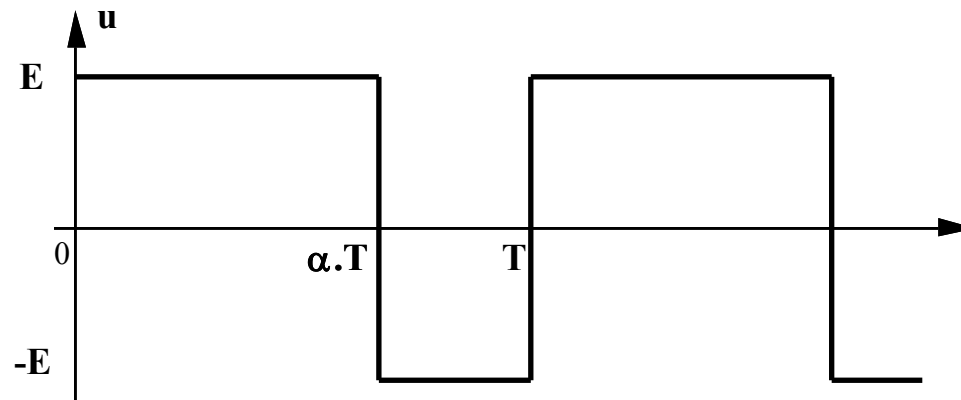
4 quadrants chopper

Inverters

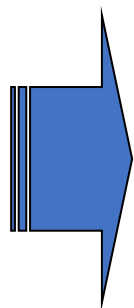
Closed loop chopper control

Energy storage based chopper

4 quadrants chopper:
Synchronous complementary control (Average output voltage)



$$\langle u \rangle = (2\alpha - 1).E$$



High voltage ripple: $2.E$
Low switching frequency

Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

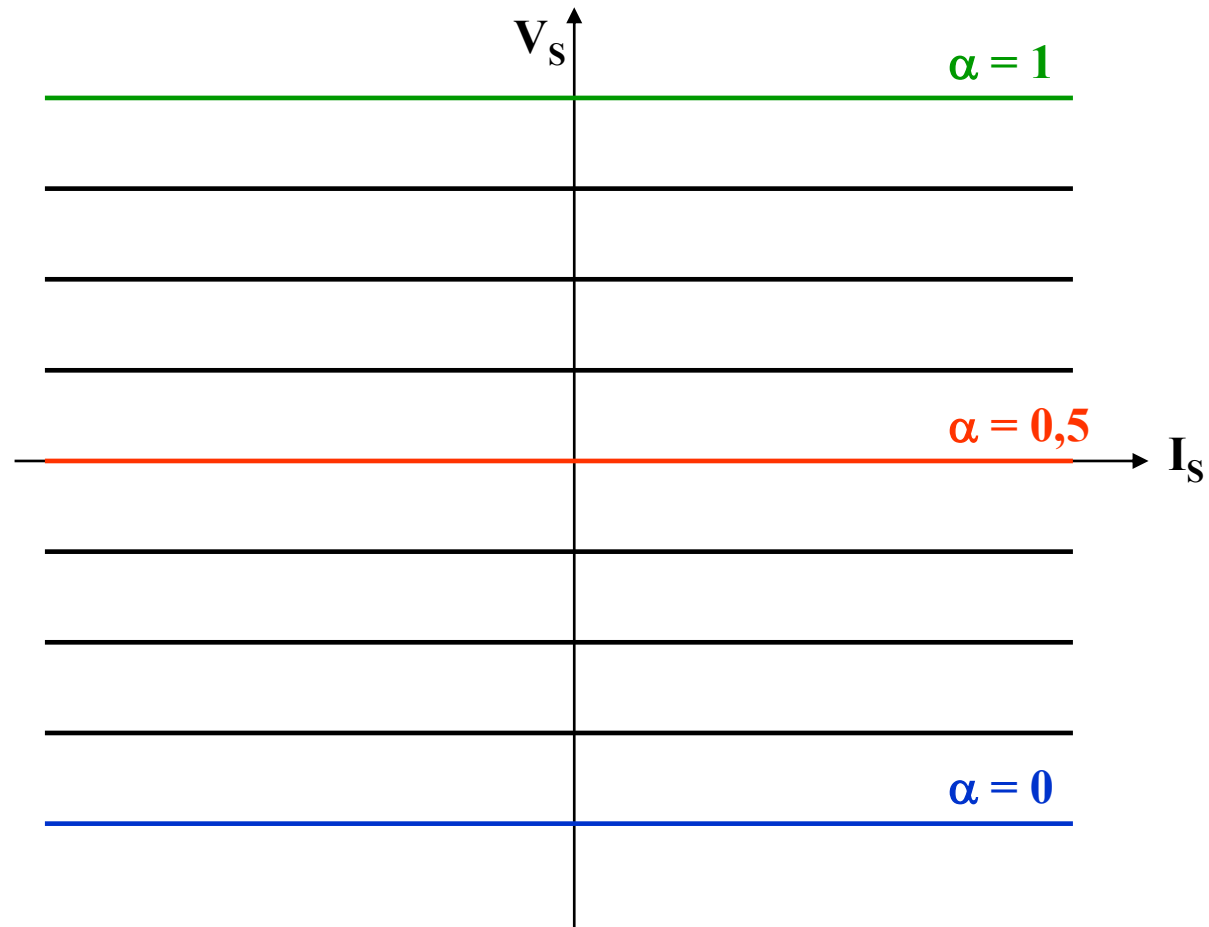
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Output characteristics



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

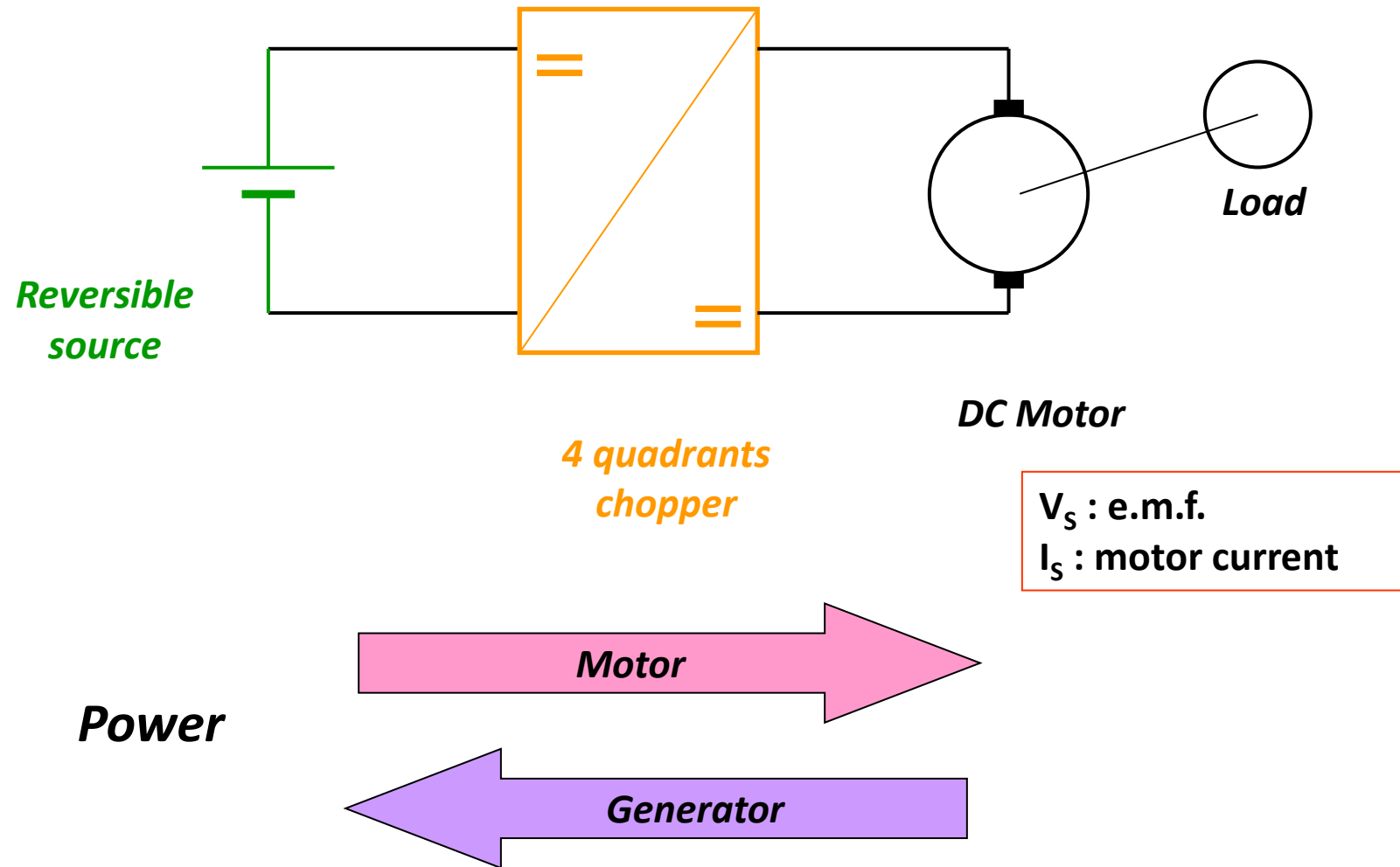
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: control of a DC motor



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

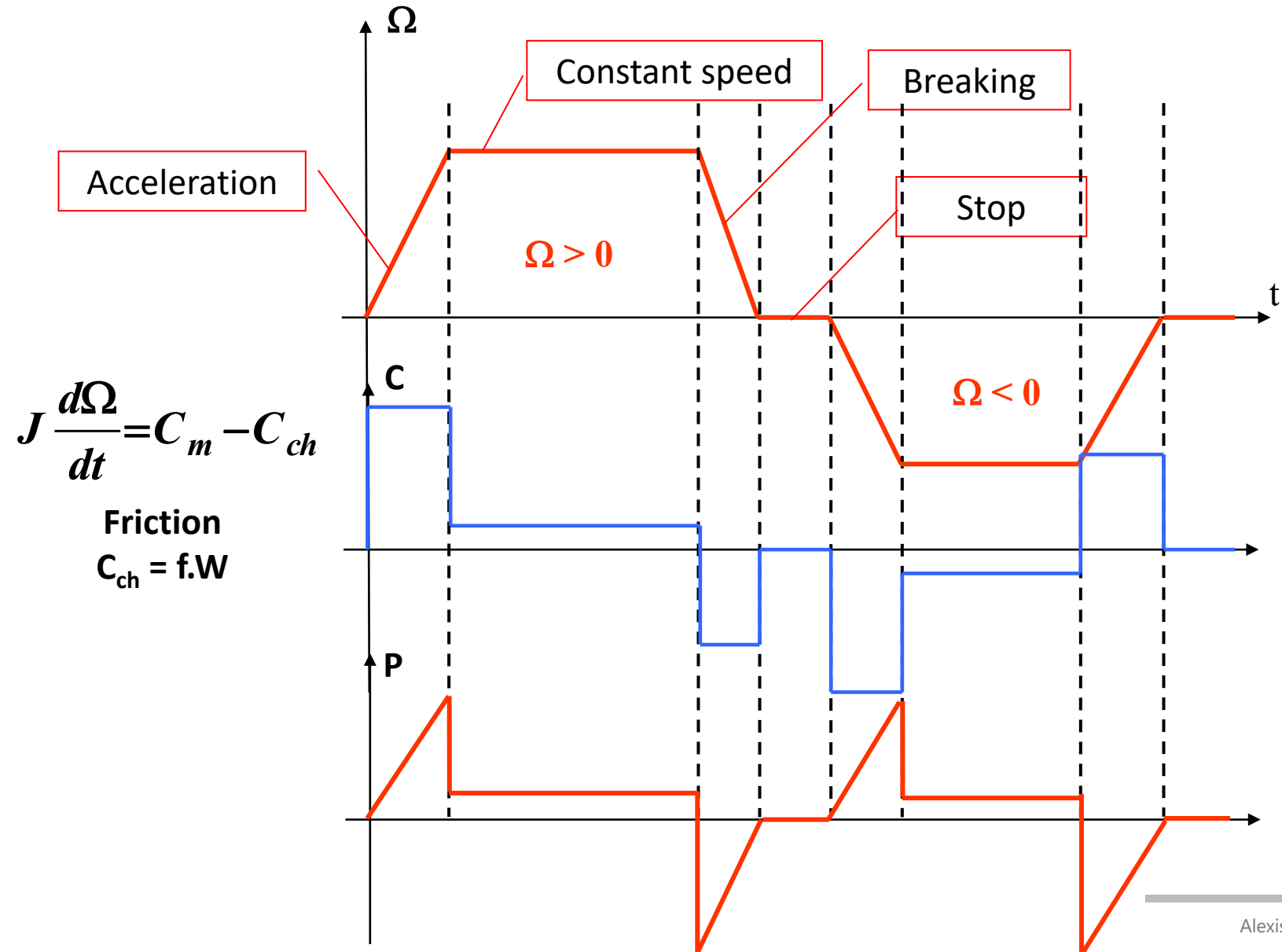
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: working cycle



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

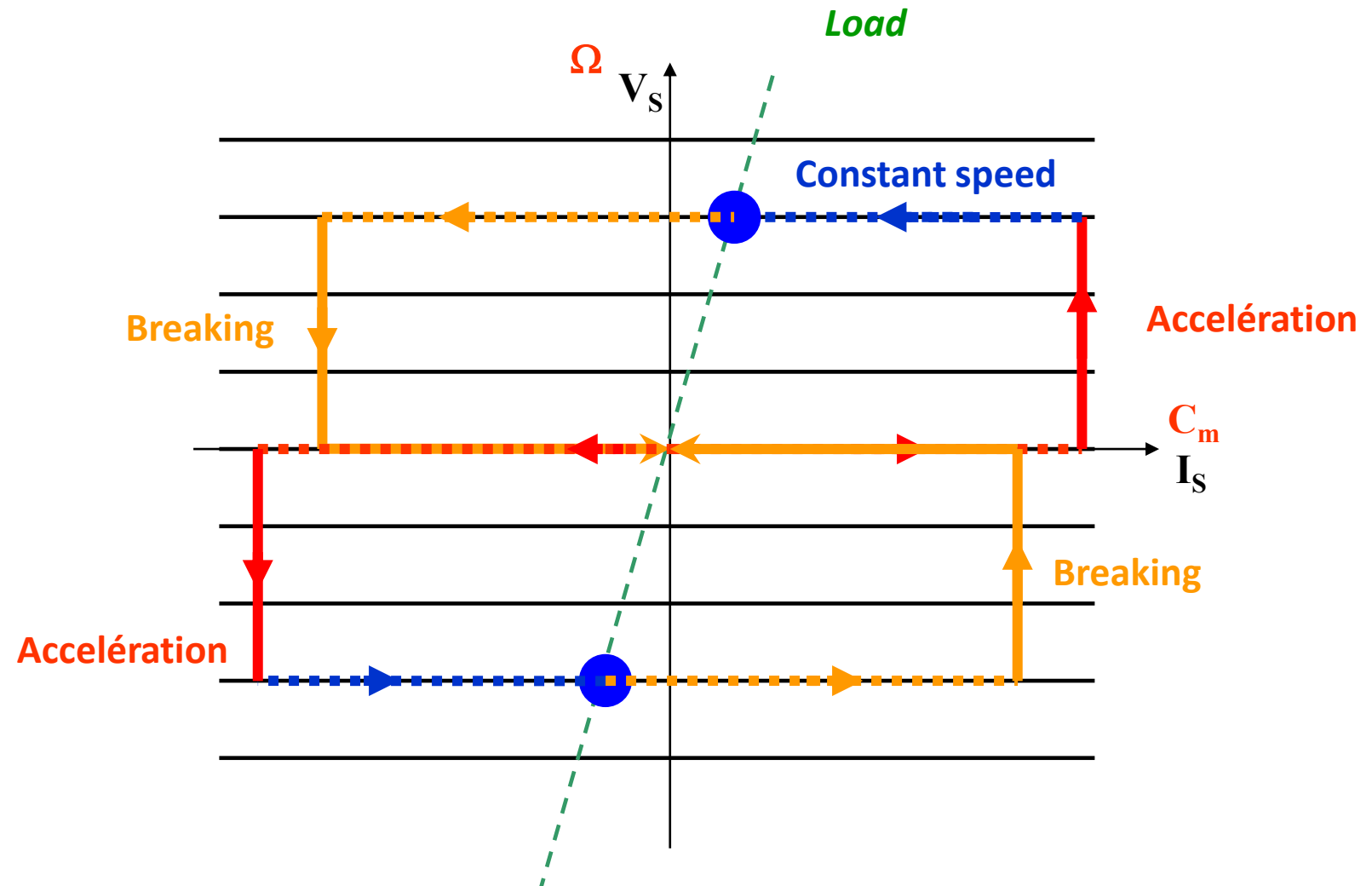
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Operating point



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

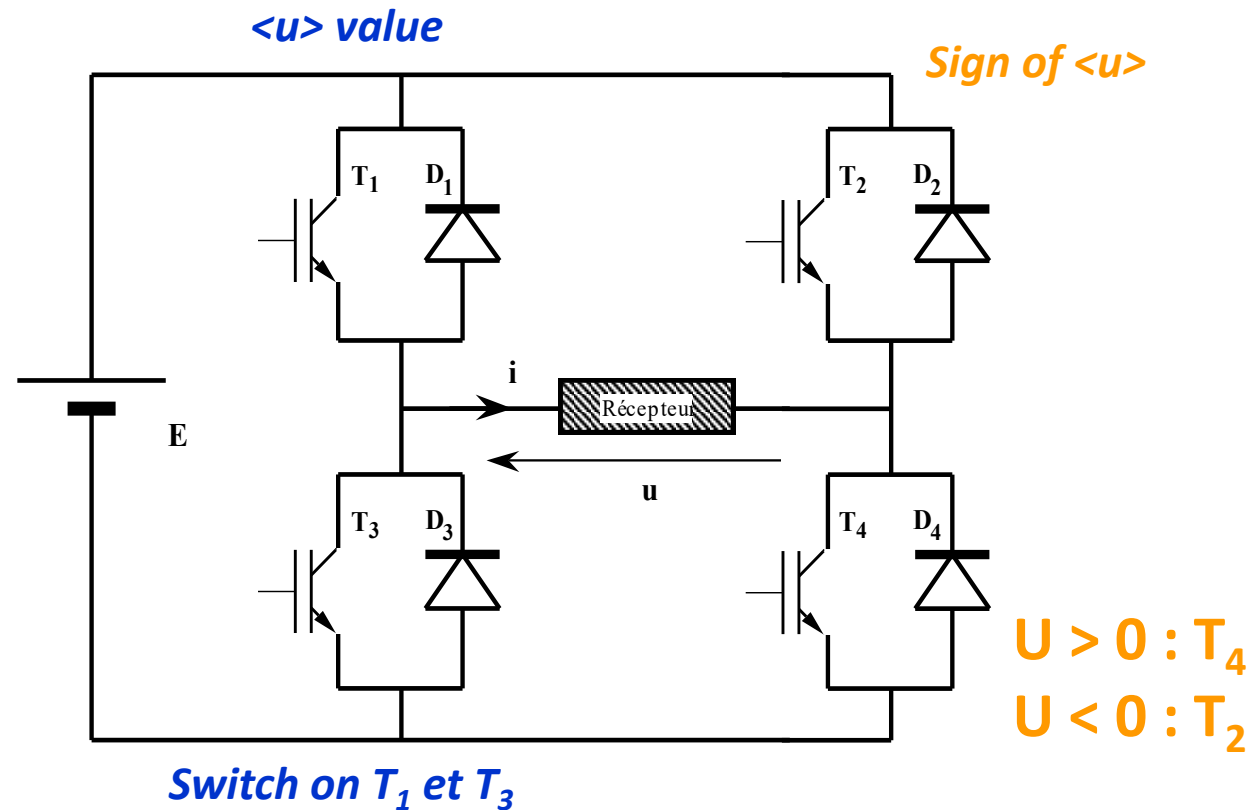
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: additional control



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

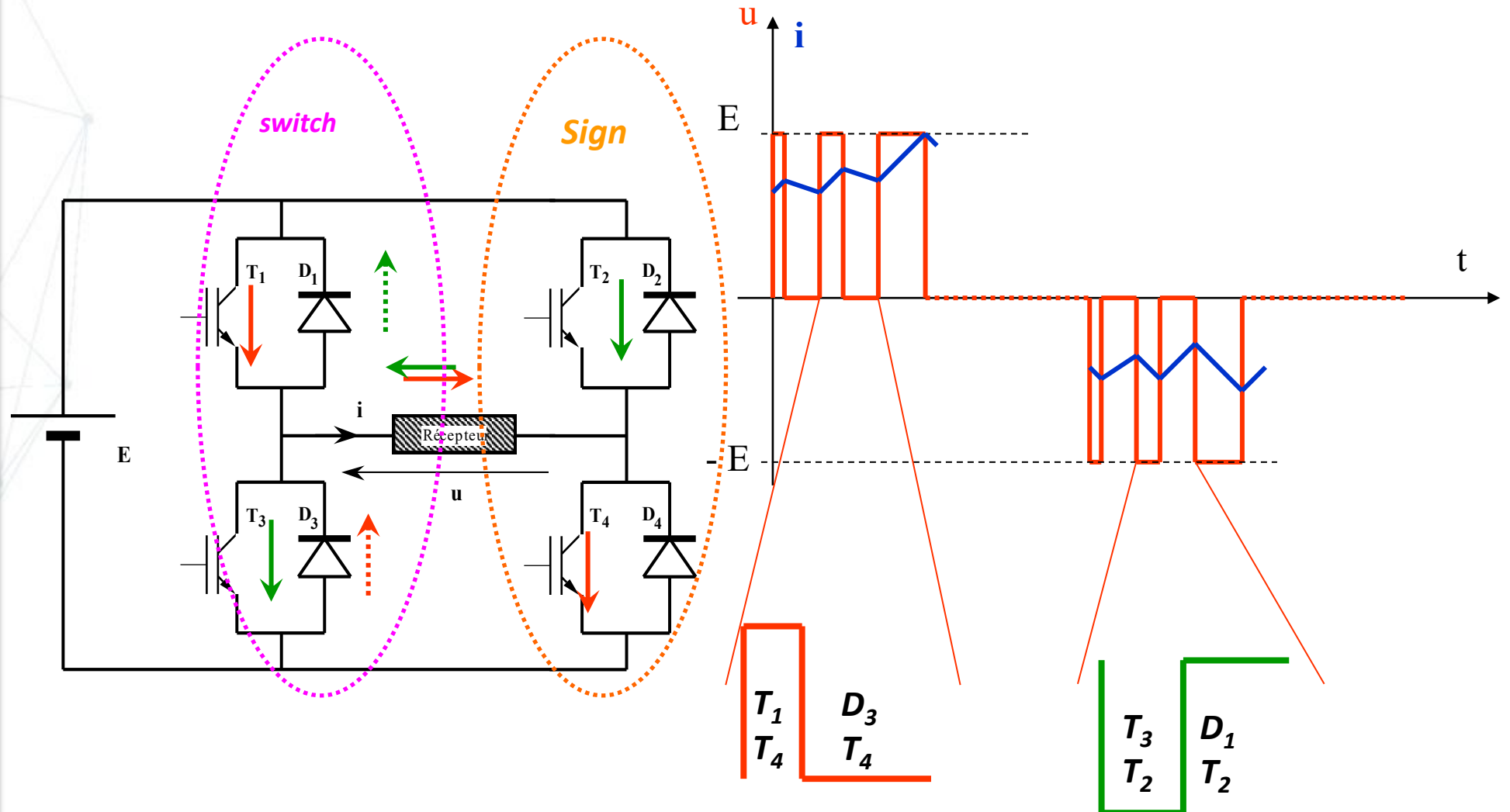
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: additional control



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

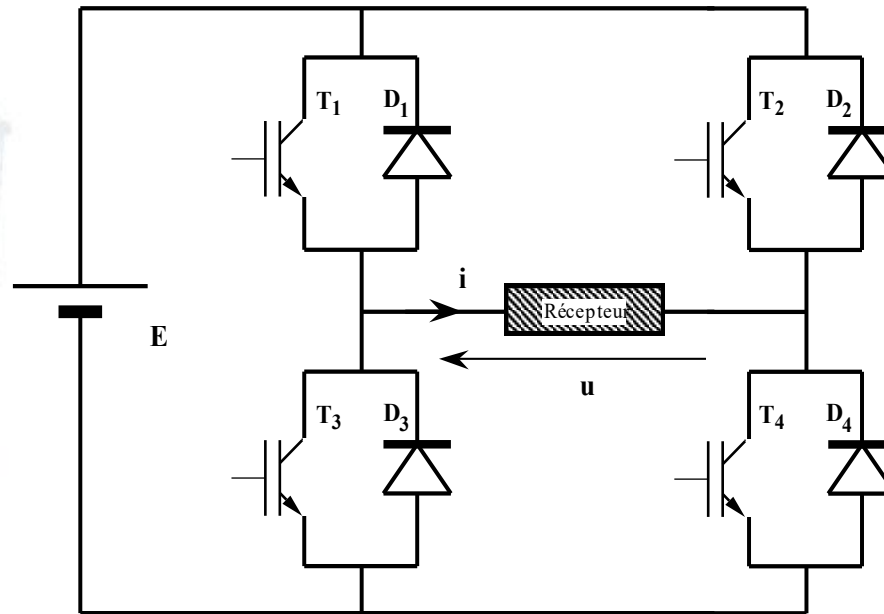
4 quadrants chopper

Inverters

Closed loop
chopper control

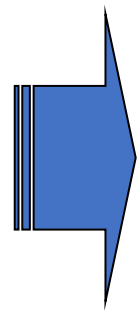
Energy storage
based chopper

4 quadrants chopper: additional control



$$U = +/ - a.E$$

Average value during
a switching time T_d



Voltage ripple reduced: E

Switching only on T_1/T_3 : losses reduced

Instabilities around 0V !!

Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

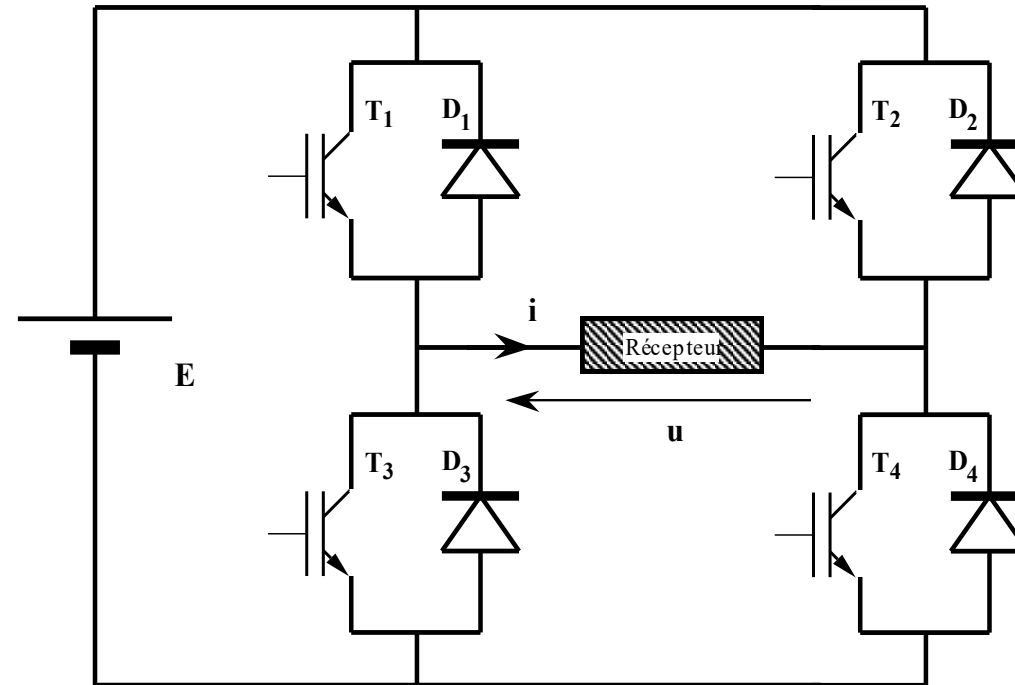
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Shifted complementary control



$$\alpha_1 = \alpha_4$$

$$\alpha_2 = \alpha_3$$

$$\alpha_3 = 1 - \alpha_1$$

$$\alpha_4 = 1 - \alpha_2$$

Switching on T_1 and T_3 at $t = 0$

Switching on T_2 and T_4 at $t = T/2$

Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

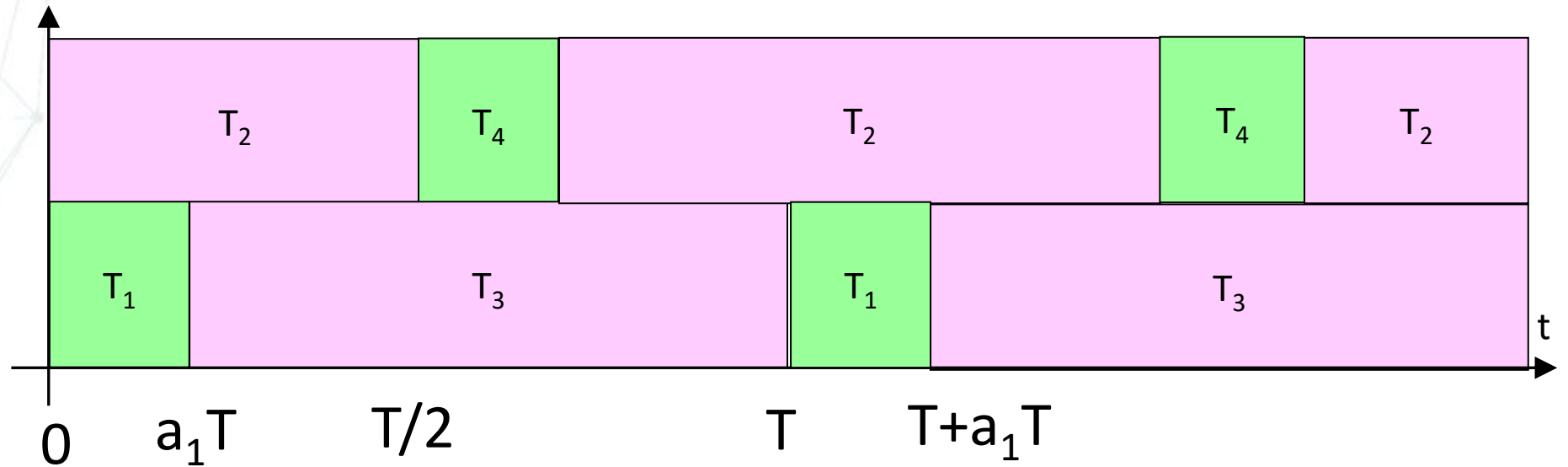
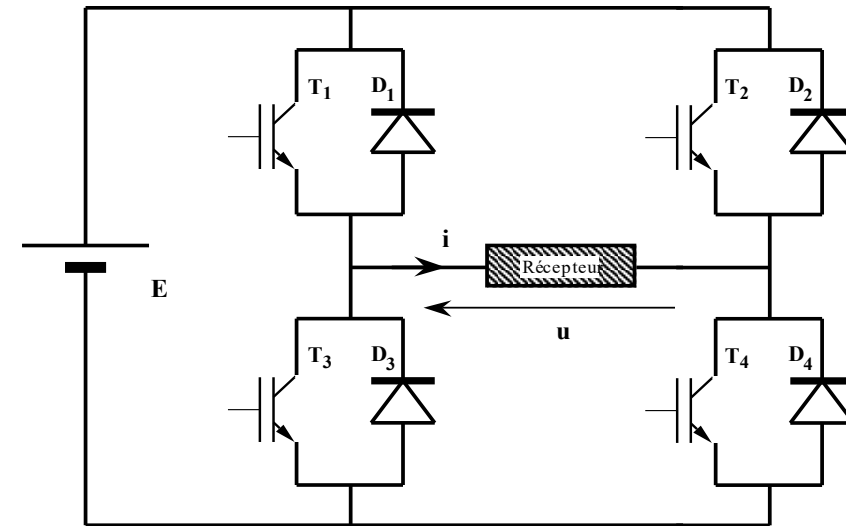
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Shifted complementary control



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

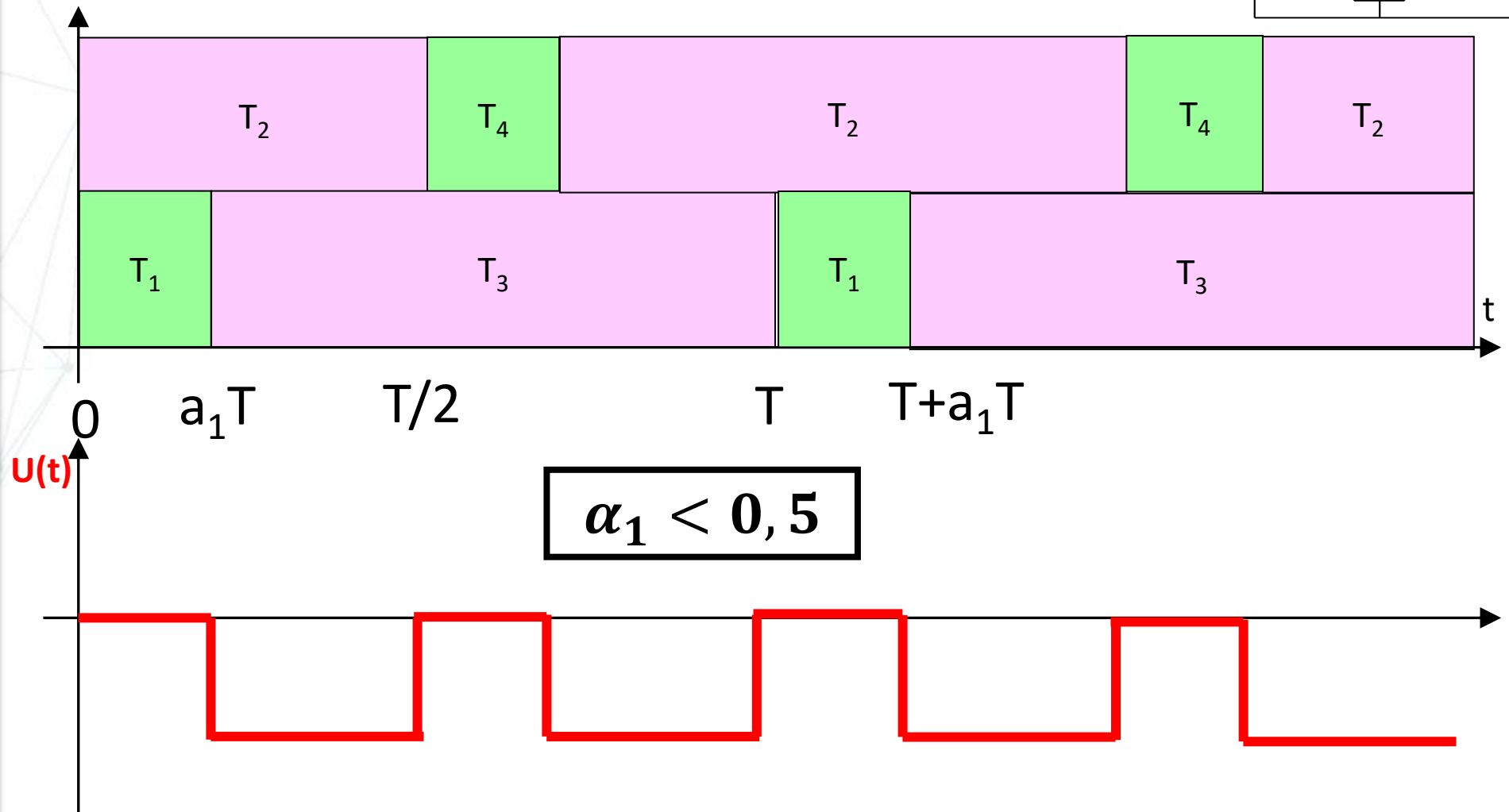
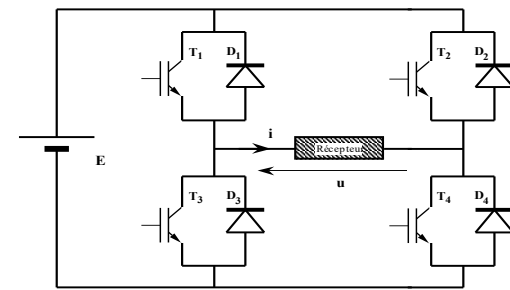
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Shifted complementary control



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

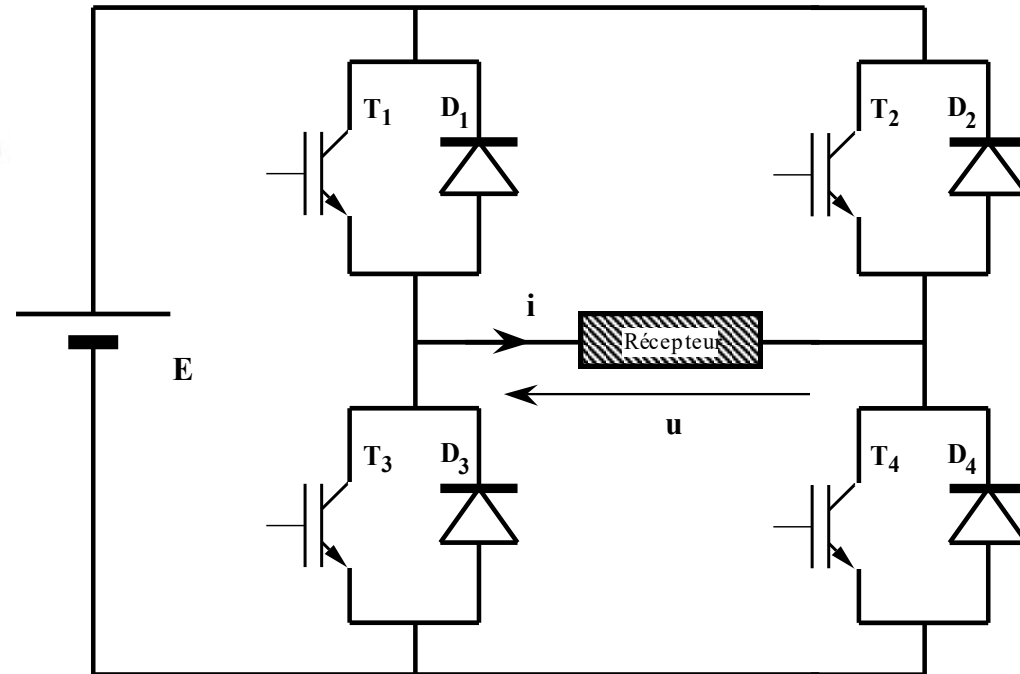
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Shifted complementary control



$$U = (2a-1).E$$



Voltage ripple reduced: E
High switching frequency
=> PWM

Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

4 quadrants chopper

Inverters

Closed loop
chopper control

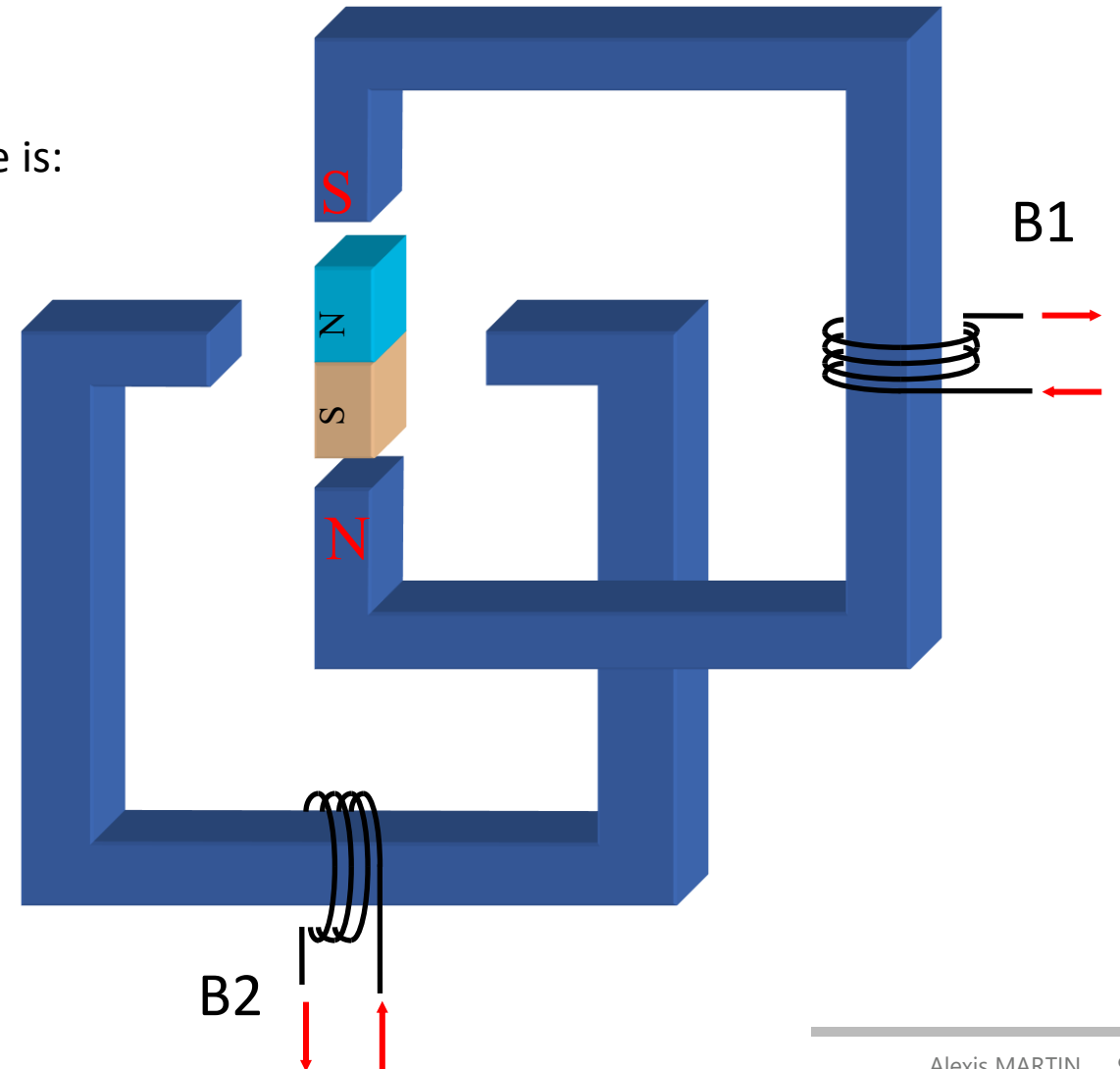
Energy storage
based chopper

4 quadrants chopper: Stepper motor control

Operation principle: full step

For a clockwise rotation, the sequence is:

B1	B2
+I	0
0	+I
-I	0
0	-I



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

4 quadrants chopper

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Closed loop
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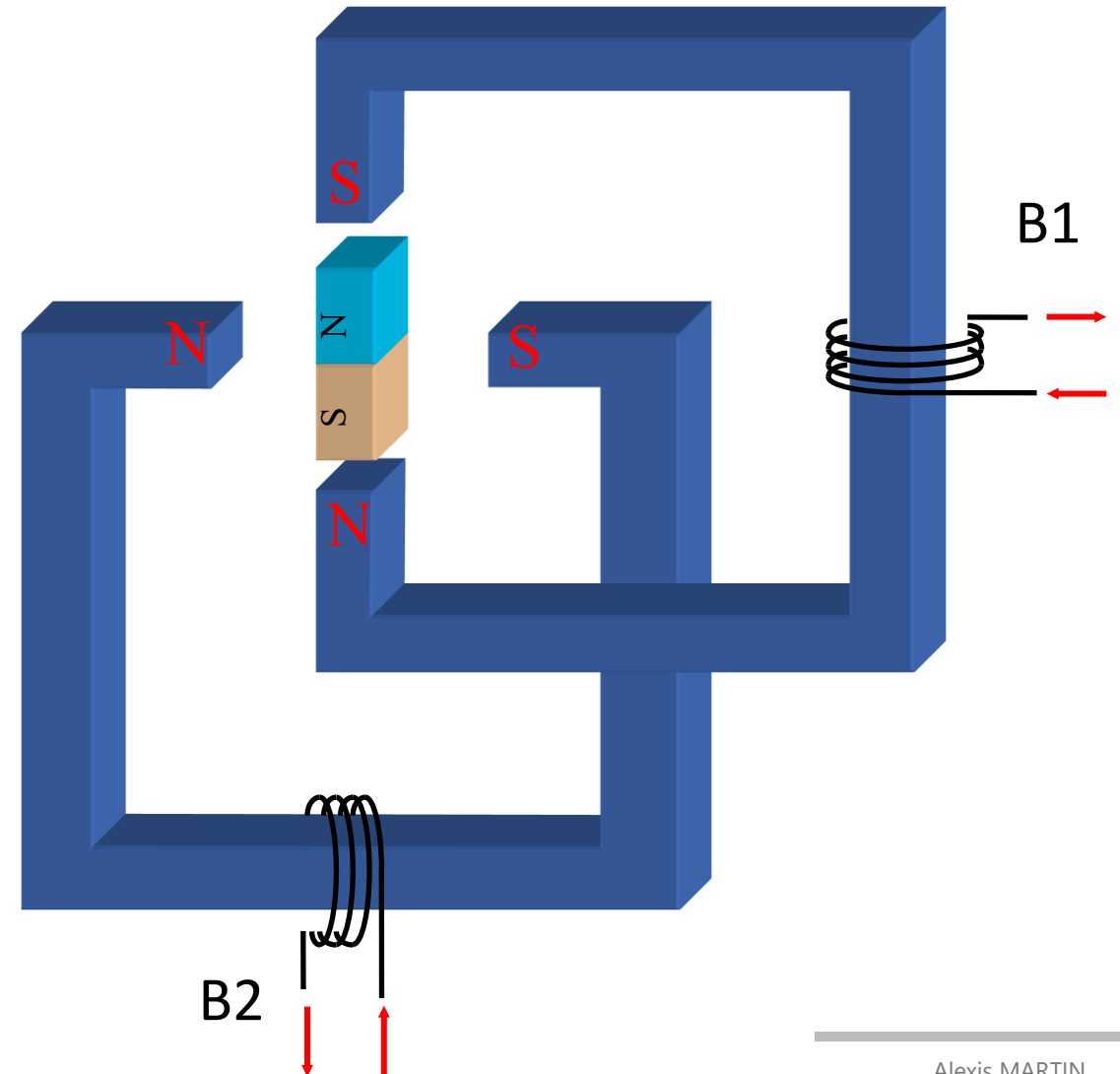
Energy storage
based chopper

4 quadrants chopper: Stepper motor control

Operation principle: half step

For a clockwise rotation:

B1	B2
I+	0
I+	I+
0	I+
I-	I+
I-	0
I-	I-
0	I-
I+	I-



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2 quadrants chopper

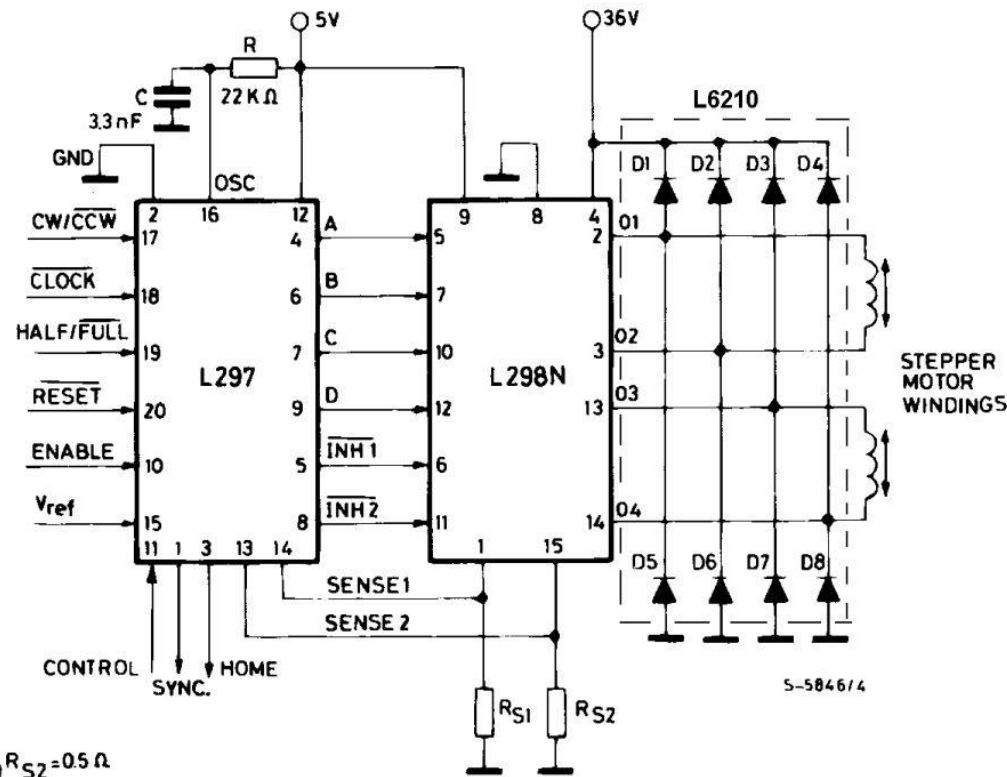
4 quadrants chopper

Inverters

Closed loop
chopper control

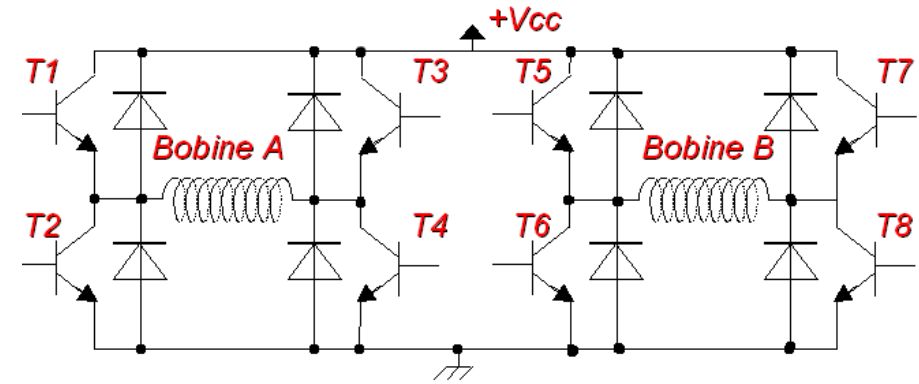
Energy storage
based chopper

4 quadrants chopper: Stepper motor control



$R_{S1}R_{S2} = 0.5 \Omega$

D1 to D8 = 2A FAST DIODES



Introduction

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Energy storage
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Inverter

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Direct conversion

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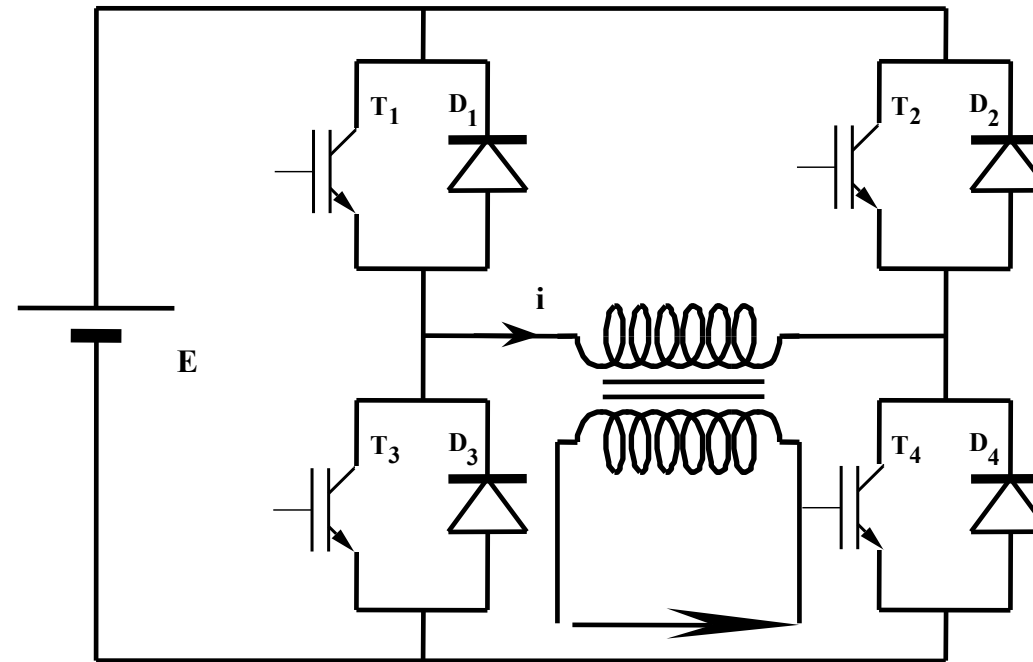
Single phase inverter

Three phase inverter
Three phase motor
control

Closed loop
chopper control

Energy storage
based chopper

Single phase inverter



$$\alpha = \frac{1}{2} + \delta_{\alpha} \sin(\omega t)$$

Introduction

Direct conversion

Reversible
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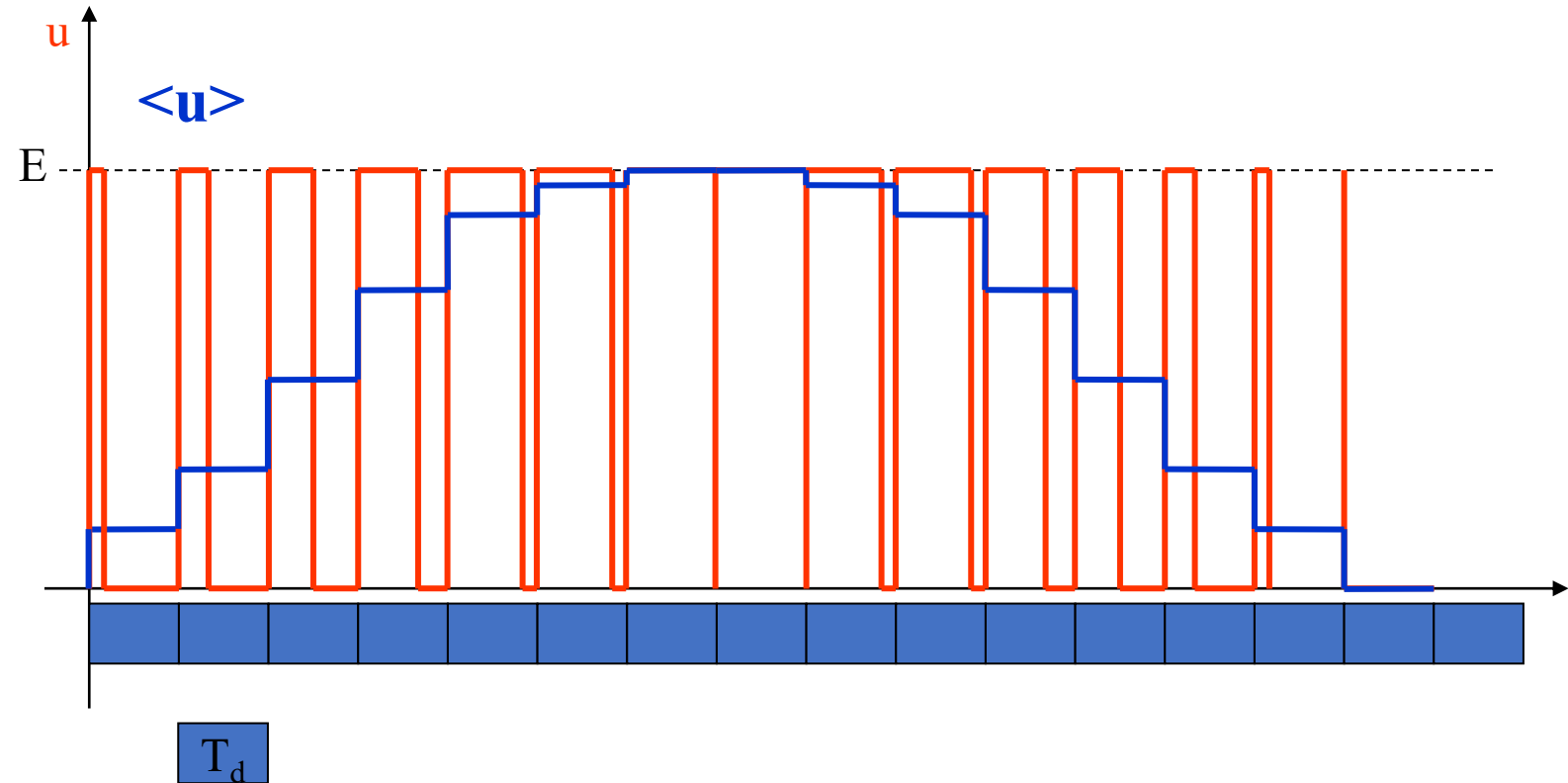
Single phase inverter

Three phase inverter
Three phase motor
control

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chopper control

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Single phase inverter: PWM control



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Reversible
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Single phase inverter

Three phase inverter

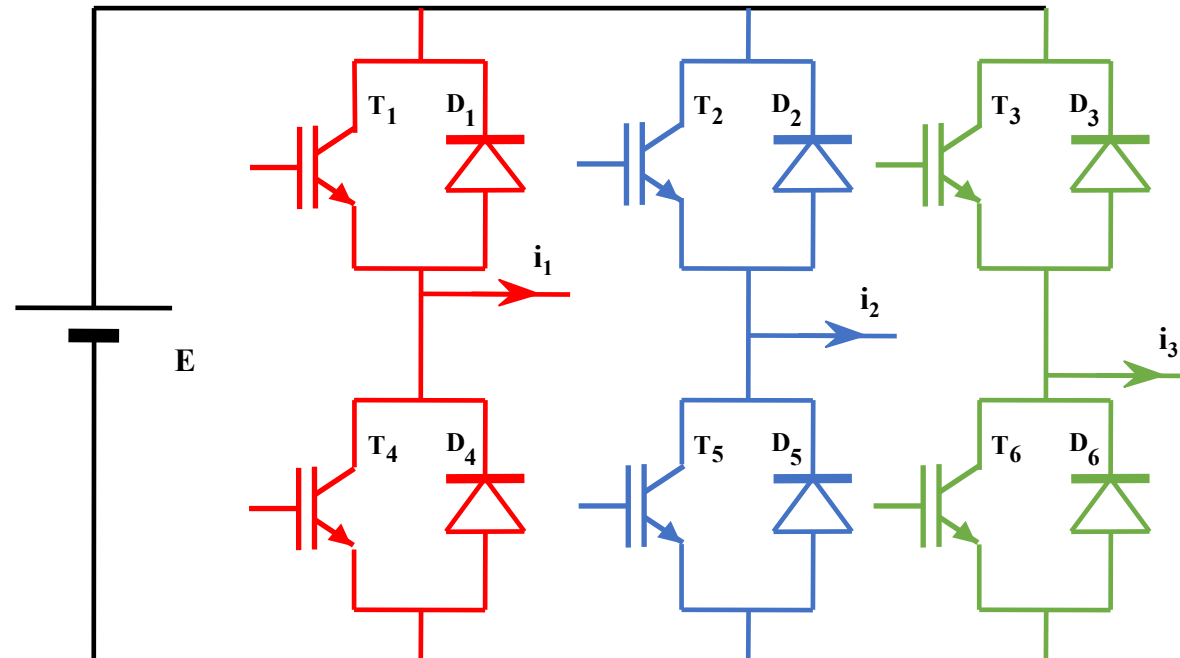
Three phase motor
control

Closed loop
chopper control

Energy storage
based chopper

Three phase inverter

$$\alpha_2 = \frac{1}{2} + \delta_\alpha \sin(\omega t - \frac{2\pi}{3})$$



$$\alpha_1 = \frac{1}{2} + \delta_\alpha \sin(\omega t)$$

$$\alpha_3 = \frac{1}{2} + \delta_\alpha \sin(\omega t + \frac{2\pi}{3})$$

Introduction

Direct conversion

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Single phase inverter

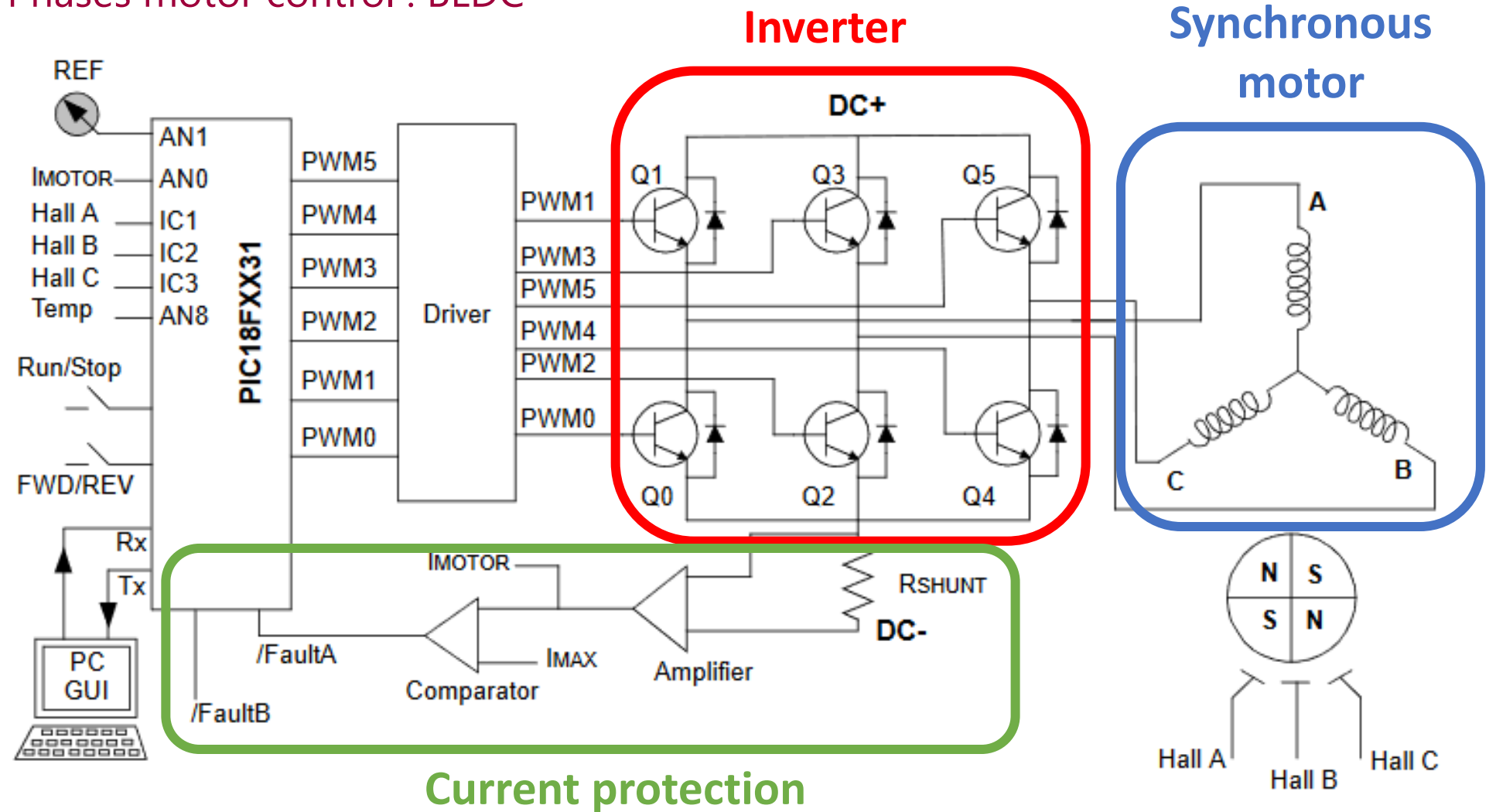
Three phase inverter

Three phase motor
control

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chopper control

Energy storage
based chopper

3 Phases motor control : BLDC



AN899 (Microchip) : Brushless DC Motor Control Using PIC18FXX31 MCUs

Introduction

Direct conversion

Reversible choppers

Inverters

Single phase inverter

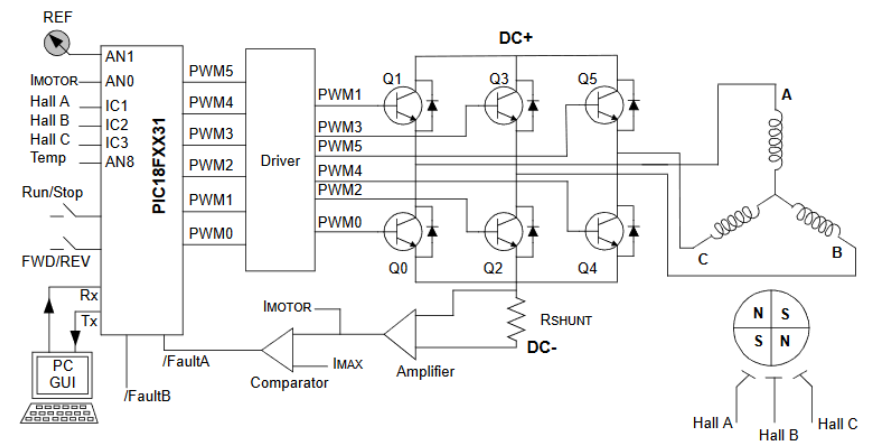
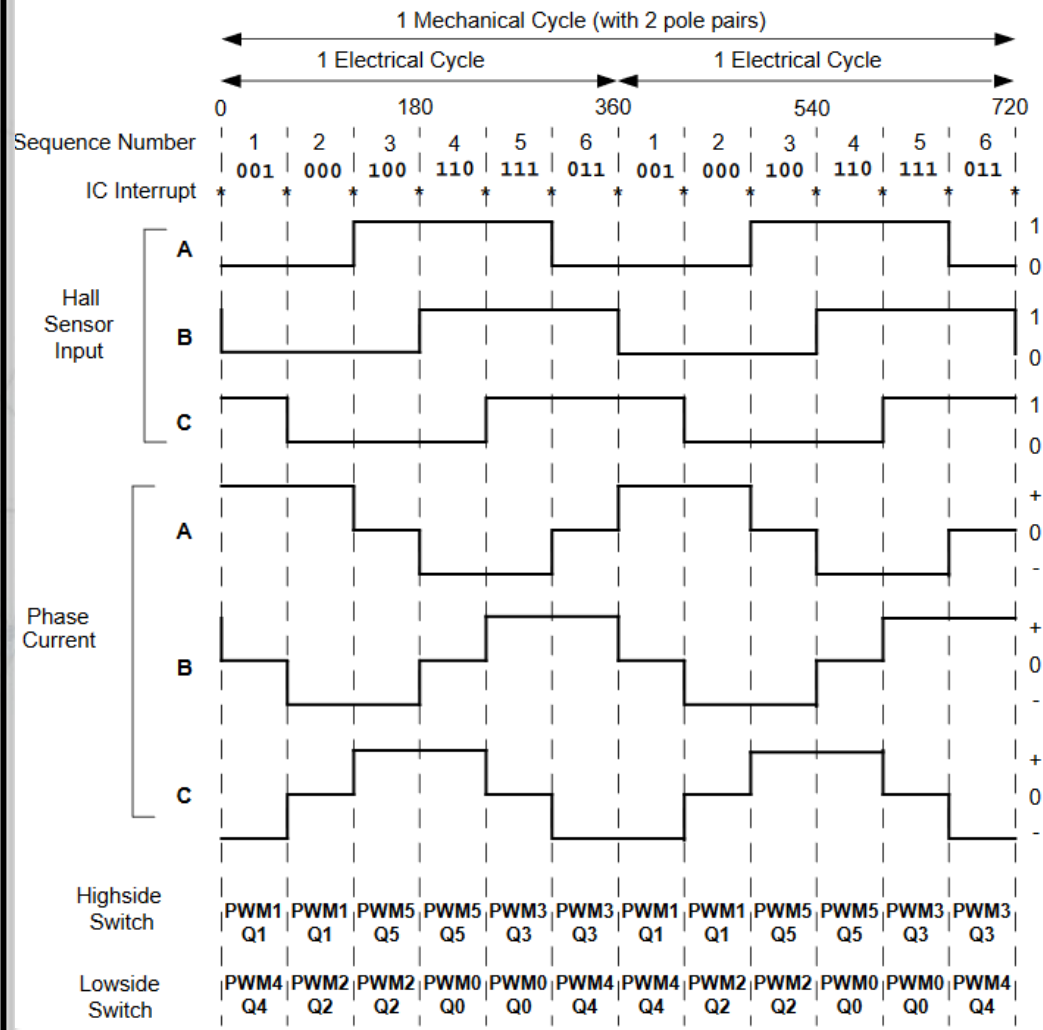
Three phase inverter

Three phase motor control

Closed loop chopper control

Energy storage based chopper

3 Phases motor control : BLDC, 6 Step



Sequence Number	Hall Sensor Input			Active PWMs		Phase Current		
	A	B	C			A	B	C
1	0	0	1	PWM1(Q1)	PWM4(Q4)	DC+	Off	DC-
2	0	0	0	PWM1(Q1)	PWM2(Q2)	DC+	DC-	Off
3	1	0	0	PWM5(Q5)	PWM2(Q2)	Off	DC-	DC+
4	1	1	0	PWM5(Q5)	PWM0(Q0)	DC-	Off	DC+
5	1	1	1	PWM3(Q3)	PWM0(Q0)	DC-	DC+	Off
6	0	1	1	PWM3(Q3)	PWM4(Q4)	Off	DC+	DC-

AN899 (Microchip) : Brushless DC Motor Control Using PIC18FXX31 MCUs

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**Closed loop
chopper control**

Energy storage
based chopper

Closed loop chopper control

Introduction

Direct conversion

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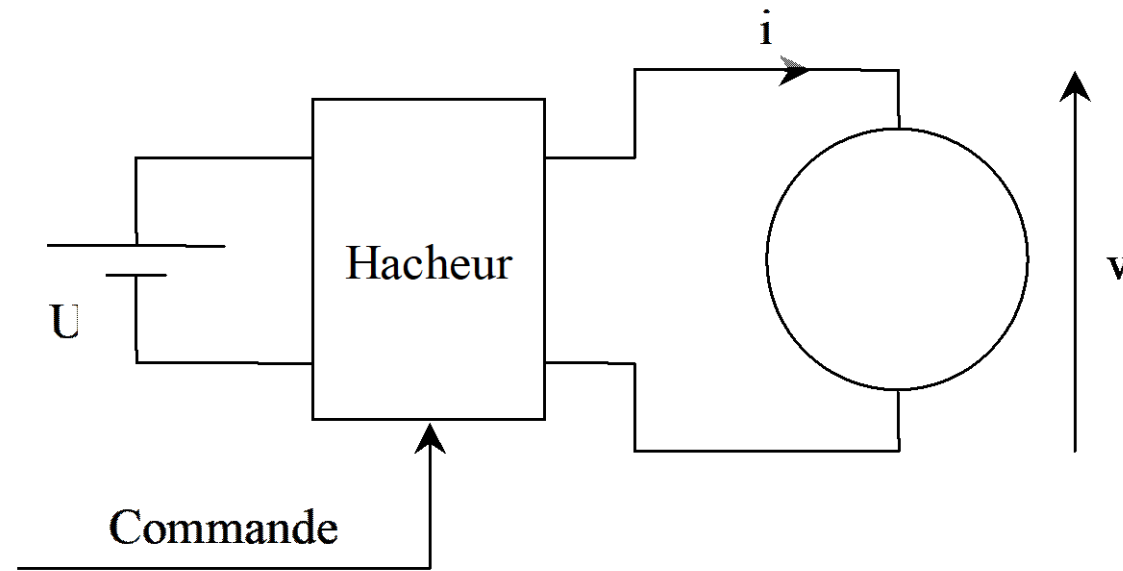
Inverters

**Closed loop
chopper control
Models**

Current control
Speed control

Energy storage
based chopper

Closed loop chopper control



Possible control : speed / torque

Voltage / current

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Direct conversion

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choppers

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Current control
Speed control

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based chopper

DC motor model

Electrical equation : $u = e + r.i + L.\frac{di}{dt}$

Mechanical equation : $J\frac{d\Omega}{dt} + f\Omega = \gamma_m - \gamma_r$

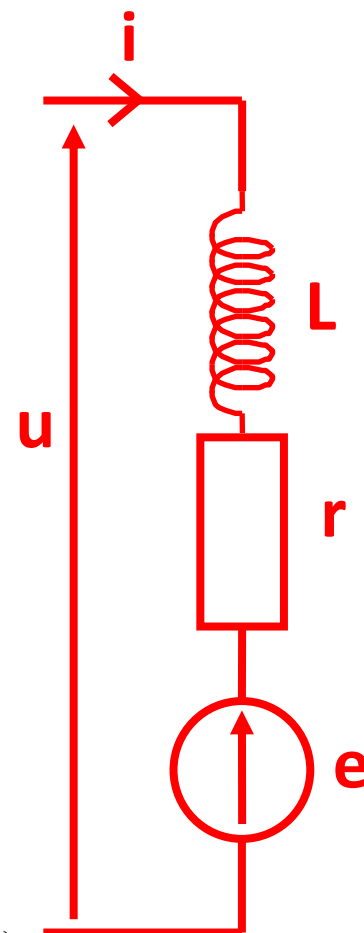
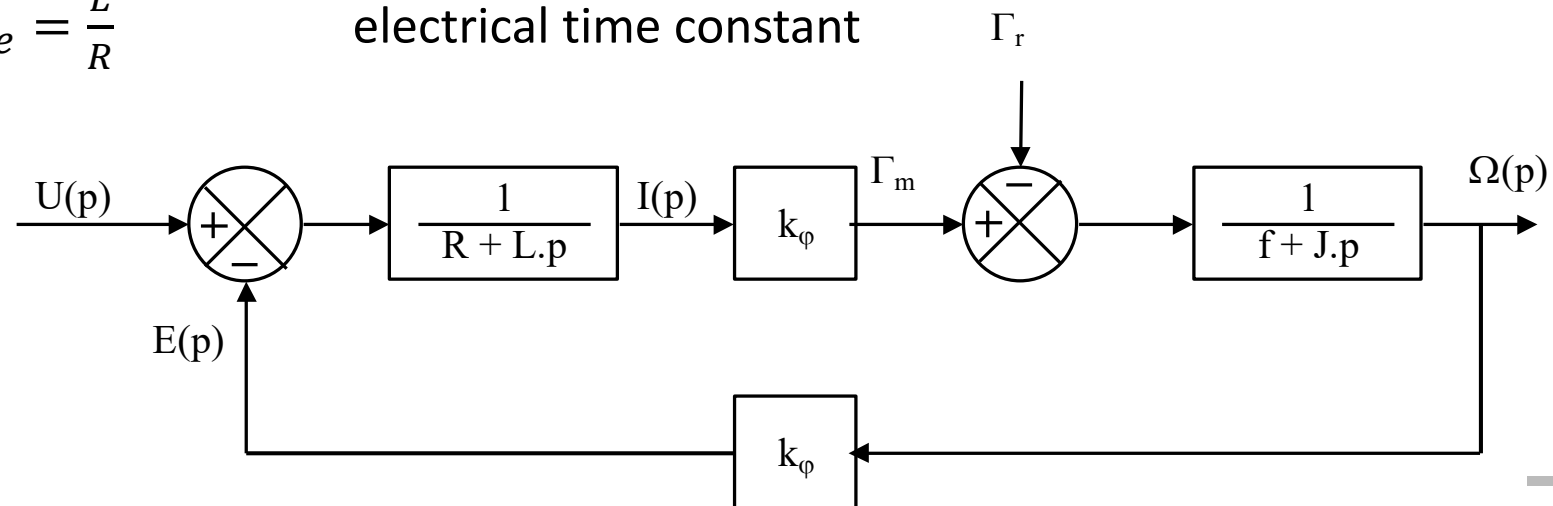
Coupling equations : $e = k_\phi.\Omega$ $\gamma_m = k_\phi.i$

$$\tau_{em} = \frac{J.R}{k_\phi^2 + R.f}$$

electromechanical time constant

$$\tau_e = \frac{L}{R}$$

electrical time constant



Introduction

Direct conversion

Reversible
choppers

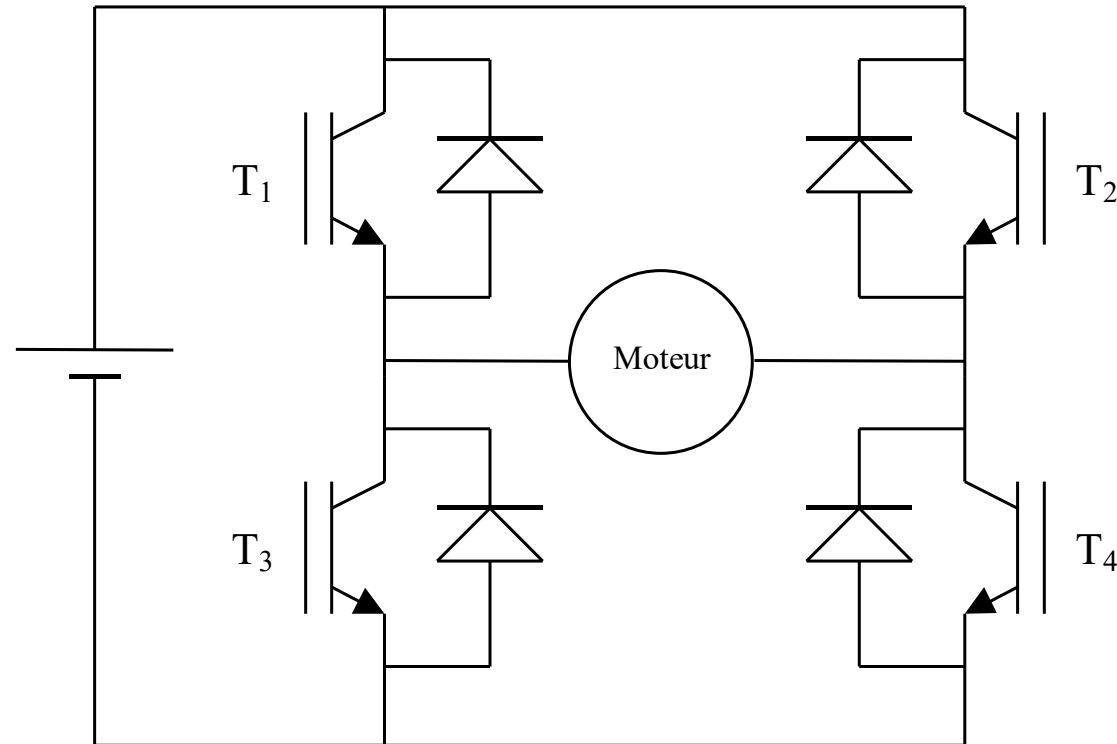
Inverters

**Closed loop
chopper control
Models**

Current control
Speed control

Energy storage
based chopper

Reversible structure



Introduction

Direct conversion

Reversible
choppers

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**Closed loop
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Speed control

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Chopper model

$$G_h(p) = \frac{U(p)}{\alpha(p)} = 2 \cdot V_{DC} \cdot e^{-\frac{T_d}{2} \cdot p}$$

(Average delay $T_d/2$)

$$G_h(p) \approx \frac{G_0}{1 + \tau_h \cdot p} \quad \text{with} \quad \tau_h = \frac{T_d}{2} \quad \text{and} \quad G_0 = 2 \cdot V_{DC}$$

(linearization)

Introduction

Direct conversion

Reversible
choppers

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**Closed loop
chopper control**

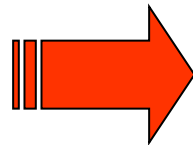
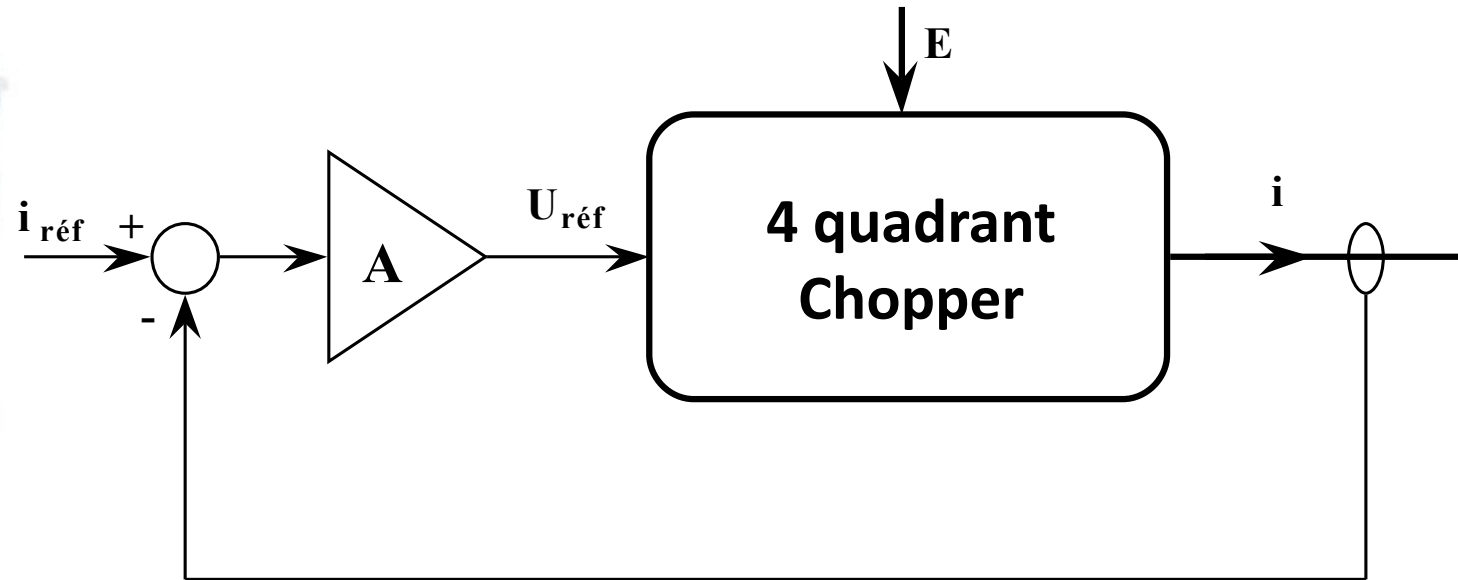
Models

Current control

Speed control

Energy storage
based chopper

Current control



Torque control of a motor

Introduction

Direct conversion

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Inverters

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chopper control

Models

Current control

Speed control

Energy storage
based chopper

Current loop

Armature current regulation

- *Current regulation*
- *Torque control ($G = k_{\varphi} \cdot I$)*
- *Better transient response*
- *Need a high performance sensor*
- *Simple Regulation (P.I.)*

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Direct conversion

Reversible
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Models

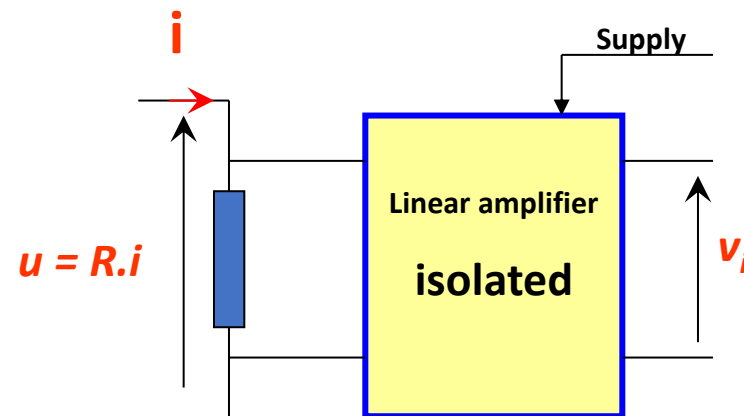
Current control

Speed control

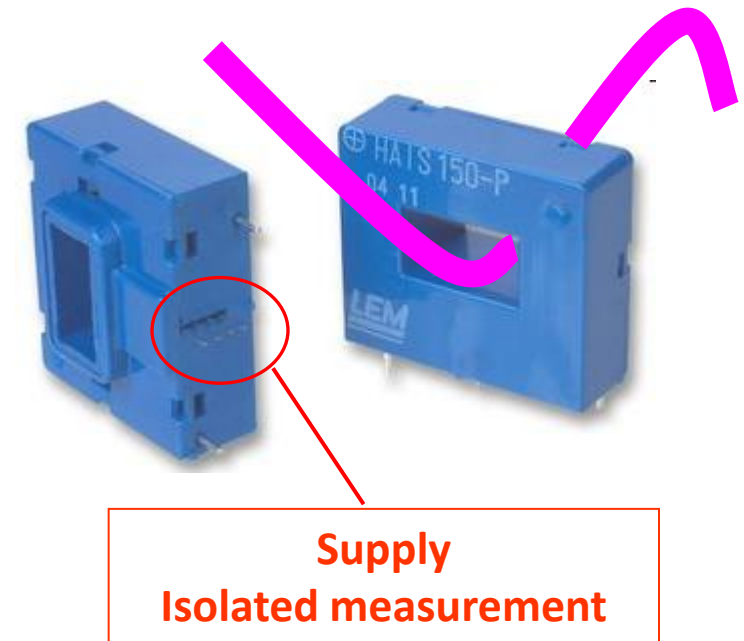
Energy storage
based chopper

Current loop: Current sensor

Resistor = shunt



Hall effect sensor



Introduction

Direct conversion

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**Closed loop
chopper control**

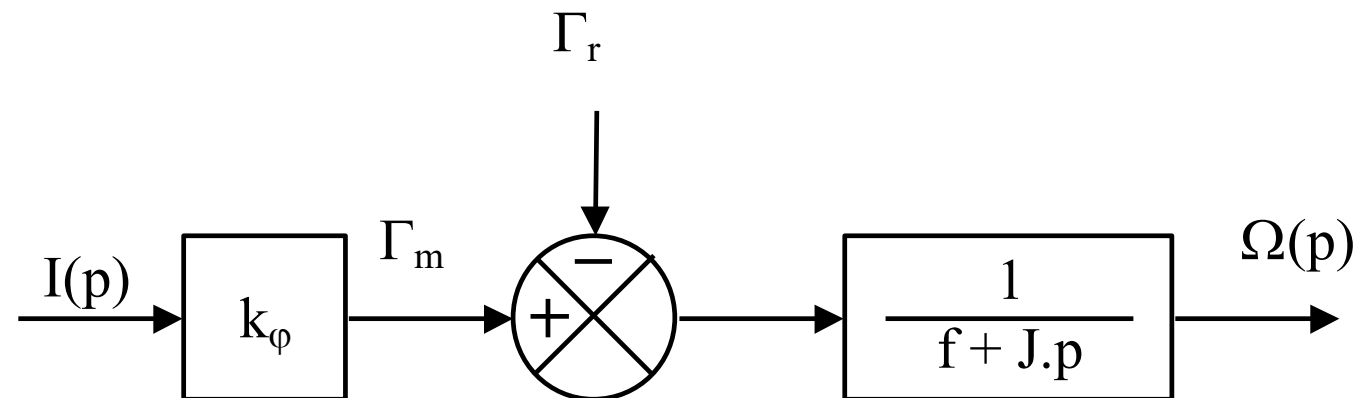
Models

Current control

Speed control

Energy storage
based chopper

DC motor: Current control



$$G_i(p) = \frac{\Omega(p)}{I(p)} = \frac{k_\phi}{f + J.p}$$

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Direct conversion

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choppers

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**Closed loop
chopper control**

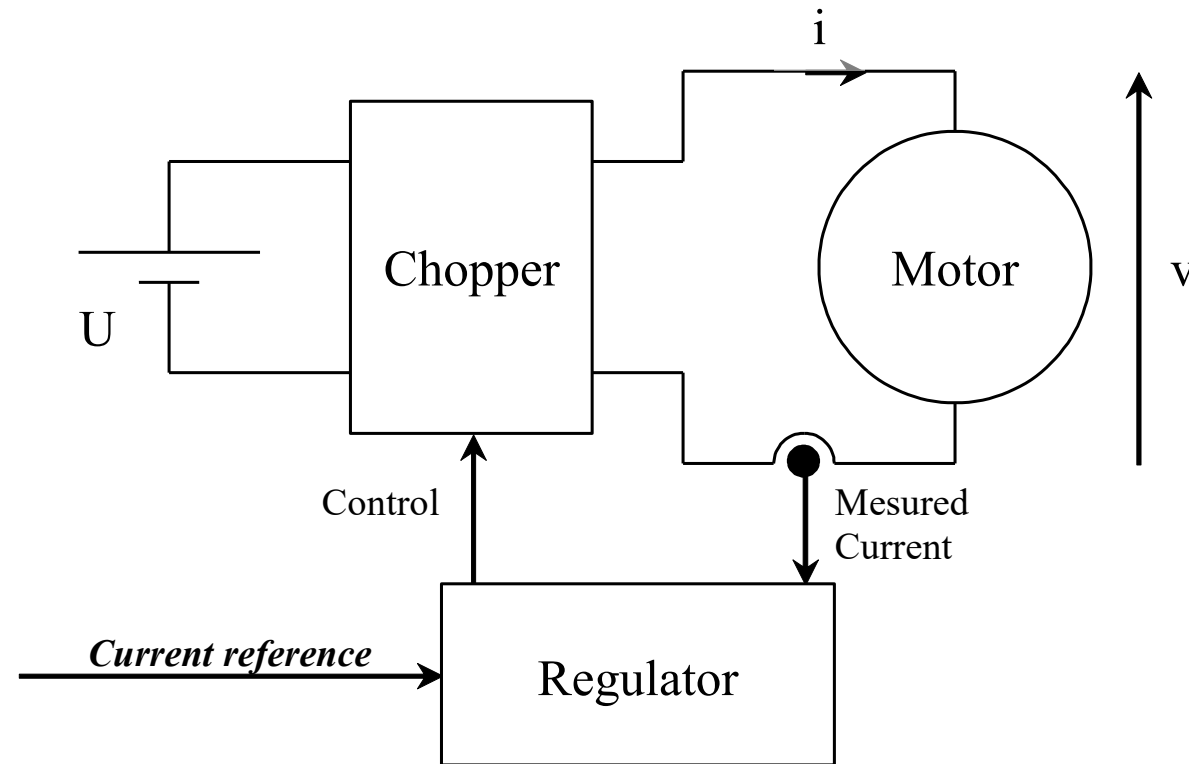
Models

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Current loop



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Models

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Speed control

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Control criteria

Accuracy : no static error

Speed : time response enhancement

Unconditional stability (high margin)

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Direct conversion

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Closed loop
chopper control

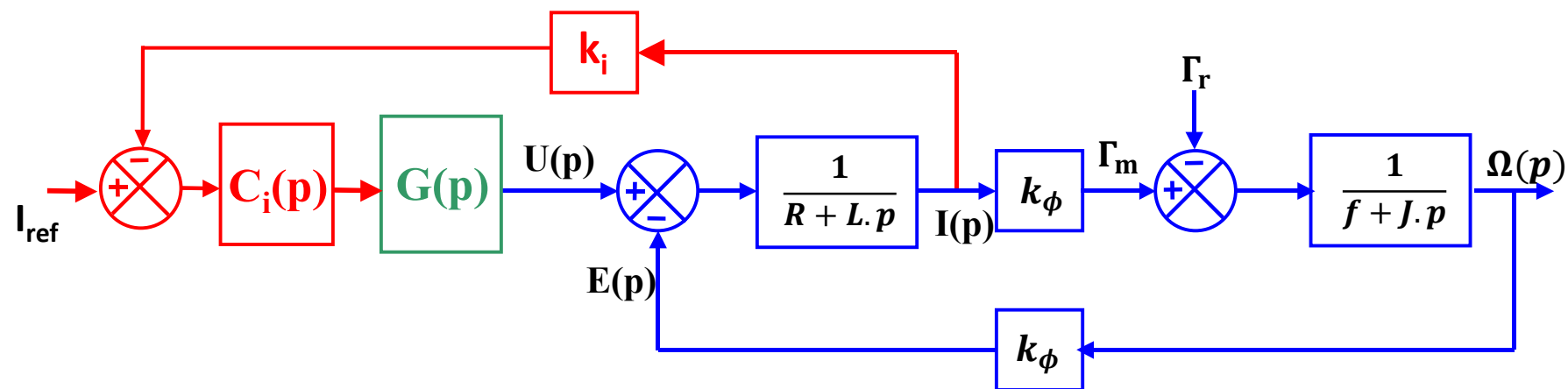
Models

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Current Control



Controller

Chopper

DC motor

= > *Controller choice: $C_i(p)$*

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Direct conversion

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**Closed loop
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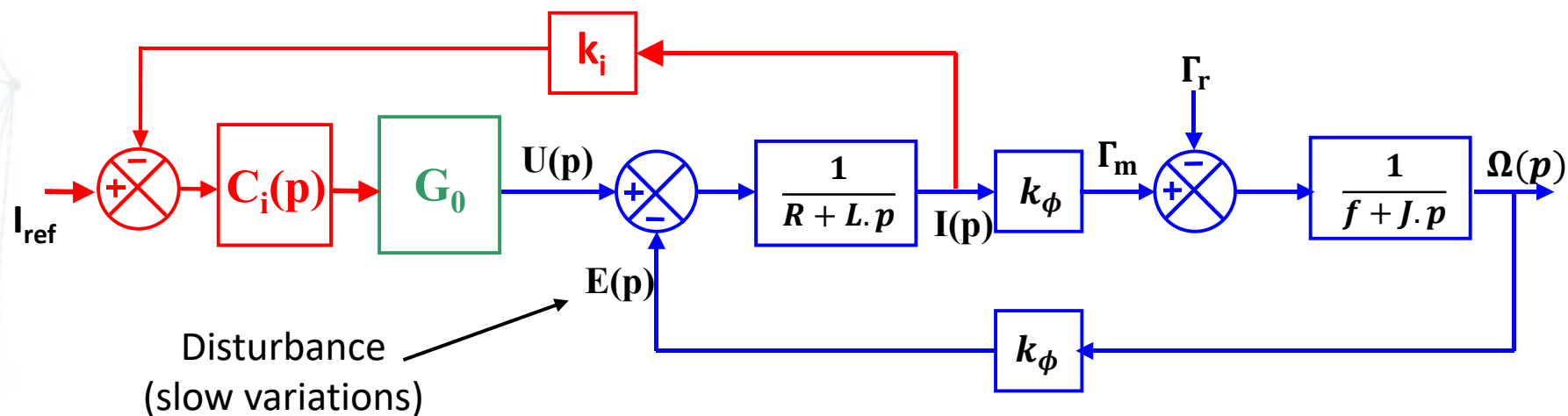
Models

Current control

Speed control

Energy storage
based chopper

P.I. control: simplified chopper



$$C_i(p) = k_i \cdot \frac{1 + \tau_i p}{\tau_i p}$$

$$H_{NCOL}(p) = \frac{G_0 / R}{1 + \tau_e p}$$

$$H_{OL}(p) = \frac{k_i \cdot (1 + \tau_i p) G_0 / R}{\tau_i p (1 + \tau_e p)}$$

$$\Rightarrow \tau_i = \tau_e$$

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based chopper

P.I. control: simplified chopper

$$H_{CL}(p) = \frac{k_i \cdot G_0 / R}{k_i \cdot G_0 / R + \tau_e p} = \frac{1}{1 + \tau_{BF} p}$$

$$\tau_{BF} = \frac{\tau_e}{\frac{k_i G_0}{R}}$$

1st order without error, faster

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Direct conversion

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**Closed loop
chopper control**

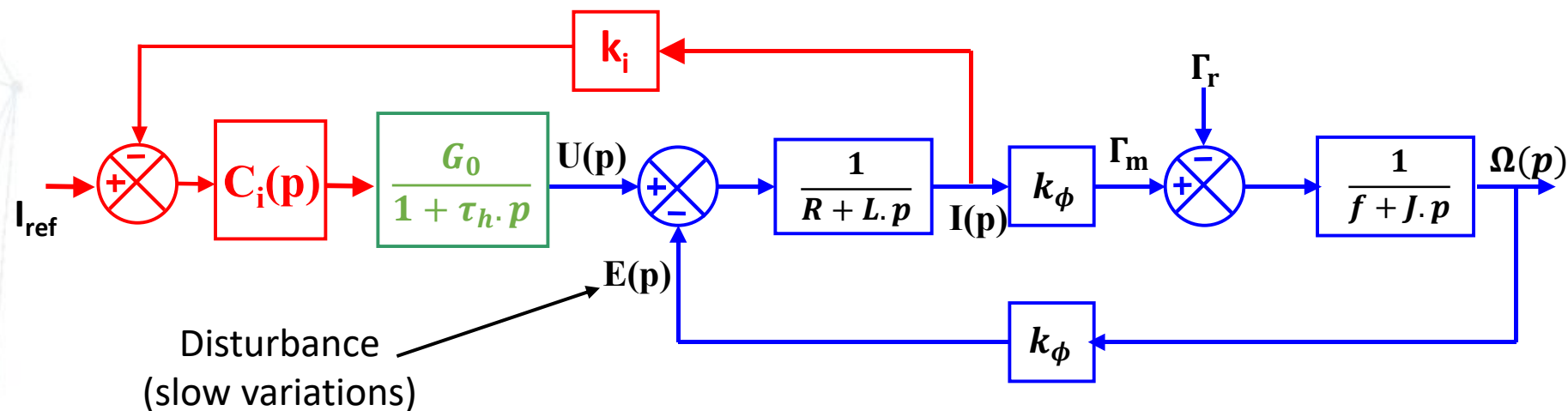
Models

Current control

Speed control

Energy storage
based chopper

P.I. control: first order chopper



$$C_i(p) = k_i \cdot \frac{1 + \tau_i p}{\tau_i p}$$

$$H_{OL}(p) = \frac{G_0 / R}{(1 + \tau_h p) \cdot (1 + \tau_e p)}$$

$$H_{OL}(p) = \frac{k_i \cdot (1 + \tau_i p) G_0 / R}{\tau_i p (1 + \tau_h p) \cdot (1 + \tau_e p)}$$

$$\Rightarrow \tau_i = \max(\tau_h, \tau_e)$$

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**Closed loop
chopper control**

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based chopper

Closed loop

$$H_{OL}(p) = \frac{k_i G_0 / R}{\tau_{he} p (1 + \tau_{he})}$$

$$H_{CL}(p) = \frac{1}{1 + 2m \frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}}$$

Second ordre without error

Choose of m : damping $[0,7 .. 1]$

Choose of ω_0 : bandwidth

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Models

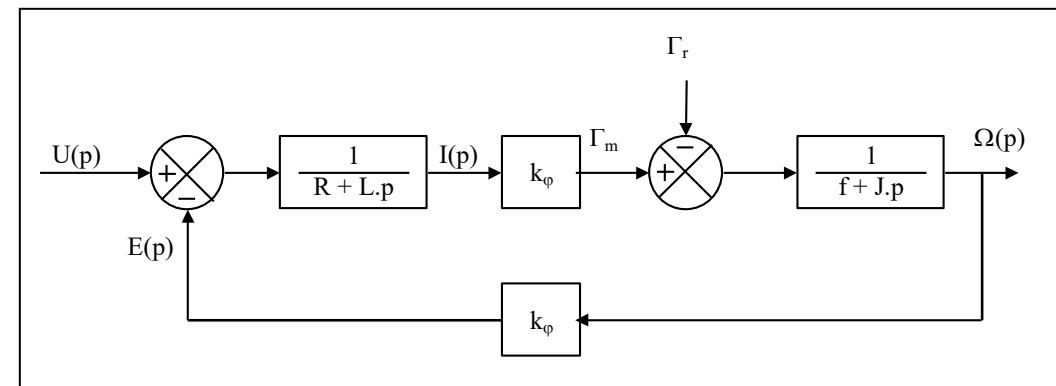
Current control

Speed control

Energy storage
based chopper

DC motor: Speed / voltage transfer function

$$G'_u(p) = \frac{\Omega(p)}{U(p)} = \frac{k_\phi}{k_\phi^2 + (R + L.p).(f + J.p)}$$



$$G_u(p) = \frac{\Omega(p)}{U(p)} = \frac{k_\phi}{k_\phi^2 + Rf} \cdot \frac{1}{(1 + \tau_e p).(1 + \tau_{em} p)}$$

$$\tau_{em} = \frac{J.R}{k_\phi^2 + R.f}$$

electromechanical time constant

$$\tau_e = \frac{L}{R}$$

electrical time constant

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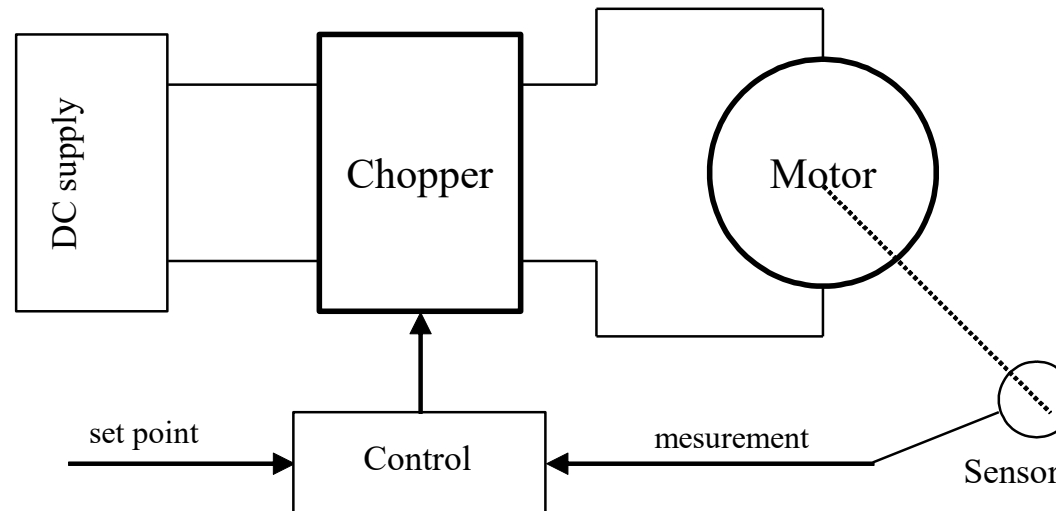
Models

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Speed control

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Speed loop



$$G_{hv}(p) = \frac{V_{\Omega mes}(p)}{V_{\Omega ref}(p)} = \frac{G_{0\Omega}}{1 + 2m \frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}}$$

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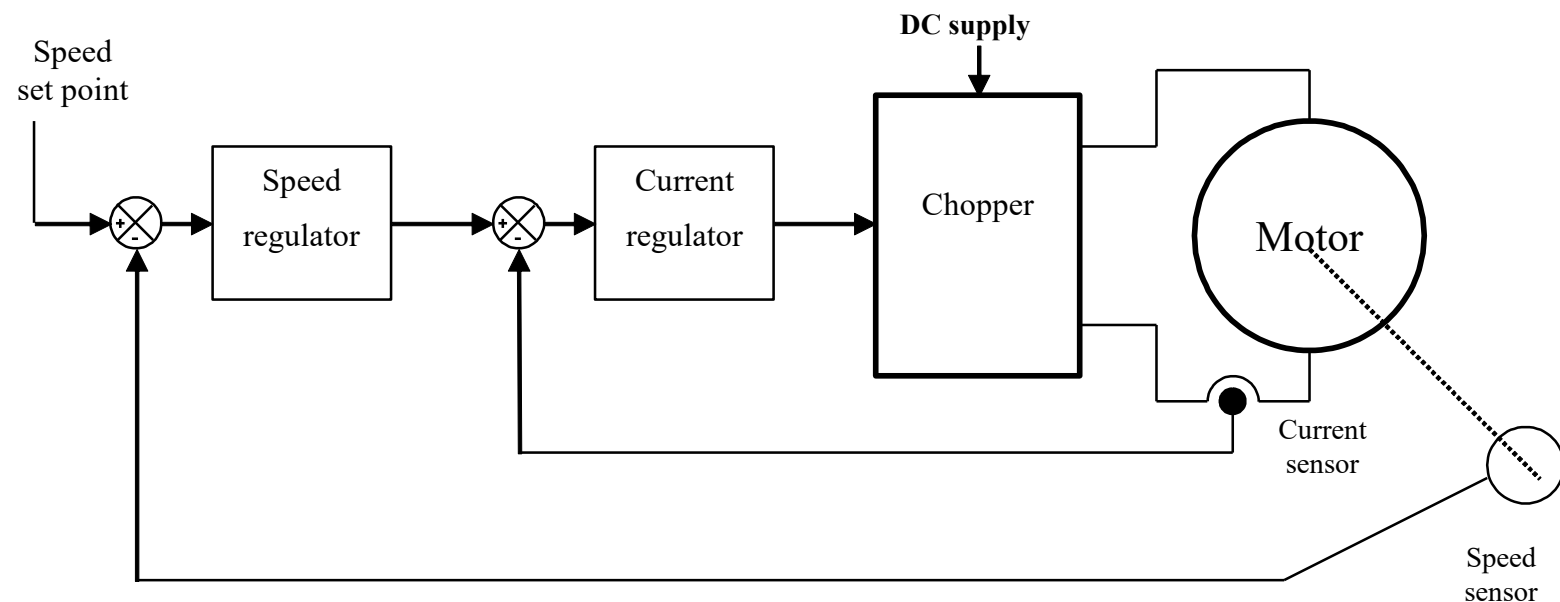
Models

Current control

Speed control

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Nested loop



Internal loop faster than external loop

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Indirect Choppers

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Direct conversion

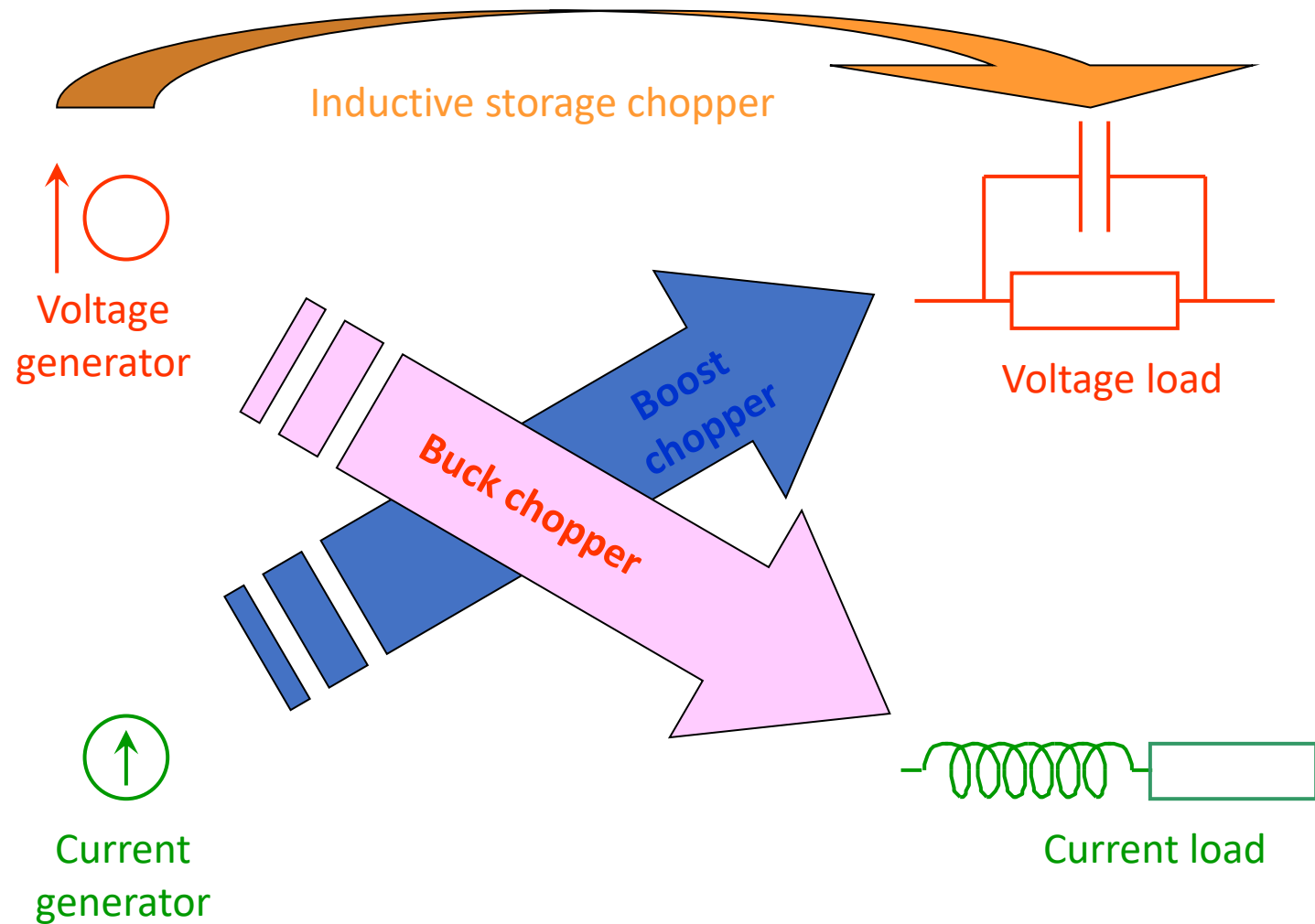
Reversible
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Indirect chopper: principle



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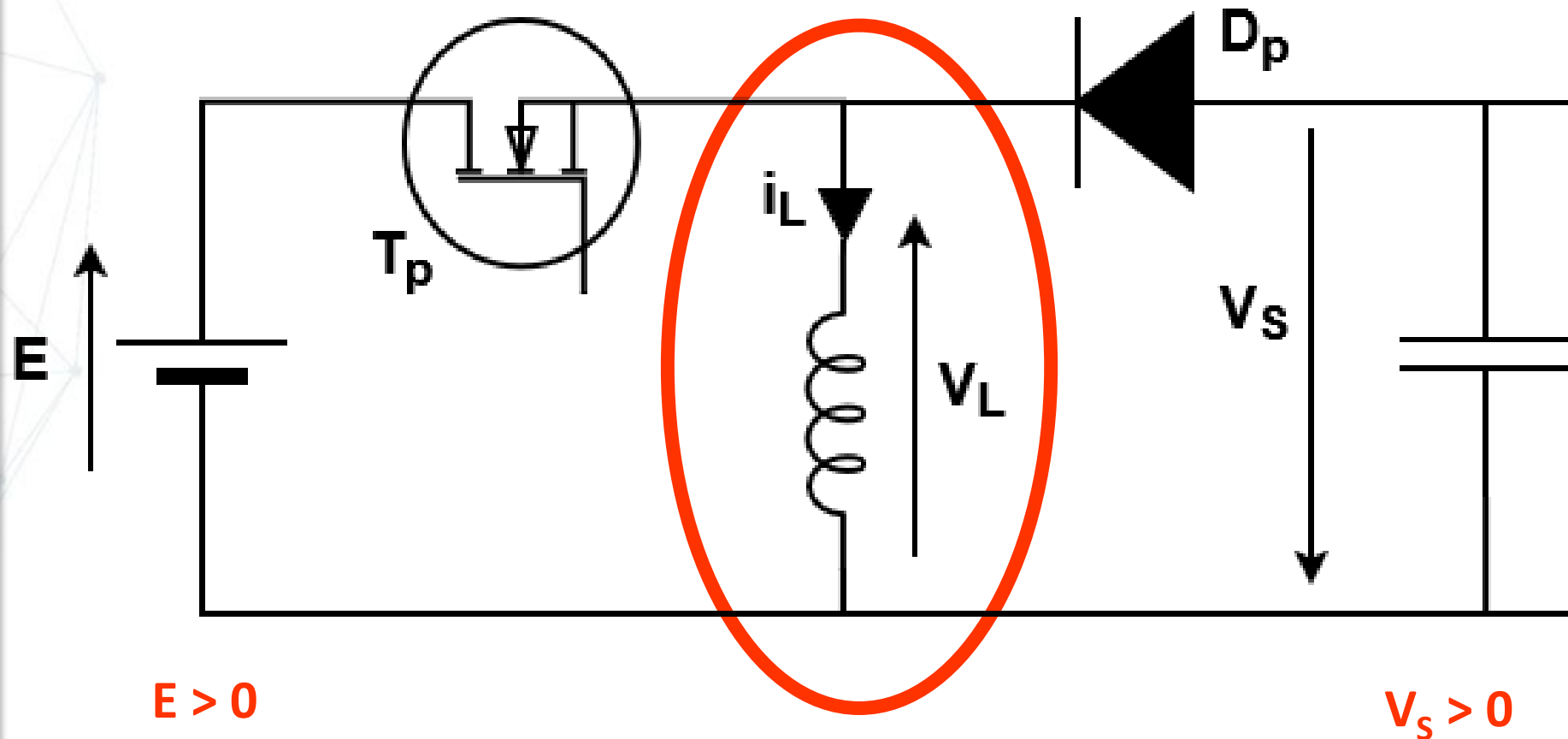
Energy storage
based chopper

Buck-Boost

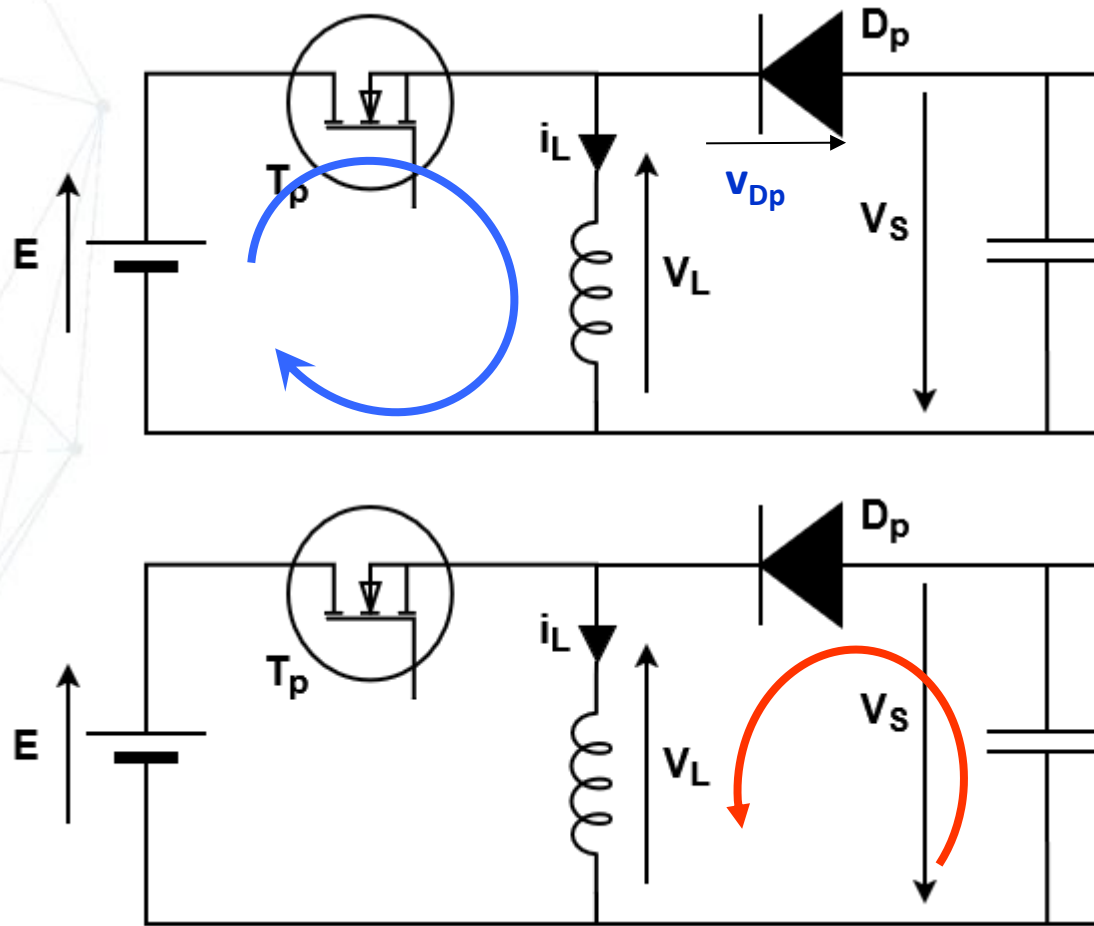
Cuk

SEPIC

Buck-Boost structure



Buck-boost: Continuous mode ($i_L > 0$)



$0 < t < \alpha T : T_p$ closed

$v_{Dp} = -E - V_S \Rightarrow D_p$ reverse

$v_L = E = L \cdot di_L/dt \Rightarrow i_L$ increasing > 0

$\alpha T < t < T : T_p$ opened

$i_L > 0 \Rightarrow D_p$ forward

$v_L = -V_S = L \cdot di_L/dt \Rightarrow i_L$ decreasing

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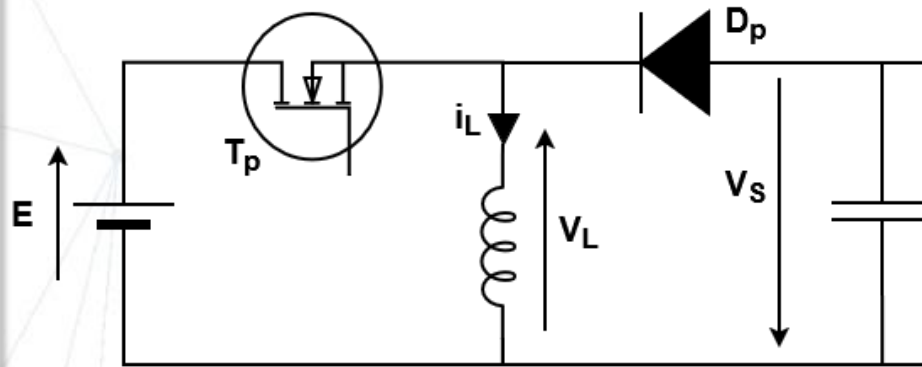
Energy storage
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Buck-Boost

Cuk

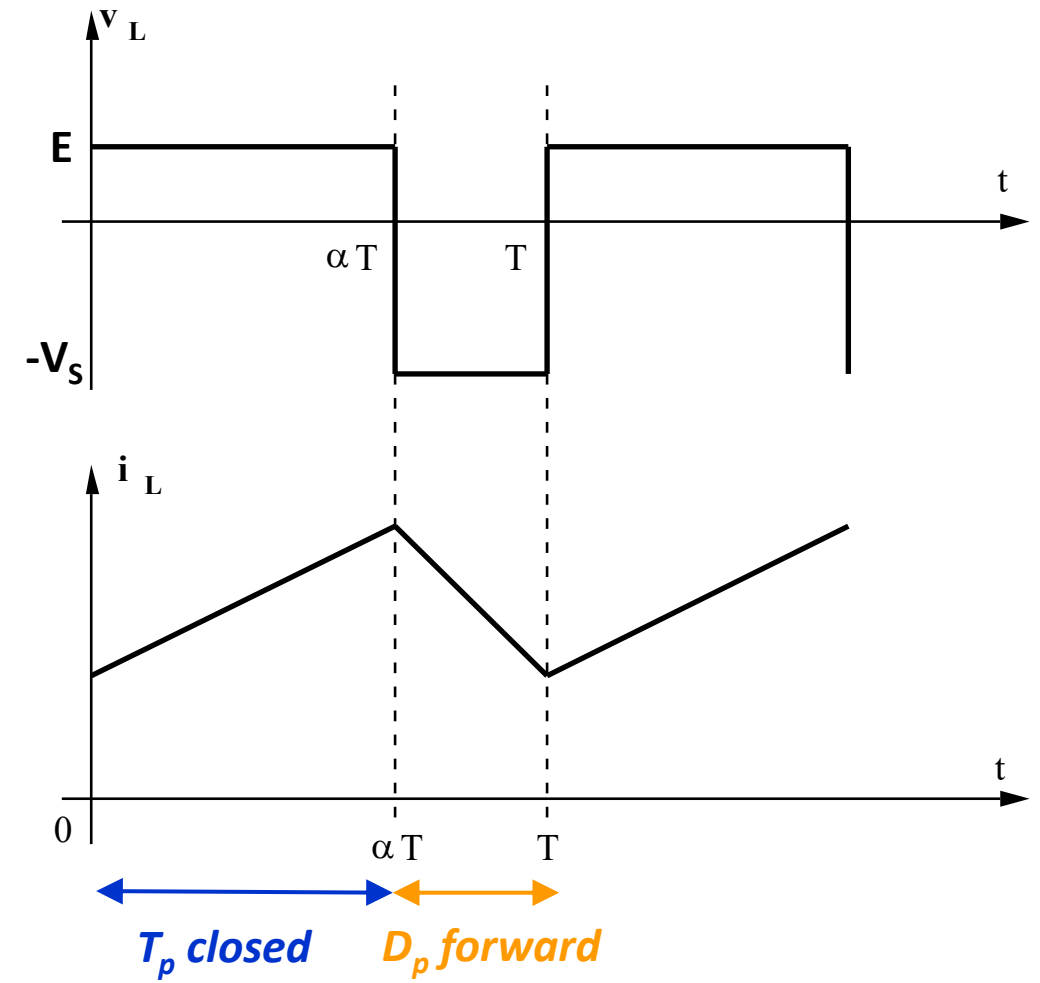
SEPIC

Buck-boost: Continuous mode (chronograms)



$$\langle v_L \rangle = 0$$

Inductive storage chopper



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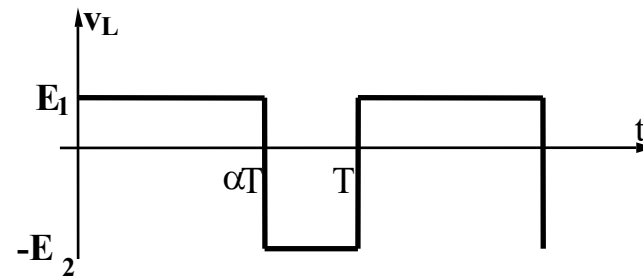
Energy storage
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Buck-Boost

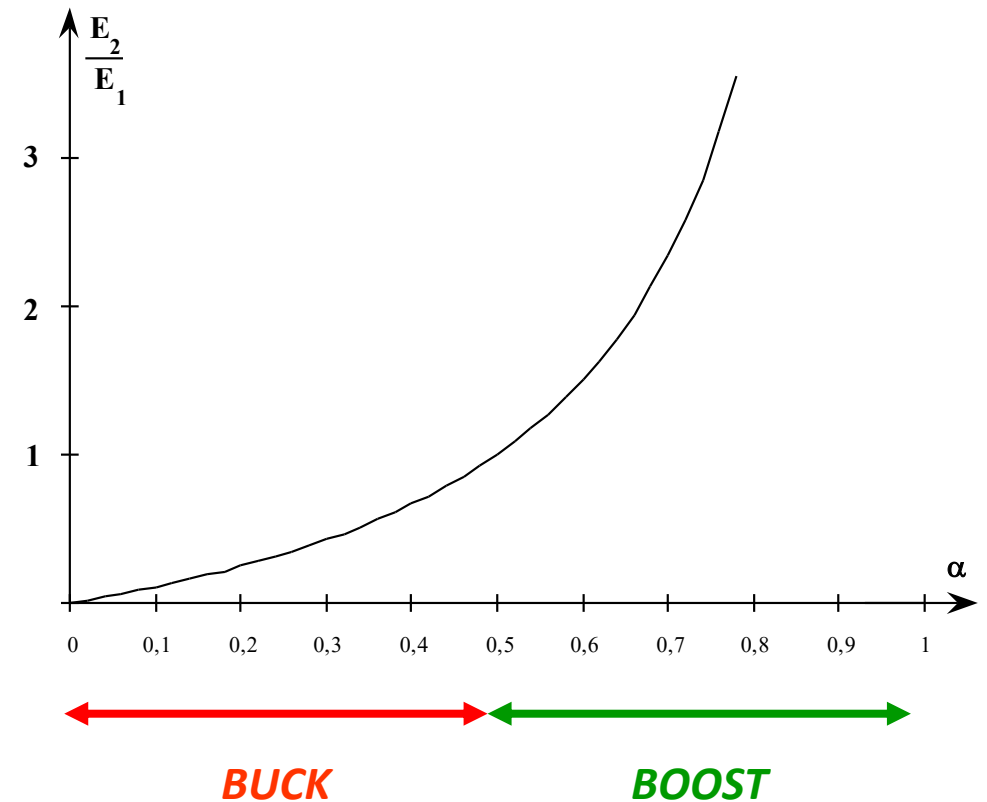
Cuk

SEPIC

Buck-boost: Continuous mode (Transfer characteristics)



$$\langle v_L \rangle = 0$$



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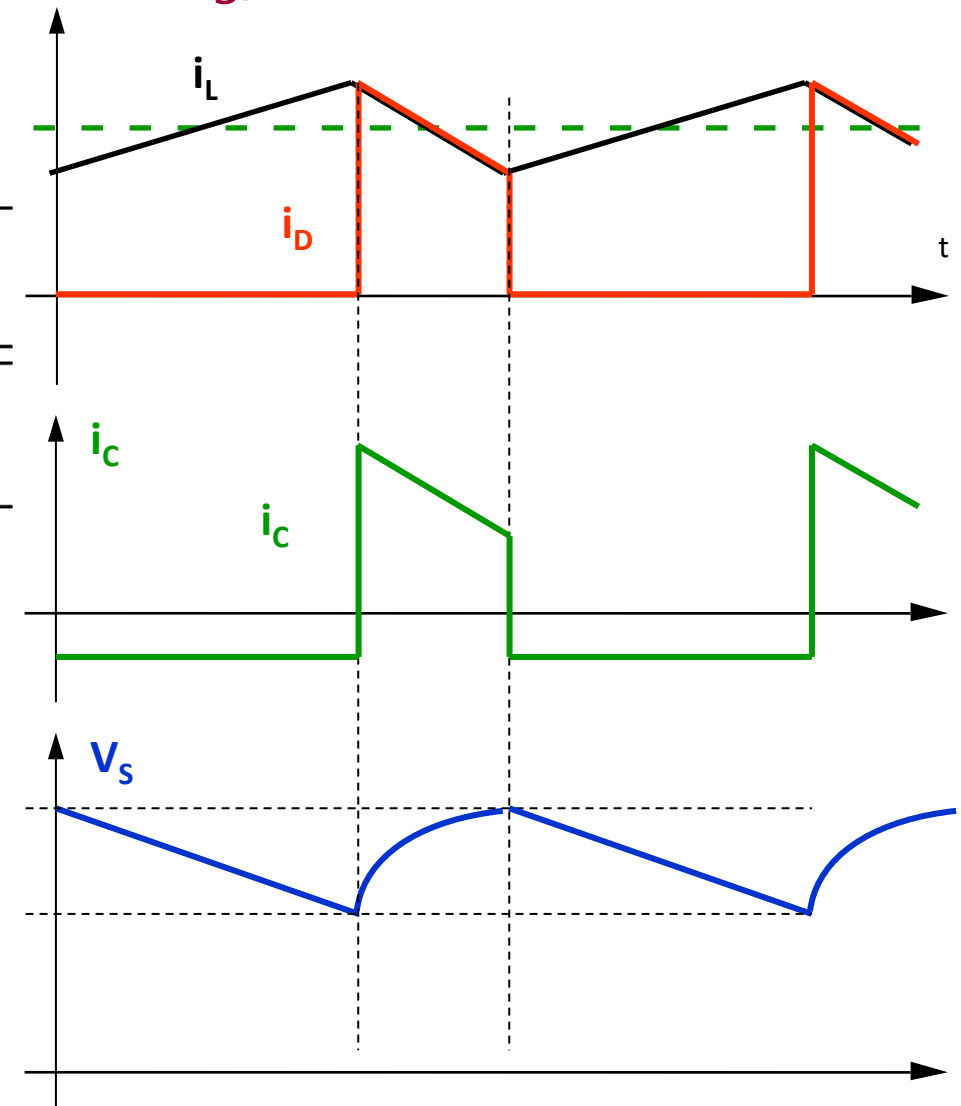
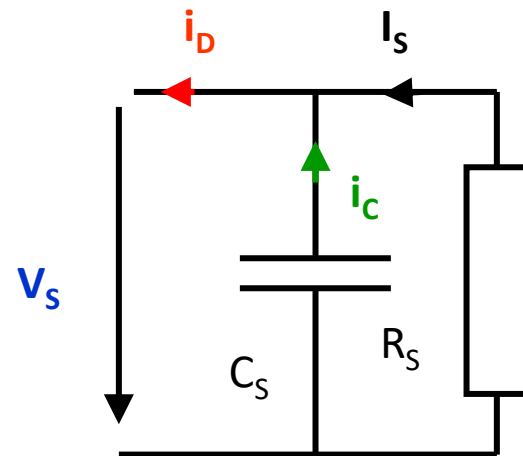
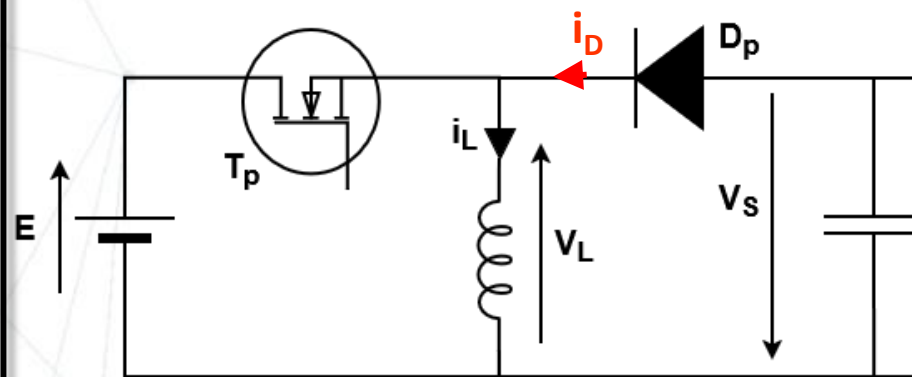
Energy storage
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Buck-Boost

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Buck-boost: Continuous mode (output filtering)



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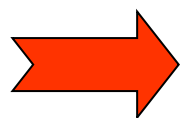
SEPIC

Buck-boost: Continuous mode (Output filtering)

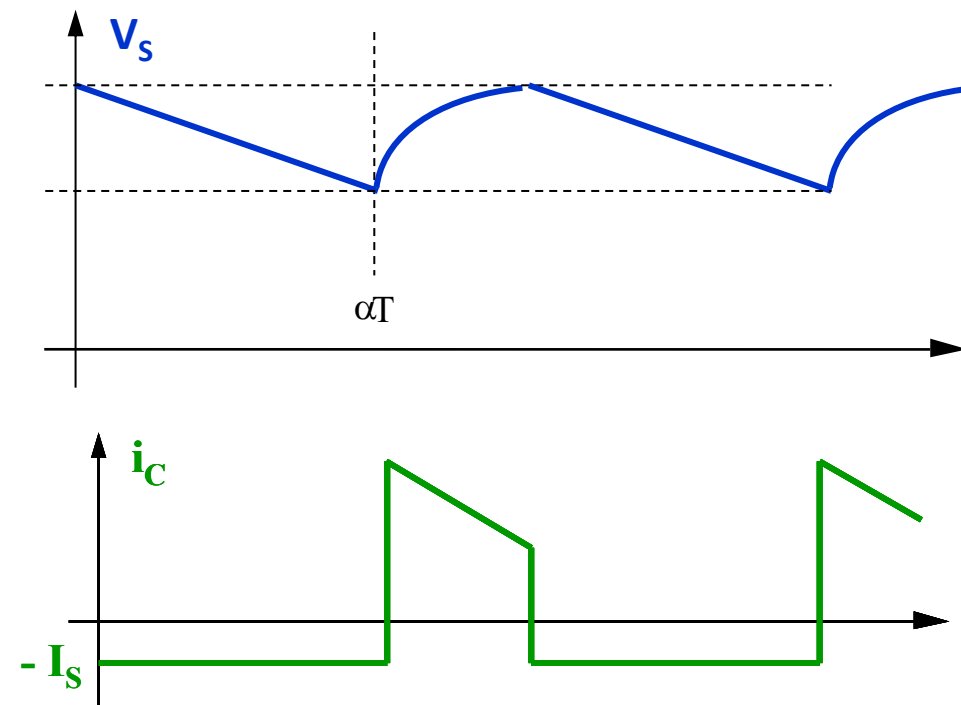
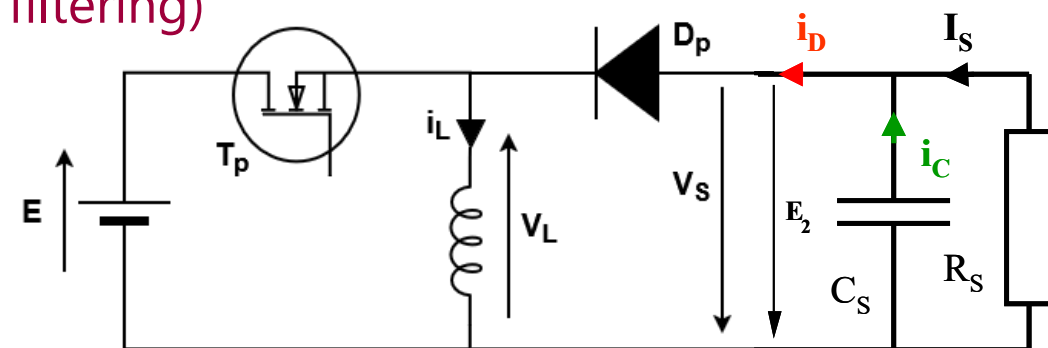
$$\Delta V_S = \frac{1}{C_S} \cdot \int_0^{\alpha T} i_C(t) \cdot dt = \frac{I_S \cdot \alpha T}{C_S}$$

$$I_S = \frac{V_S}{R_S} = \frac{\alpha \cdot E}{(1 - \alpha) R_S}$$

$$\Delta V_S = \frac{ET}{R_S C_S} \cdot \frac{\alpha^2}{1 - \alpha}$$



$$C_S = \frac{\alpha^2 \cdot E \cdot T}{(1 - \alpha) \cdot R_S \cdot \Delta V_S}$$



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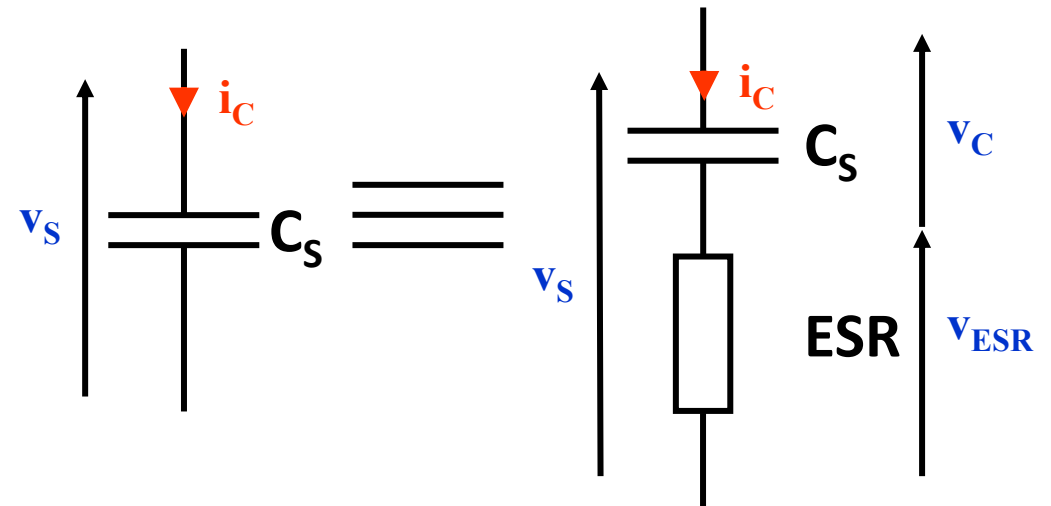
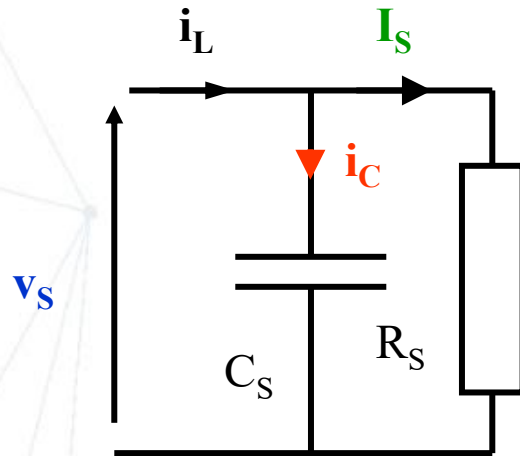
Energy storage
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Buck-Boost

Cuk

SEPIC

Buck-boost: Continuous mode (Output filtering)

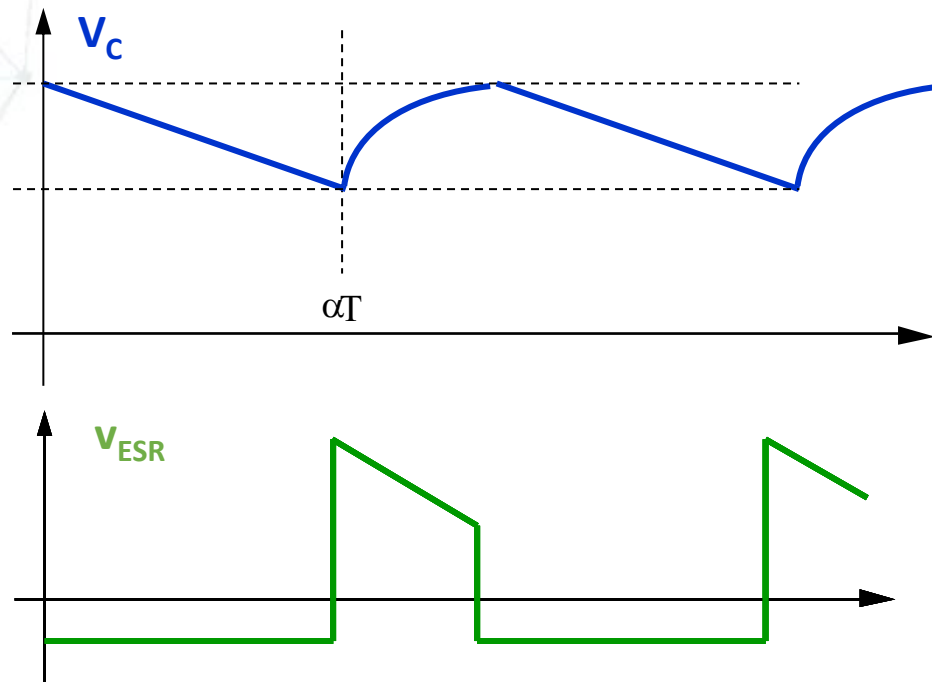


$$v_S = v_C + v_{ESR}$$

$$\Delta v_S < \Delta v_C + \Delta v_{ESR}$$

For high value of C_S ,
 $\Delta v_S \approx \Delta v_{ESR} = ESR \cdot \Delta i_L$

$$ESR < \frac{\Delta v_S}{\Delta i_L}$$



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Buck-Boost

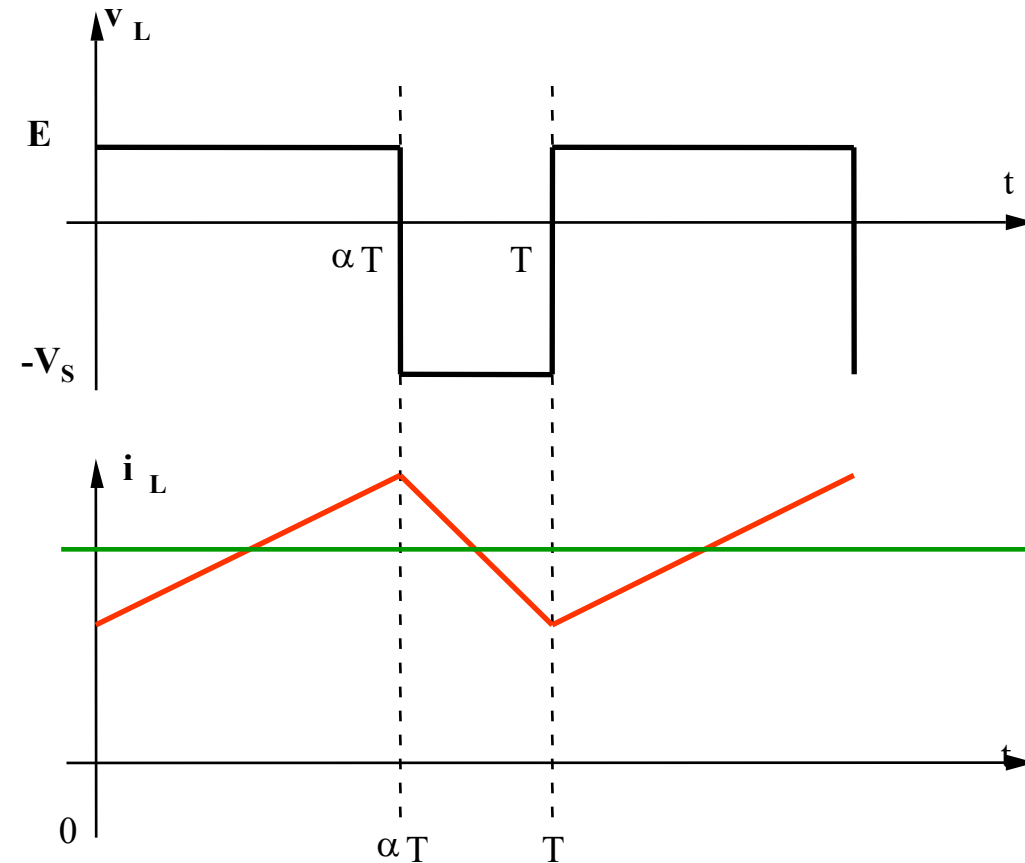
Cuk

SEPIC

Buck-boost: "decreasing" load



I_L



=> Boundary mode

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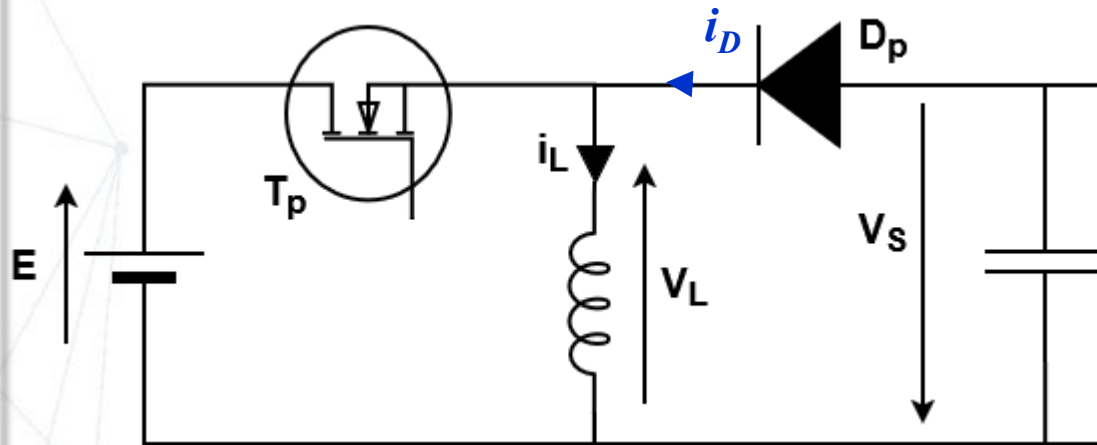
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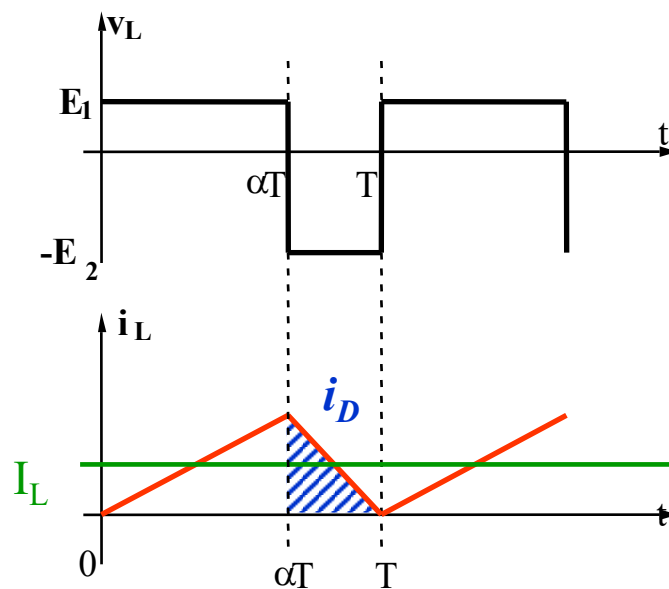
SEPIC

Buck-boost: Boundary mode



$$V_S = E \cdot \frac{\alpha}{1 - \alpha} \Rightarrow \alpha = \frac{V_S}{E + V_S}$$

$$I_{Lc} = \frac{E}{L} \frac{\alpha T}{2} = \frac{E \cdot V_S}{E + V_S} \cdot \frac{T}{2L} = \frac{I_{Lmax}}{2}$$



Average boundary mode $I_S = \langle i_D \rangle$

$$I_S = \frac{I_{Lmax}}{2} (1 - \alpha) = \frac{E^2 \cdot V_S}{(E + V_S)^2} \frac{T}{2L}$$

➡ $V_S = f(I_S)$

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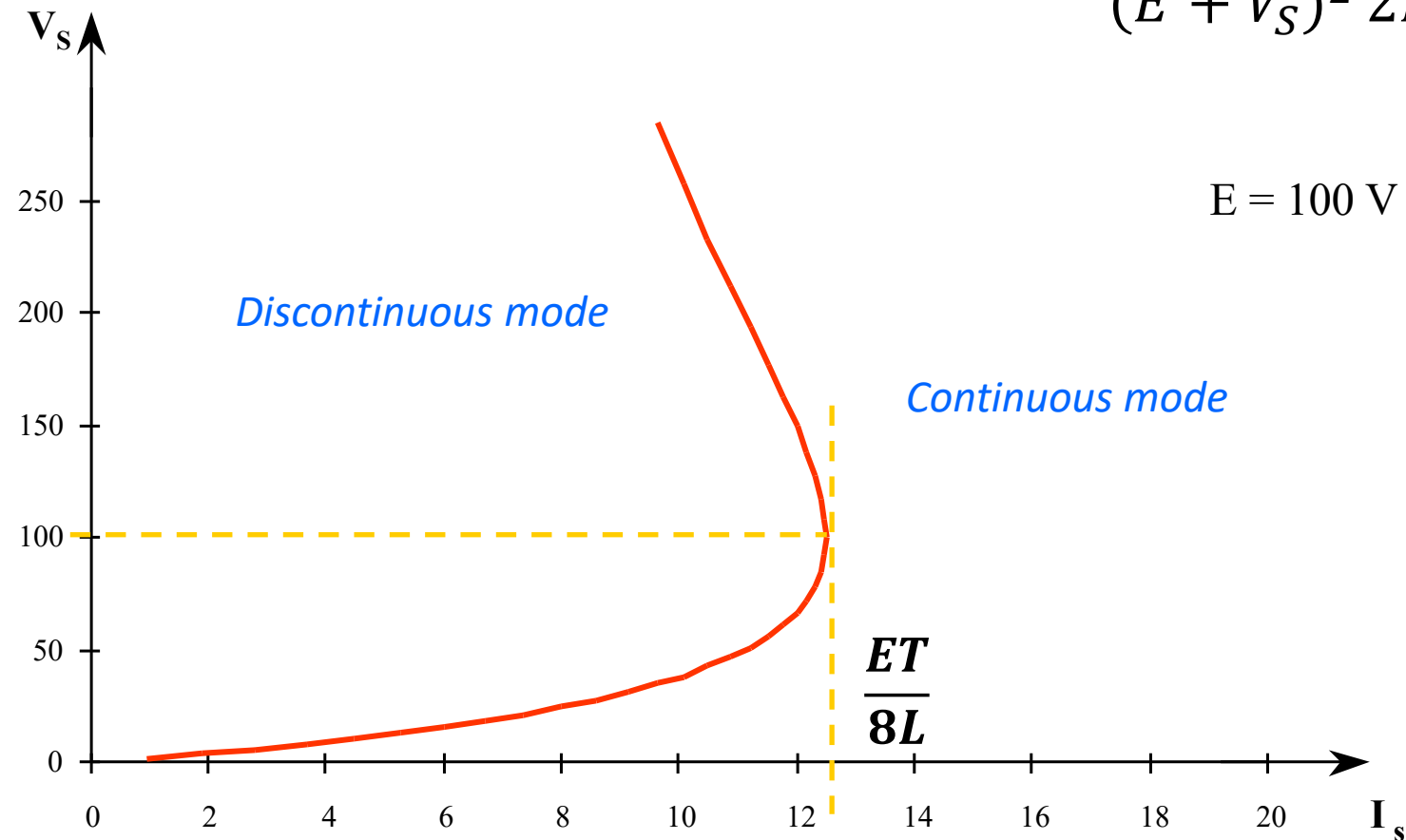
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Buck-boost: Boundary mode (characteristics)

$$I_S = \frac{E^2 \cdot V_S}{(E + V_S)^2} \frac{T}{2L}$$

$$E = 100 \text{ V}$$



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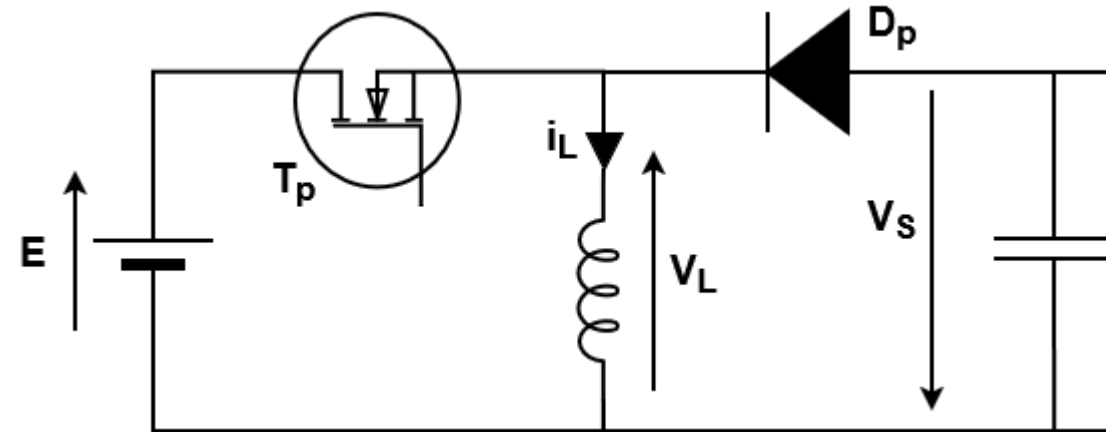
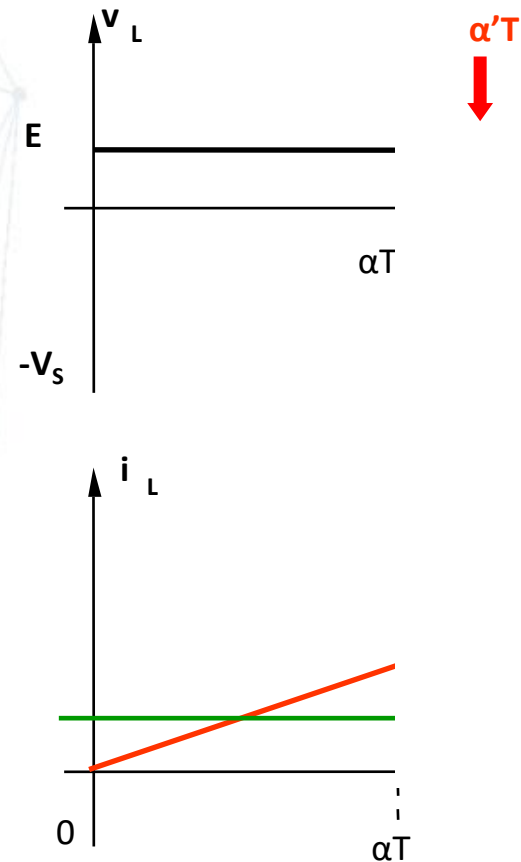
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Buck-boost: Discontinuous mode



Full study: Switched-mode power supply

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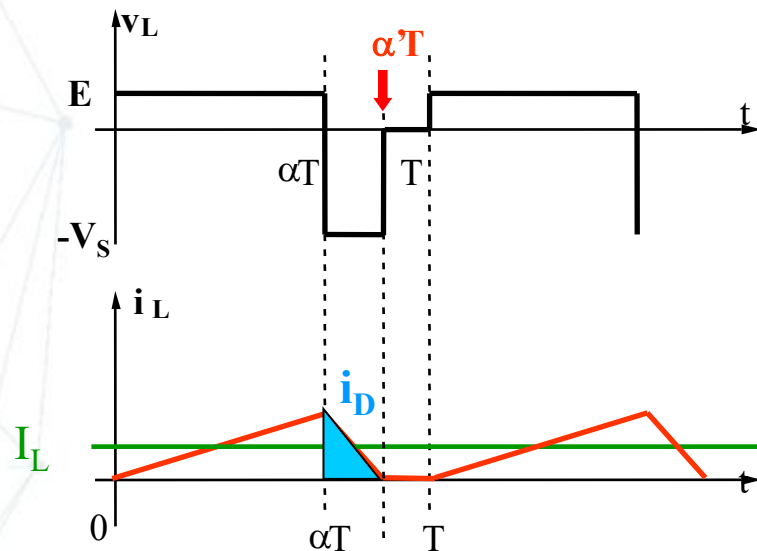
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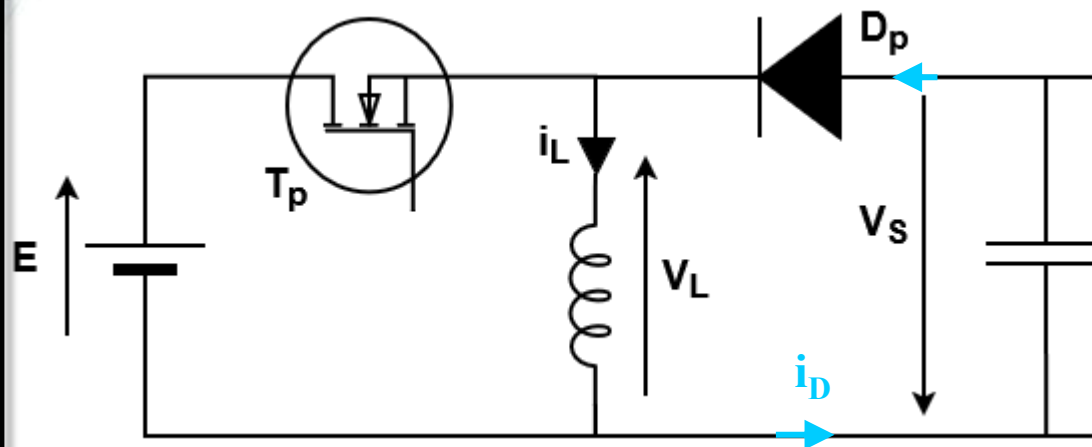
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Buck-boost: Discontinuous mode



$$\langle v_L \rangle = 0 = \alpha \cdot E - (\alpha' - \alpha) \cdot V_S \Rightarrow \alpha' = \alpha \frac{E + V_S}{V_S}$$

$$I_S = \langle i_D \rangle = (\alpha' - \alpha) \cdot \frac{I_{L\max}}{2} = (\alpha' - \alpha) \left(\alpha \frac{ET}{2L} \right)$$



$$I_S = \frac{\alpha^2 \cdot E^2 \cdot T}{2L \cdot V_S}$$

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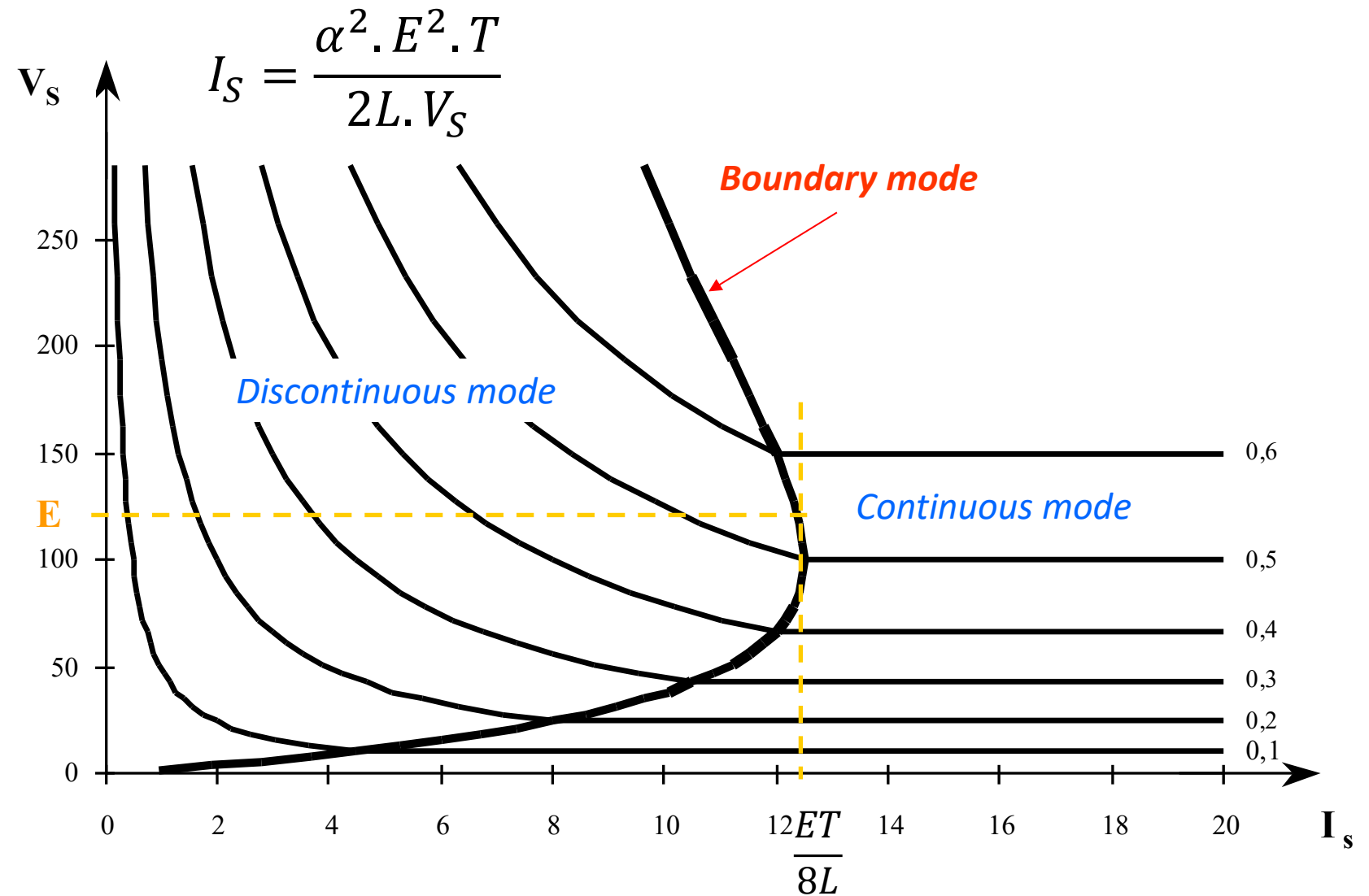
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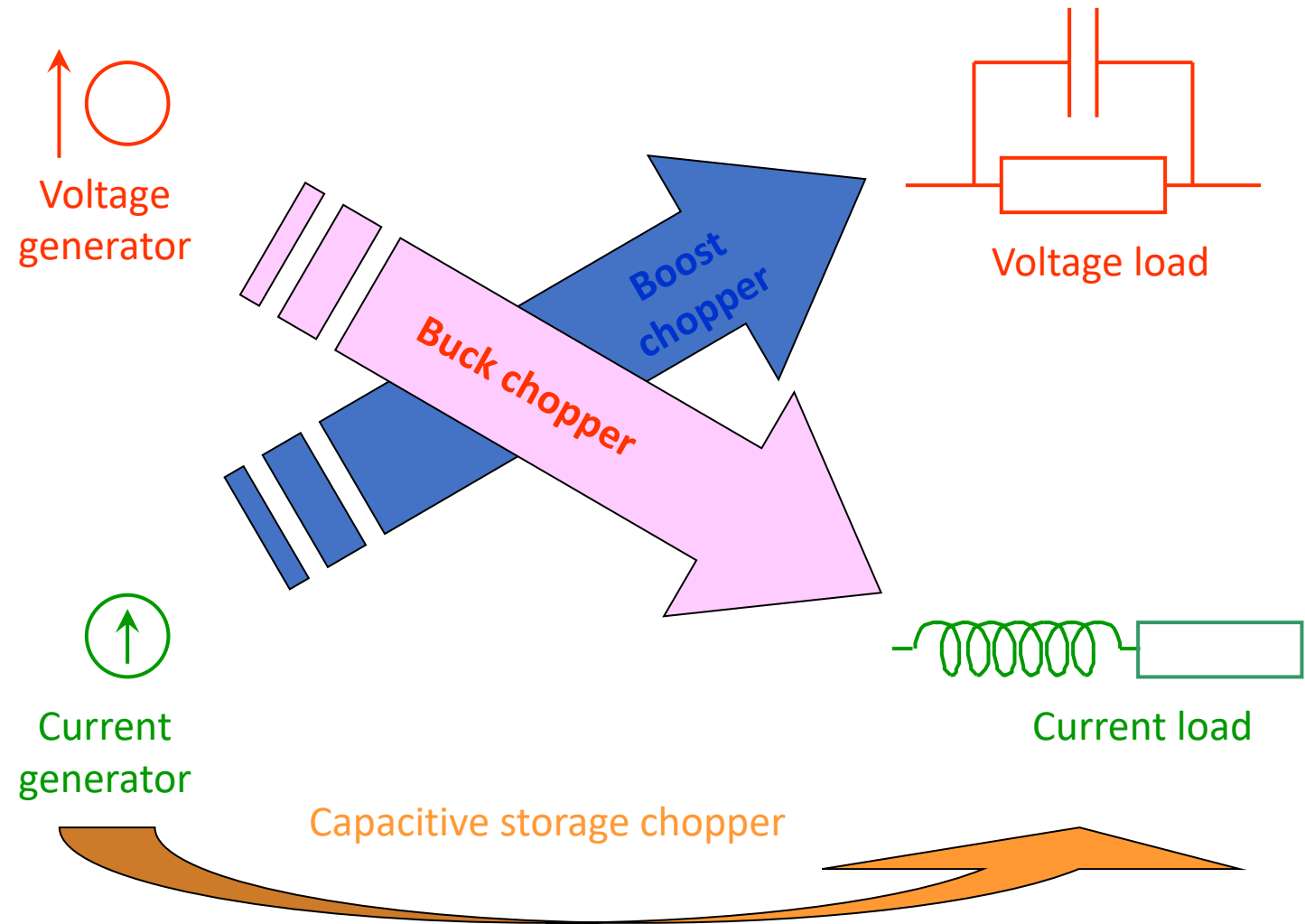
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Buck-boost: Output characteristics



Capacitive storage chopper



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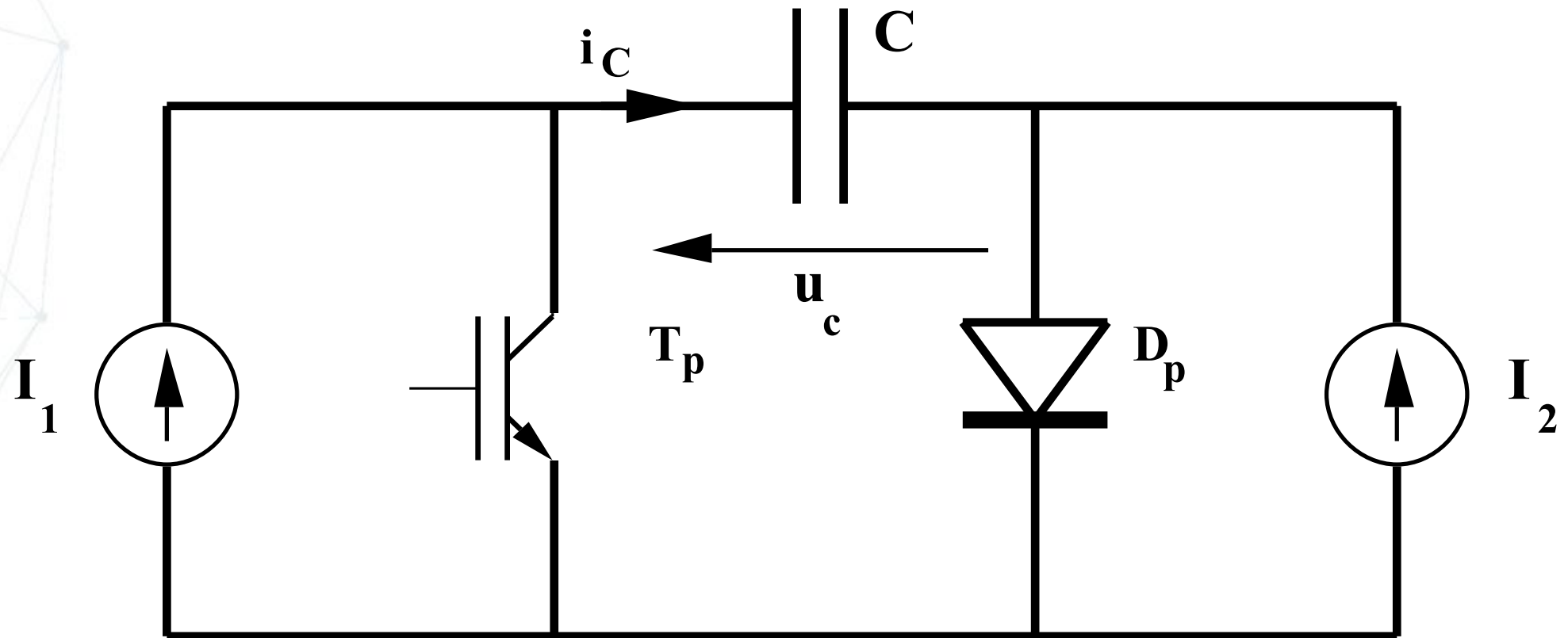
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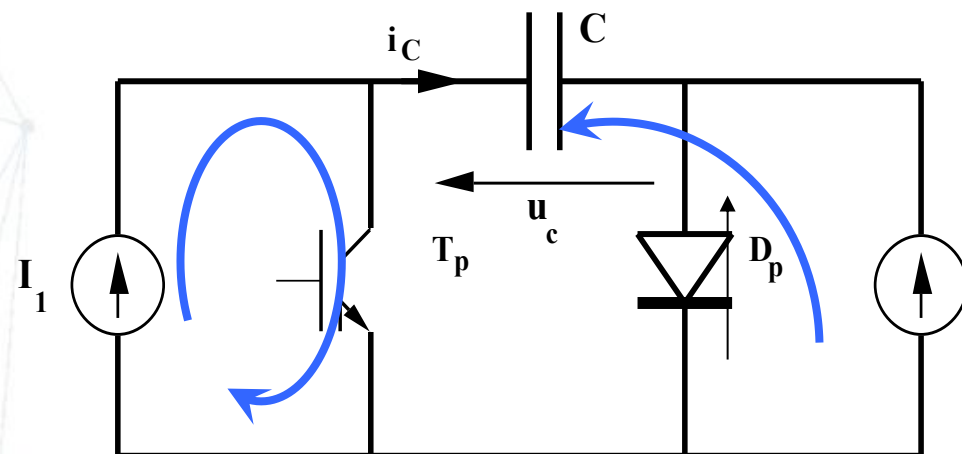
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CUK structure



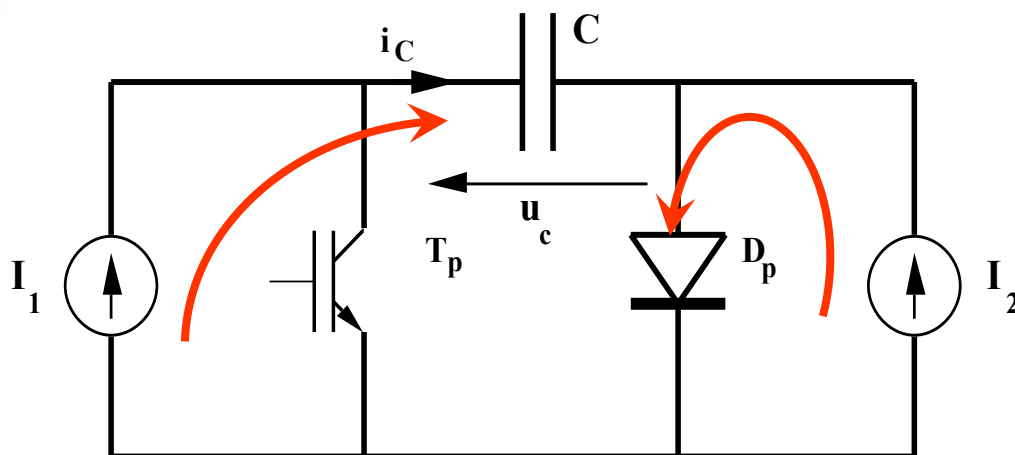
CUK structure: continuous mode ($u_c > 0$)



$0 < t < \alpha T : T_p \text{ closed}$

$v_{Dp} = -u_c \Rightarrow D_p \text{ reverse}$

$i_c = -I_2 = C \cdot du_c/dt \Rightarrow u_c \text{ decreases}$



$\alpha T < t < T : T_p \text{ opened}$

$i_c = I_1 = C \cdot du_c/dt \Rightarrow u_c \text{ increases}$

$i_D = I_1 + I_2 \Rightarrow D_p \text{ forward}$

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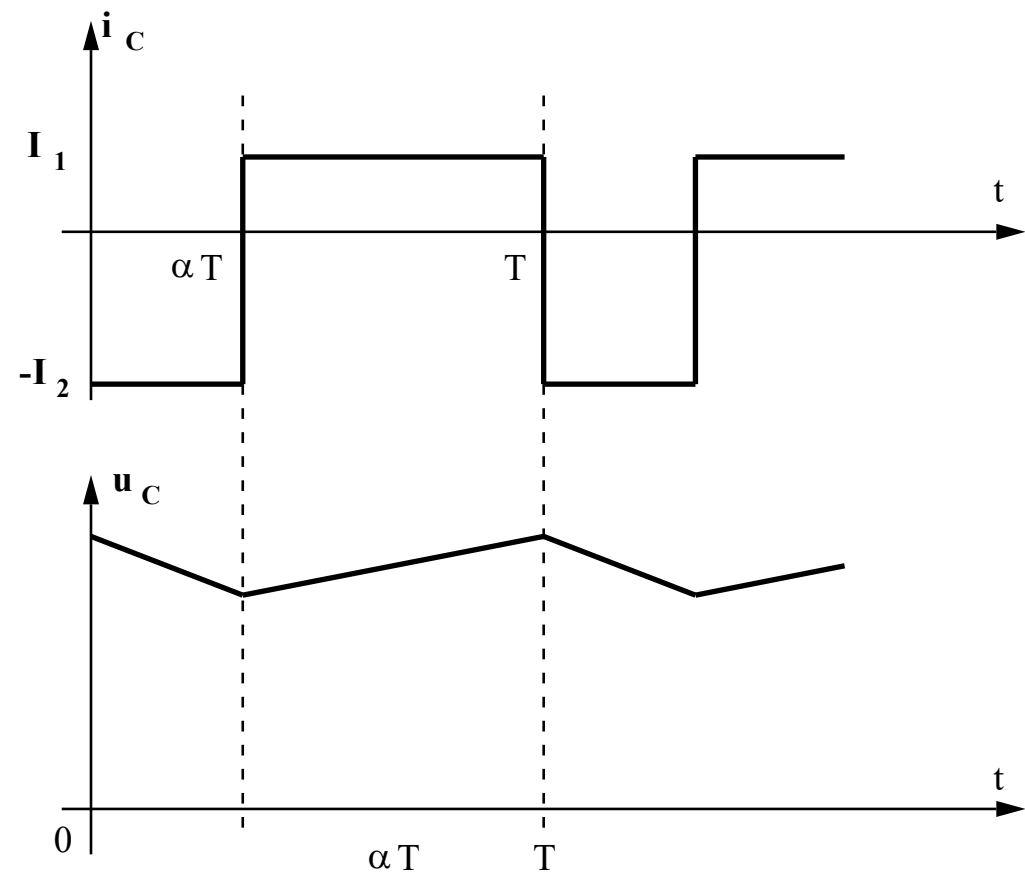
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CUK structure: Chronograms

$$\langle i_C \rangle = 0$$

$$I_2 = I_1 \cdot \frac{1 - \alpha}{\alpha}$$



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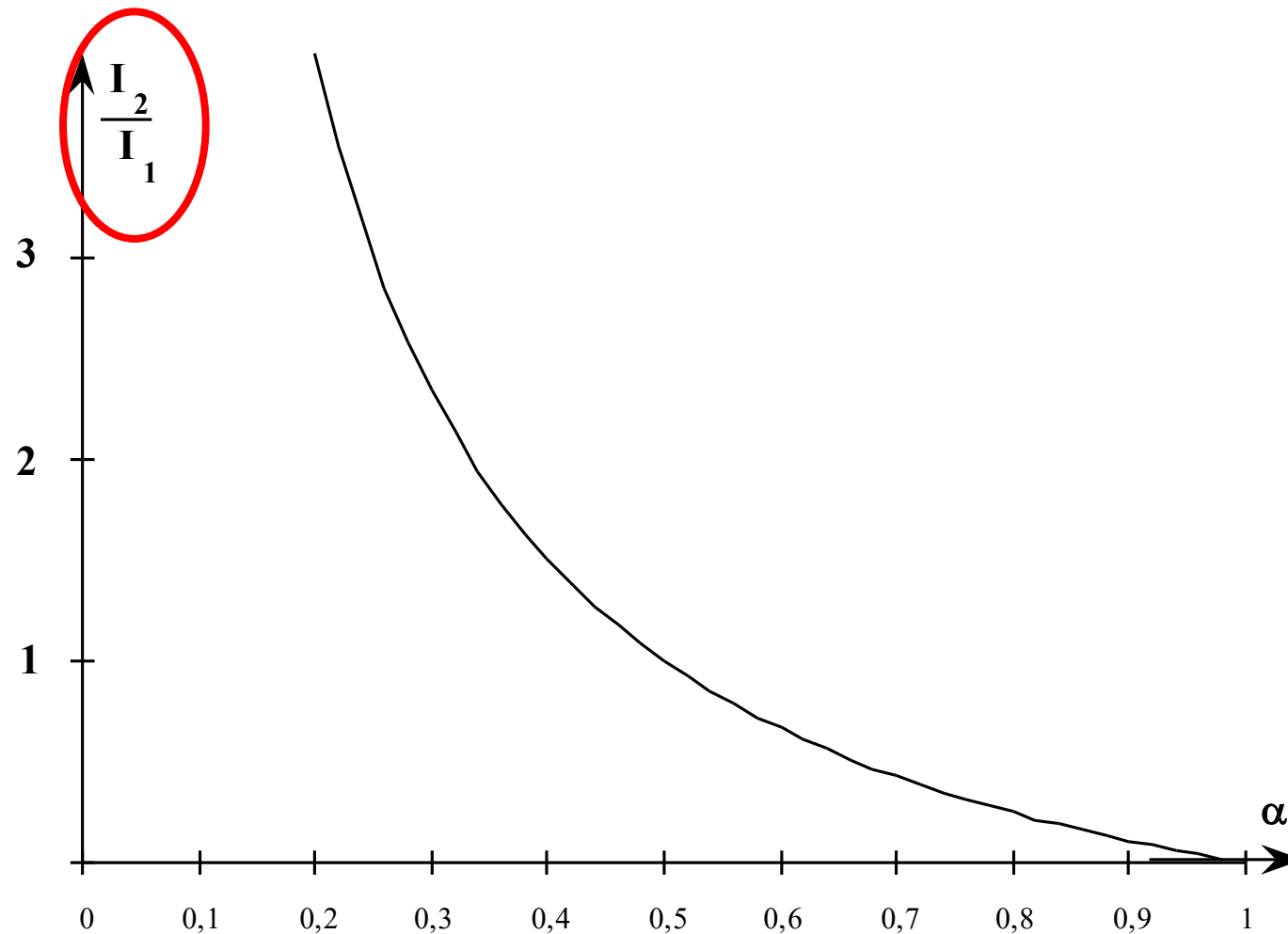
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CUK structure: Transfer characteristic



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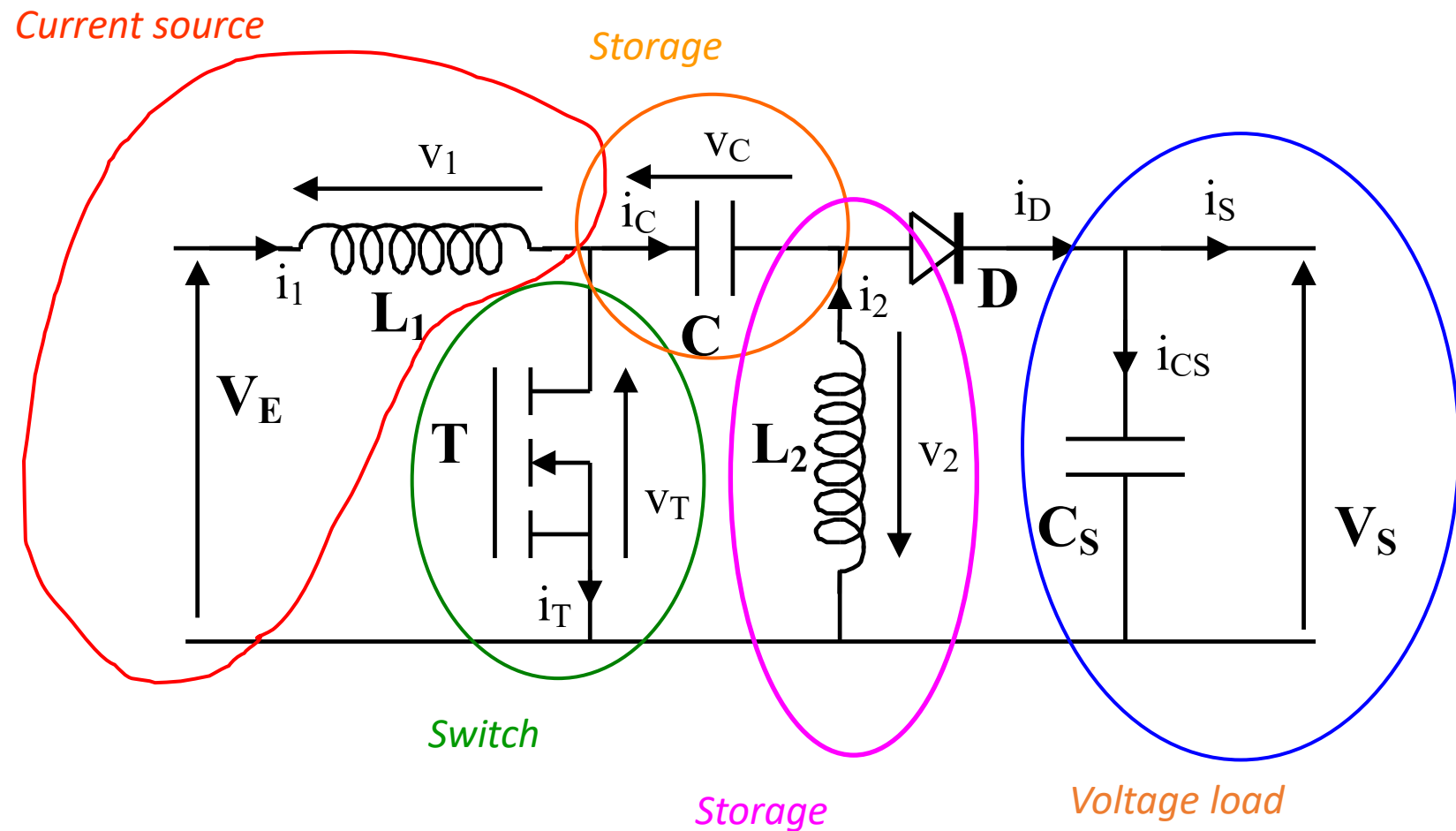
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SEPIC: Single Ended Primary Inductor Convertor



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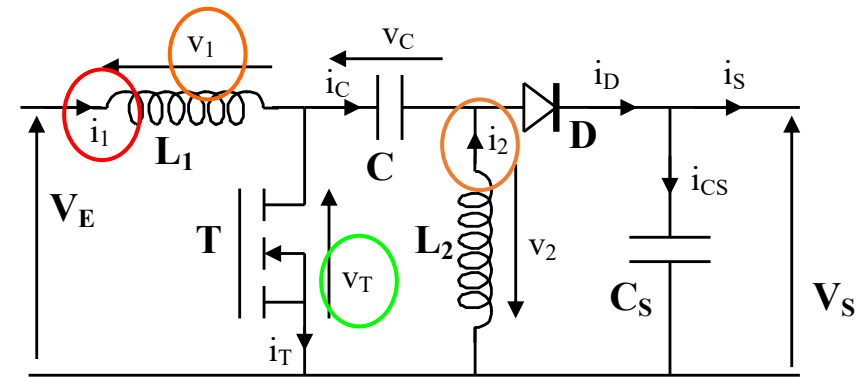
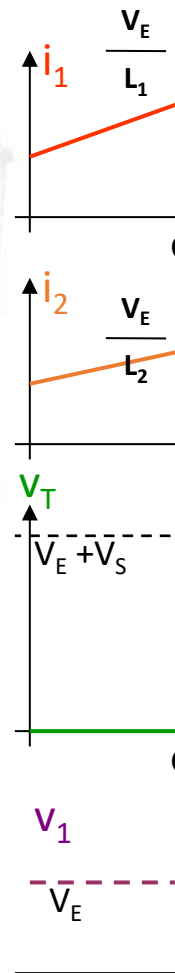
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SEPIC: Continuous mode (voltage transfer)



$$\langle v_1 \rangle = 0 \text{ et } \langle v_2 \rangle = 0 \Rightarrow \langle v_C \rangle = V_E = \text{Constant}$$

$$V_S = V_E \cdot \frac{\alpha}{1 - \alpha}$$

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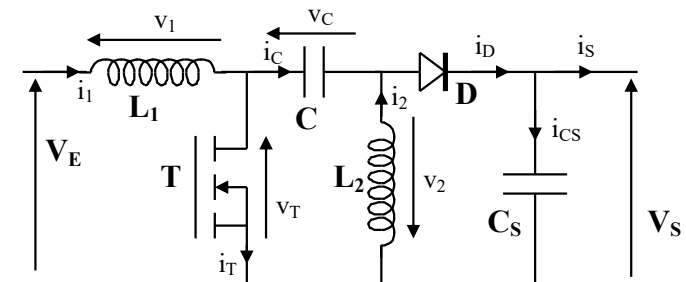
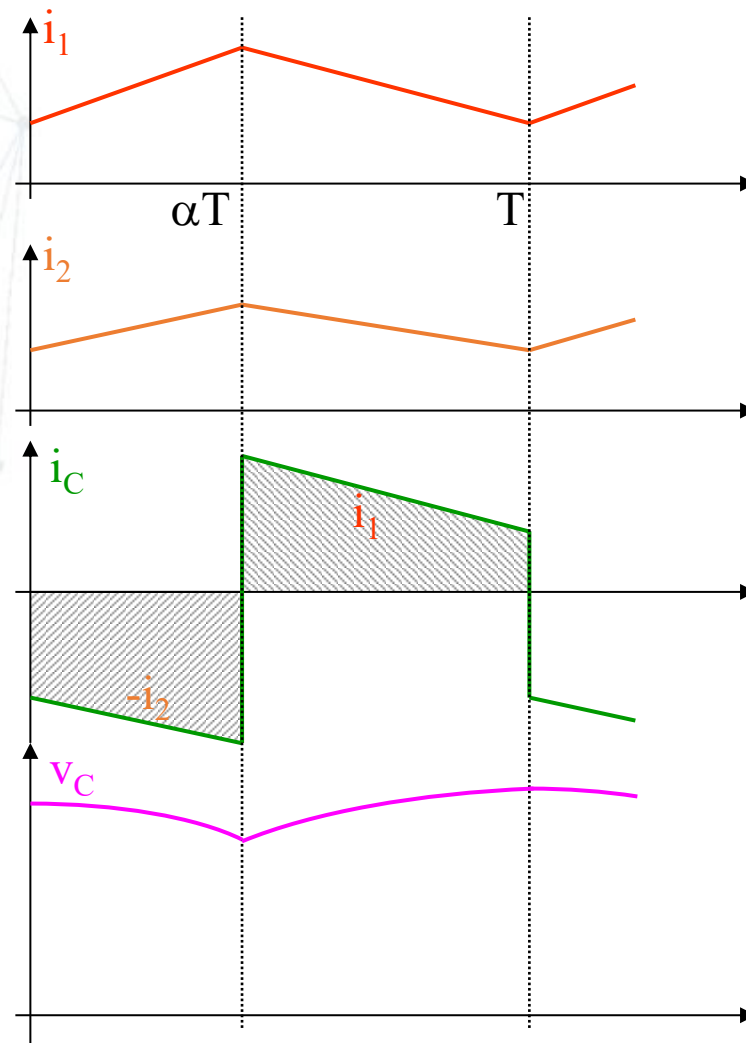
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SEPIC: Continuous mode (coupling capacitor)



$$I_2 = I_S \quad I_S = I_E \cdot \frac{1-\alpha}{\alpha}$$

$$\Delta V_C = \frac{\alpha \cdot I_S}{C \cdot f}$$

$$I_{Ceff} = I_S \cdot \sqrt{\frac{\alpha}{1-\alpha}}$$

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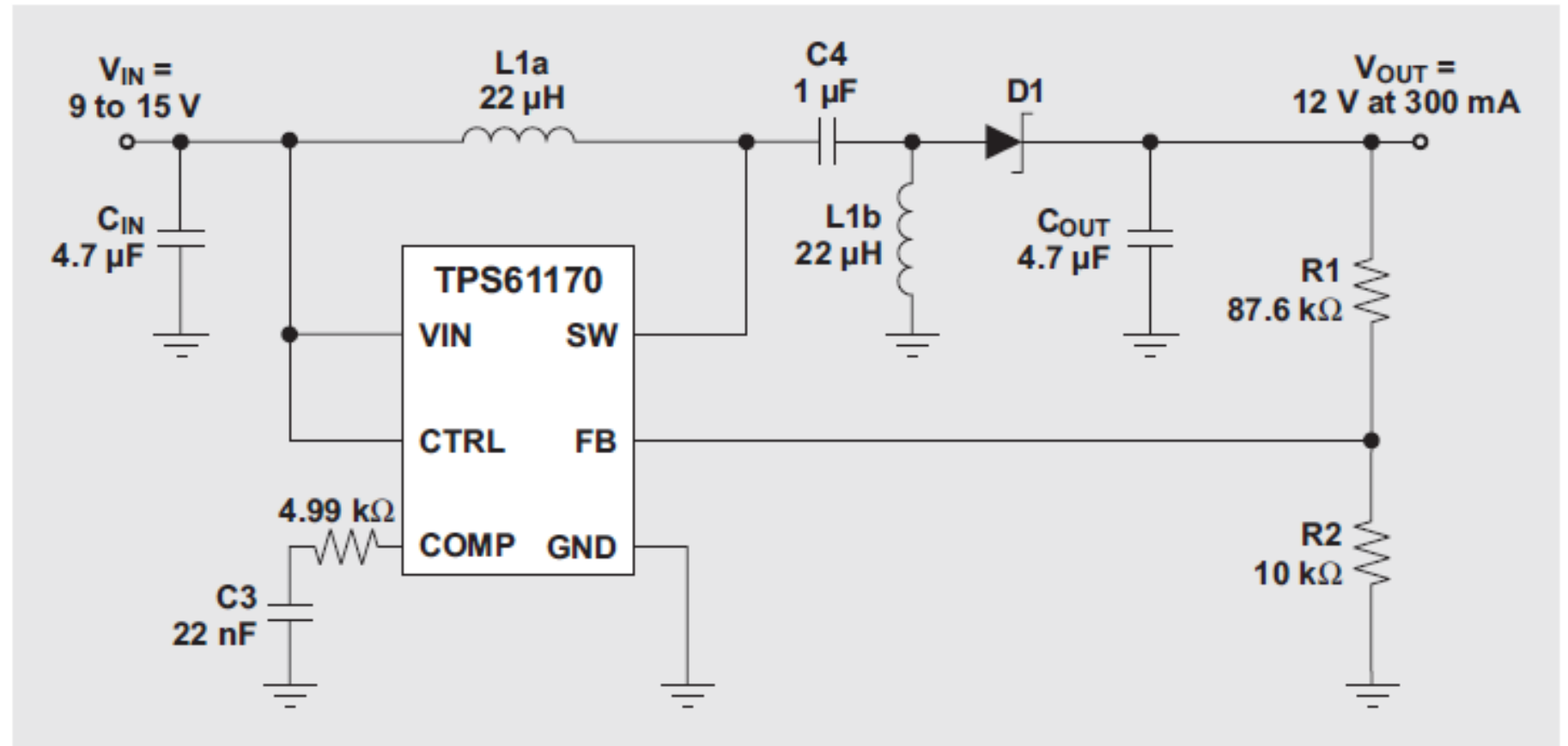
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SEPIC: circuit example
TPS 61170 – Texas Instrument



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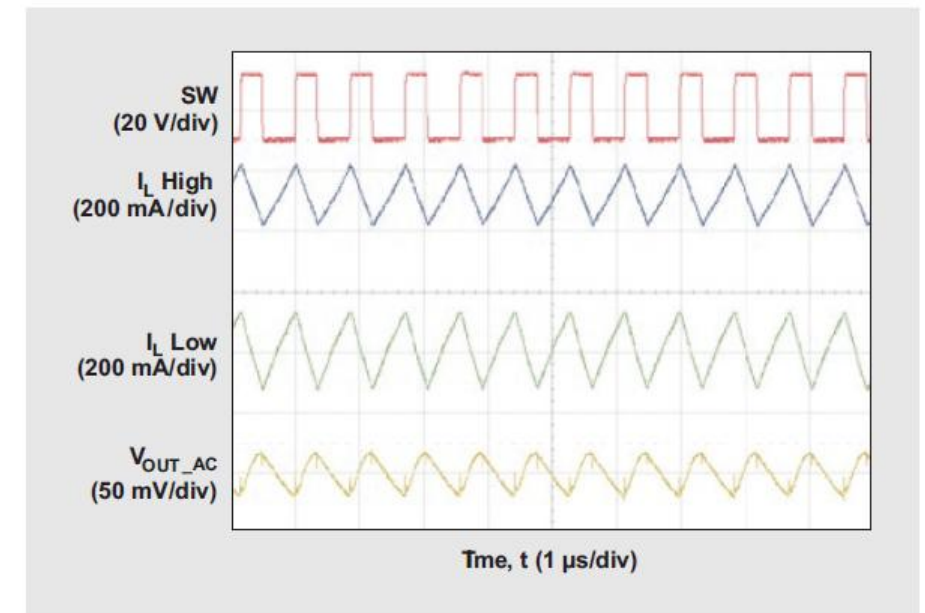
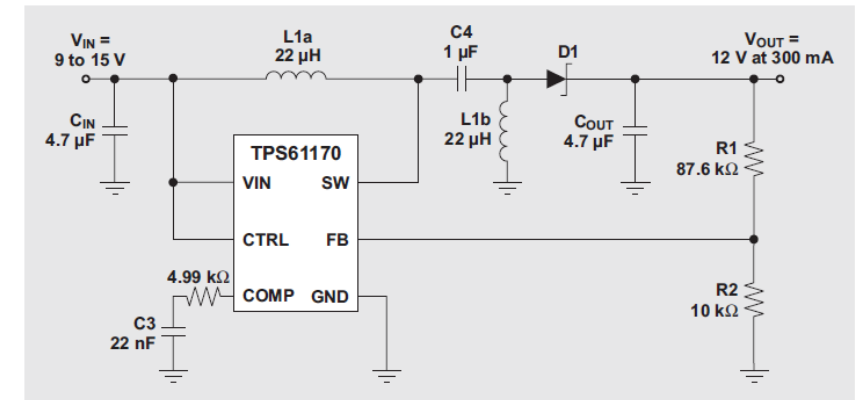
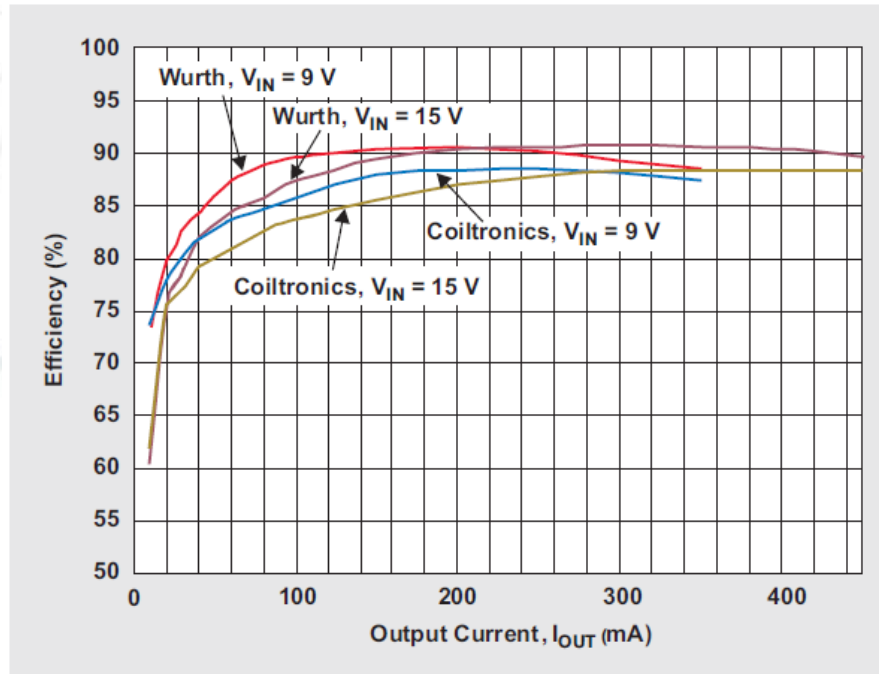
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SEPIC: circuit example
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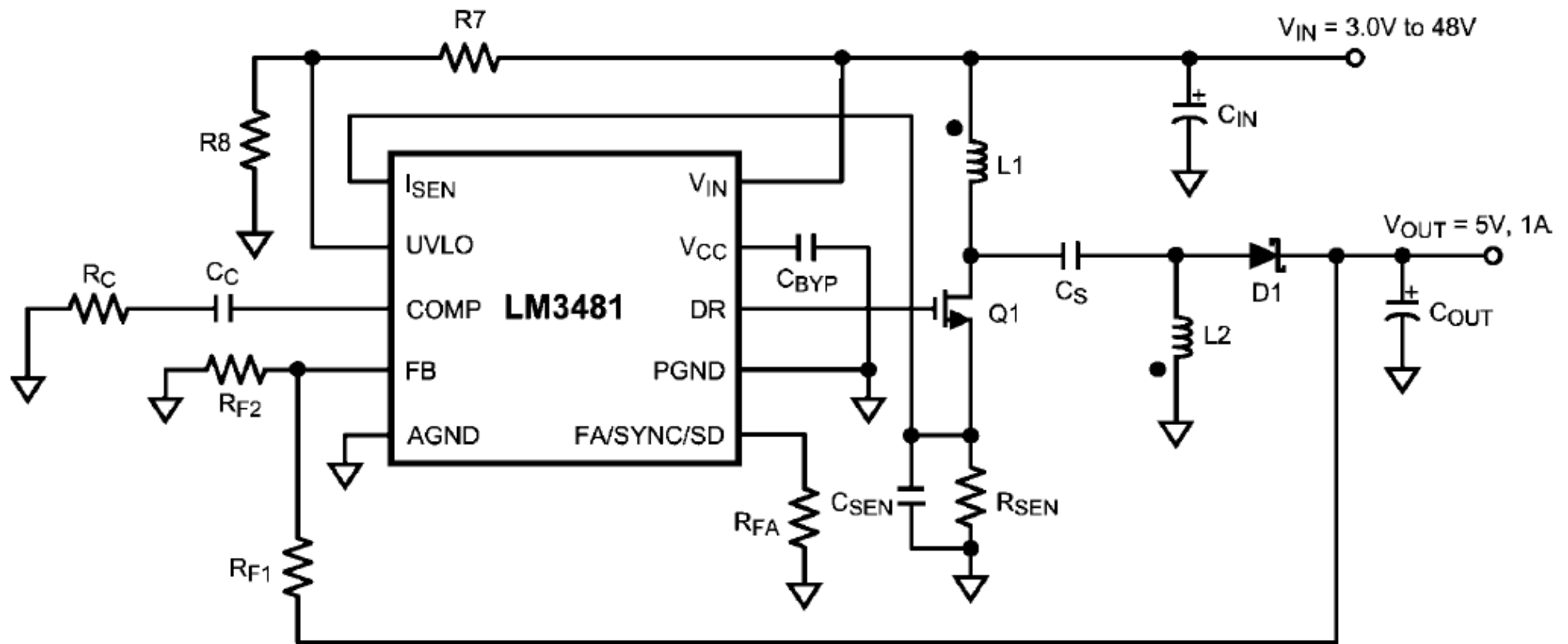
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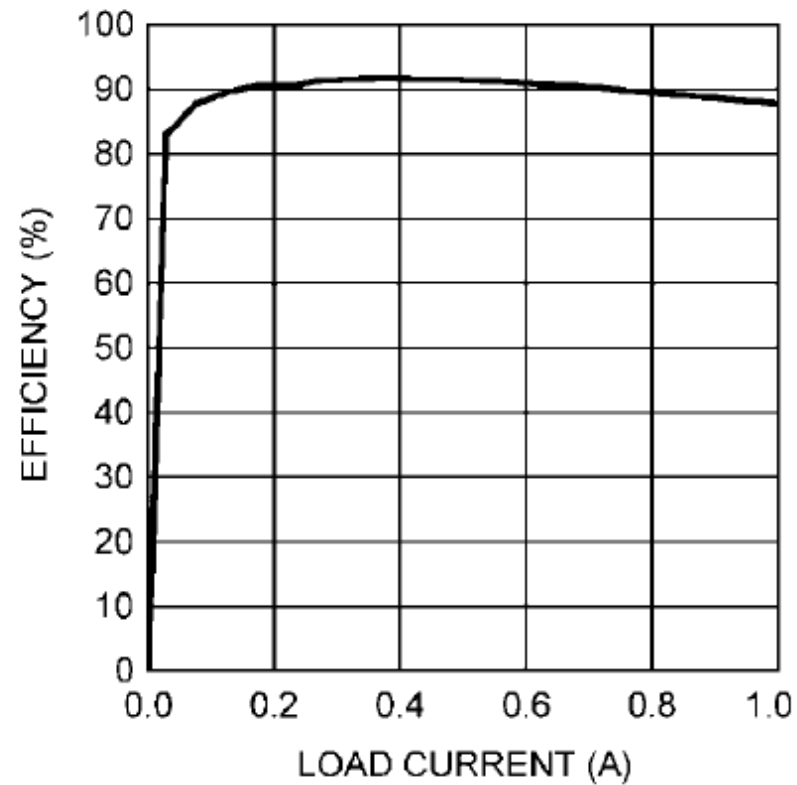
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Efficiency vs. Load Current ($3.3V_{IN}$ and $12V_{OUT}$)



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LM3481: internal schematic

